

AIrtist: Agentic Concept Development for Contemporary-Art Installations

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Abstract. Early-stage contemporary-art installation development often begins from an under-specified artist-authored problem rather than a resolved artifact. We introduce AIrtist, an agentic system that develops such briefs into five complete project hypotheses through persistent proposal lineages, domain-structured review, contestable intervention, and separate terminal ranking. Each incumbent is reviewed into an opportunity state that distinguishes what should be preserved from what should change; Focused and Structural challengers then compete with the unchanged incumbent rather than automatically replacing it. We evaluate AIrtist on 50 held-out commission-style briefs using exhaustive blind 5×5 candidate-pool comparisons, exact A/B presentation reversal, and a separately instantiated GPT-5.6 Sol holistic pairwise evaluator. Against a substantially matched shared-parent generic iterative control, AIrtist yields a centered complete-pool effect of +0.2352 (95% CI [+0.1472, +0.3212], Holm $p < .0001$). Across 250 lineages, 22 final states remain incumbent-identical; among 228 adopted replacements, challengers receive candidate-weighted exact-parent preference 0.6601 and a brief-balanced centered effect of +0.3347 (95% CI [+0.2143, +0.4520]). Terminal selection is less resolved: Base has higher mean Utility@1 than Hybrid (0.6633 versus 0.6353), while Hybrid more often places an externally highest-utility candidate first (0.32 versus 0.26); neither ranking policy establishes superiority. These results separate candidate-set construction, intervention quality, and attention allocation as distinct empirical problems. The evidence concerns textual proposals under model-mediated evaluation; it does not establish professional curatorial judgment, realized-artwork quality, or human co-creative benefit.

Keywords: contemporary art; installation art; concept development; creative agents; computational creativity; human-AI co-creativity; agentic AI.

1 Introduction

A consequential part of contemporary-art practice occurs before there is an artifact to fabricate, render, or exhibit. An artist may begin with a pressure rather than an image: a social relation that should become physically operative, a site whose existing use should not be erased, a material restriction, an institutional contradiction, or a conviction that several obvious artistic solutions would be wrong. At this stage, the artist may know what matters without yet knowing what the work should be. The problem is not how to execute a resolved concept, but how to develop alternative propositions for what the artwork could become.

Recent generative models make this stage newly accessible to computational support. They can produce plausible project descriptions, variations, references, images, and technical elaborations rapidly. Yet abundance of plausible completions does not by itself solve early concept development. In a contemporary-art installation, the same premise can lead to projects with radically different artistic operations: a concern may be represented symbolically, made causal through a material process, embedded in a spatial condition, delegated to an institutional procedure, or enacted through a spectator's position. A proposal can become clearer, longer, more interactive, or more technically sophisticated while becoming artistically weaker. Several incompatible directions may also remain viable at once. The computational problem is therefore not simply to generate more material, but to construct useful alternatives, decide what in an existing proposal should survive intervention, and recognize which resulting possibilities deserve attention.

Installation practice makes these difficulties particularly visible because the identity of a work can depend on relations among material, bodies, space, duration, site, and institutional framing

rather than on a detachable artifact alone [1, 2]. We use artistic operation for the relation or mechanism through which a proposal enacts its stakes: what the work actually does through its material, spatial, temporal, spectator, or institutional organization. This does not define artistic quality, but it gives concept development a more discriminating computational object than topic, style, or visual appearance. A system working at this stage must reason not only about whether a proposal is coherent, but about which relation gives it identity and what kinds of change would preserve, sharpen, or destroy that relation.

Recent artistic practice already shows generative systems participating while the identity of a future artwork is still being articulated, including Kuran’s Errorism [3], Lund’s In the Middle of Nowhere [4], and Cai Guo-Qiang’s cAI [5, 6]. Section 3 examines these precedents in detail.

These practices establish the use case but remain artist-specific procedures embedded in particular oeuvres and projects. Adjacent computational research, meanwhile, provides sophisticated machinery for early ideation, including professional concept design and scientific hypothesis generation, iterative review, and multi-candidate development [7, 8, 9]. What remains largely unformalized is the task between them: taking a previously unseen, under-specified contemporary-art problem and systematically developing several complete proposals for what the artwork itself could become.

We introduce AItist, an agentic system for this task. An artist supplies a rough contemporary-art installation brief describing the pressure they want to investigate, conditions that must remain consequential, directions they want to avoid, and—where relevant—site, material, institutional, or production constraints. We use concept development for this domain-facing task; computationally, AItist represents it as search over persistent proposal states. The system treats the brief as a provisional problem specification rather than as a description of a desired artifact, develops five complete project hypotheses, allows existing states to survive critique, and finally ranks the resulting five-proposal set.

To our knowledge, AItist is the first evaluated computational system whose primary task is to develop an under-specified artist-authored contemporary-art installation brief into multiple complete hypotheses for what the artwork itself could become. The computational object is the artwork proposition itself: an artistic operation together with its material, spatial, spectator, field, and institutional realization.

A held-out example illustrates the distinction. Brief 053 asks for an installation about contested maintenance using canvas, galvanized mesh, and hand-mixed pigment, with every component repairable in public during opening hours. In the parent proposal, Echoing Gutter, a six-meter galvanized-mesh curtain spans a corridor. Pigmented water is released onto the mesh, and a visitor uses a broad brush to spread the wet streak back across the surface until the concentrated mark becomes less visible. The cycle makes damage and repair public, but it is largely reversible: each intervention returns the mesh toward its previous appearance, while canvas—although required by the brief—remains peripheral.

One Structural challenger generated by AItist, Layered Ledger, reorganizes the installation as mesh panels with pigment-receptive canvas behind them. A fresh pigment line is again released onto the mesh. Brushing now has two simultaneous effects: it redistributes the concentrated wet streak across the front while driving part of the pigment through the mesh openings toward the canvas, where it is absorbed and leaves a persistent ghost trace. Each cycle therefore returns the front toward visual uniformity while irreversibly adding to the record behind it. The artistic operation changes from damage → repair → reset to damage → repair → {front returns toward uniformity, history accumulated}. The intervention changes what maintenance physically does. In the evaluation, the challenger passed eligibility but narrowly lost the lane comparison to the incumbent; it is used here to illustrate an operation-level transformation rather than successful selection.

This example also exposes why generic iterative refinement is an incomplete model of the task. A critique may correctly identify a weakness while the proposed repair destroys the strongest relation in the parent. Conversely, a proposal may contain a valuable operation even when some part of its realization should be replaced. Additional inference should therefore be able to test a possible transformation without making transformation mandatory. Likewise, producing several proposals creates a second problem: a strong direction may already exist in the candidate set while a ranking policy fails to surface it. Concept development, intervention, and selection are consequently related but empirically separable problems.

AItist operationalizes this view through five persistent proposal lineages. Each incumbent first receives heterogeneous specialized review. The resulting evidence is compiled into an opportunity state that identifies what should survive, the dominant liability, a familiar field

logic to escape, an unused material or spatial opportunity, a decisive transformation to attempt, and additive fixes that should not count as progress. Two challengers are then generated independently from the same unchanged parent. A Focused reconstruction rebuilds the diagnosed weak part while preserving the incumbent property marked for protection; a Structural escape has broader permission to replace the dominant operation or realization and must produce a materially distinct alternative rather than an elaboration of the incumbent. Crucially, the incumbent remains eligible. Generating a challenger therefore does not require accepting it. After one state survives in each lineage, a separate terminal selector ranks the five resulting proposals.

The evaluated system is intentionally autonomous within this bounded stage: the artist provides the brief once and receives five ranked project hypotheses without intervening during the run. AItist is therefore a concept-development backend rather than a complete conversational co-creative partner. Negotiation over diagnoses, criteria, branches, and longer-term artistic memory is left to future artist-facing systems.

The experimental design follows the same decomposition. RQ1 asks whether domain-structured development produces a stronger five-proposal set than raw generation, a generic self-refinement pass, and a substantially matched generic iterative process. The strongest control, M5, starts from the exact same five self-refined parent proposals as AItist, maintains the same number of lineages and challengers, keeps the incumbent eligible, and shares challenger gating and lane comparison, while removing AItist’s specialized review, opportunity representation, and differentiated intervention roles. RQ2 follows proposal identity through development and asks whether replacements adopted by the evaluated policy are externally preferred to the exact parent states they displace. RQ3 holds the generated five-proposal set completely fixed and changes only the terminal ranking policy, isolating recognition from generation.

We evaluate the system on 50 held-out commission-style briefs. For RQ1, every AItist proposal is blindly compared with every proposal in each of three control pools, yielding complete 5×5 cross-pool matrices rather than evaluation of only system-selected winners. Each logical comparison is repeated under exact A/B presentation reversal. The primary external evaluator is a separately instantiated GPT-5.6 Sol holistic pairwise judge that does not observe system identity, lineage history, internal scores, or the external candidate utilities later derived from its judgments. Statistical inference is performed over briefs rather than individual candidate comparisons.

The results distinguish the three stages rather than producing a uniform end-to-end success story. For candidate-set construction, AItist significantly exceeds all three planned controls after Holm correction. Against the substantially matched M5 condition, the centered complete-pool effect is $+0.2352$ (95% CI $[+0.1472, +0.3212]$, Holm $p < .0001$); secondary diagnostics indicate that the gain is not confined to one oracle-best candidate. For intervention, 22 of 250 final lineage states remain content-identical to their incumbents. Among the 228 replacements actually adopted, candidate-weighted challenger preference against the exact displaced parent is 0.6601, with a brief-balanced centered effect of $+0.3347$ (95% CI $[+0.2143, +0.4520]$). Selection is less resolved. Base’s mean Utility@1 is 0.6633 versus Hybrid’s 0.6353 on identical candidate pools, while Hybrid identifies an externally highest-utility candidate first on 32% of briefs versus 26% for Base; the primary Hybrid-minus-Base difference is not statistically resolved and secondary ranking metrics are mixed. The experiment therefore provides strong evidence for higher measured candidate-set quality and for external preference toward adopted replacements, while identifying terminal recognition as a distinct unresolved problem.

These results suggest a broader diagnostic view of creative agents. A weak surfaced proposal can arise because the system failed to construct strong alternatives, because an intervention damaged an existing state, or because ranking hid useful material that had already been generated. Evaluating only a final top-ranked output collapses these failure modes. Contemporary-art concept development makes the distinction especially visible because a change in material causality, spatial organization, spectator position, or institutional procedure can alter the identity of the proposal rather than simply improve its presentation. Generative assistance can induce fixation [10] and, in other settings, improve individual outputs while narrowing collective diversity [11], reinforcing the need to distinguish candidate quality from the mere existence of multiple branches.

The paper makes four main contributions:

- 1. Contemporary-art concept development as an AI task.** We formulate and evaluate early-stage installation concept development from under-specified artist-authored briefs, where

the computational object is a set of alternative project hypotheses rather than the realization of an already resolved artwork.

2. A contestable agentic development architecture. AIRTIST maintains persistent proposal lineages, converts heterogeneous review into an explicit opportunity state, generates Focused and Structural interventions from the same incumbent, and allows the incumbent to survive. The architecture represents what should remain invariant separately from what may need to change.

3. A multi-level evaluation methodology for open-ended creative agents. We separately evaluate the quality of the sampled possibility set, adopted interventions against their exact parents, and terminal ranking over identical generated candidate sets, including a substantially matched shared-parent iterative control.

4. Empirical evidence about where agentic value and error arise. Domain-structured development improves measured candidate-set quality over matched generic iteration; adopted replacements are preferred to the exact states they replace on average; and terminal selection remains an independent recognition problem even when stronger measured alternatives already exist in the generated set.

The claims remain bounded by the evaluation setting. AIRTIST does not establish autonomous artistic agency, professional curatorial judgment, or improved human creativity. The evaluated objects are textual project hypotheses, the principal quality signal is model-mediated, and no artist negotiates with the system inside the evaluated experimental pipeline. The present study instead asks whether early contemporary-art concept development can be made into an explicit computational task and whether the proposed agentic development policy produces measurable improvements under a disclosed evaluation procedure.

Paper organization. Section 2 develops the contemporary-art formulation underlying the task, including artistic operation and the bounded working quality model. Section 3 positions AIRTIST relative to contemporary artistic practice, AI-assisted ideation, and agentic refinement. Section 4 describes the AIRTIST architecture, and Section 5 specifies the benchmark, controls, external evaluation, and inference procedure. Sections 6 and 7 report the main results and mechanism/failure analysis. Section 8 discusses implications for creative-agent design; Section 9 develops the intended artist-facing use and future human-negotiated search; Section 10 states limitations and threats to validity; and Section 11 concludes.

2 Representing Contemporary-Art Installation Ideation as a Computational Problem

AIRTIST does not generate a physical installation. It generates a proposal for a future installation: a textual account of what would be installed, how its material and spatial organization would behave, how spectators would encounter it, what institutional conditions it would depend on, and why those choices belong to the artistic proposition rather than merely to its fabrication. The computational object in this paper is therefore not the realized artwork but a sufficiently developed textual dossier representing an installation hypothesis. We use *dossier* for a proposal developed enough to support further concept development and comparative judgment, not for a fabrication-ready specification. Further artistic, curatorial, engineering, conservation, and site-specific decisions remain expected.

The evaluated pipeline is autonomous after brief submission, but its intended artist-facing role is co-creative rather than substitutive. It develops candidate directions that an artist may inspect, reject, combine, reinterpret, or develop further. The method does not assume that the model captures the artist's tacit judgment, broader practice, or all physical, material, and institutional contingencies of realization. A realized installation may therefore depart substantially from its dossier through scale, fabrication, material accident, maintenance, institutional negotiation, audience behavior, or later artistic decisions. Such departures are compatible with the intended workflow rather than failures to reproduce a fixed specification.

A separate difficulty concerns representation within the textual task. A language model can readily generate a title, describe a room and a set of objects, propose a material palette, and produce persuasive curatorial prose. None of these, separately or together, guarantees that the text specifies a coherent installation. Two dossiers may address the same topic and use almost the same materials while proposing fundamentally different works because bodies move differently through them, matter changes differently over time, access is withheld under different conditions, or the institution plays a different role. Conversely, works that look unrelated at the level of object vocabulary may depend on closely related temporal, material, spatial, perceptual,

spectator, or institutional dependencies. For installation ideation, representing the components is therefore not enough: a small change in their relations can change what the proposed work does.

The study therefore makes a limited representational assumption: a textual dossier can describe a proposed installation in enough detail to support meaningful computational reasoning about the relations through which the work is intended to operate. “Enough detail” here has a practical meaning. The dossier should state what exists, what changes, what depends on what, what a spectator can perceive or do, which material properties matter, and which production or institutional conditions are necessary. It need not—and in practice cannot—determine every property of the eventual artwork.

For proposal development, we distinguish three analytical levels:

- **Artistic stakes:** What tension, contradiction, experience, or question makes the proposed work worth pursuing?
- **Artistic operation:** Which specified dependency makes that tension occur in the proposed work rather than merely being named by it?
- **Realization:** What concrete materials, spatial arrangement, timing, technical system, and institutional conditions instantiate that operation here?

These are not claimed to be ontological components present in every artwork. They form a computational scaffold for proposal development: three aspects that the system must be able to distinguish if it is to alter one part of a proposal without inadvertently erasing another.

The section develops this representation in two stages. First, installation theory and established artistic practice motivate why space, duration, material behavior, embodied spectatorship, site, and institutional context may be constitutive rather than incidental. Second, we translate those concerns into an explicit, bounded computational surrogate of proposal quality for one task: developing contemporary-art installation proposals under museum-commission conditions. The theoretical literature motivates the kinds of relation that the representation should be able to express; it does not directly determine the computational quality model introduced later or how AIRTIST operationalizes it.

2.1 Installation as a Situated Configuration

A useful starting point is that installation is rarely exhausted by a self-contained object. Bishop foregrounds the spectator’s bodily presence within the work; Petersen describes installation as activating physical space and context, extending the work in time, and emphasizing embodied perception; and Kwon traces a transformation of site from fixed physical location toward institutional, social, and discursive conditions [1, 12, 2]. O’Doherty’s analysis of the white cube makes a complementary point: gallery space actively frames the conditions under which something is encountered as art rather than serving as a transparent container [13]. Rebentisch treats site-specific installation as engaging the interrelation of literal and social sites [14]. These positions are not interchangeable, and AIRTIST does not combine them into a single definition of installation. Morris’s account of sculpture and Burnham’s systems aesthetics make the move away from the isolated object especially explicit: placement, surrounding space, the viewer’s changing position, and relations among people and environmental components can become constitutive conditions of the work [15, 16]. For the present computational problem, this literature supports a narrower representational premise: an installation may be constituted not only by the intrinsic properties of objects but by dependencies among material, space, duration, spectatorship, and institutional context.

We do not adopt these theoretical positions as a ready-made computational definition. Their relevance is practical. They warn against representing an installation as a list of objects plus a theme. If changing a route, a temporal condition, a spectator position, a material behavior, or an institutional rule can change what the work does, those elements must be available to the computational system as first-class parts of the proposal.

Table 1 makes this representational consequence concrete across ten heterogeneous works. It does not propose a taxonomy of installation or attribute AIRTIST’s terminology to the artists. The physical-encounter column summarizes documented features of each work in enough detail to make the relevant relation intelligible without prior familiarity with the work; the artistic-operation column gives our analytical reading of one constitutive dependency relevant to the present representation rather than an exhaustive account of the work; and the final column identifies the realization conditions on which that dependency relies.

Table 1: Artistic operations and their realization across heterogeneous installations.

Work	What a visitor physically encounters	Artistic operation	Realization conditions
James Turrell, <i>Frontal Passage</i> (1994) [view]	A visitor passes through a darkened entryway into a dark chamber. Concealed light sources produce a radiant, sharply bounded field of red light across the space. The source itself is not visible; the apparent boundary is light, yet it seems to end abruptly in space as if it had density or formed a physical plane. [17]	<i>Situated viewer ↔ controlled light field</i> : under these viewing conditions, immaterial light is perceived as a bounded, apparently physical presence.	Darkened entry and chamber; concealed light sources; a sharply controlled red-light field; a viewer positioned within the constructed viewing environment.
Janet Cardiff, <i>The Forty Part Motet</i> (2001) [view]	Forty separately recorded choir voices are played simultaneously through forty individual loudspeakers arranged in eight groups of five around a large oval. Moving close to one speaker isolates an individual singer; standing between speakers produces combinations of nearby voices; from the center, the listener can hear the forty voices together as a choir. [18]	<i>Listener position ↔ spatially separated voices</i> : movement through the speaker field changes the audible relation between individual singers, local combinations of voices, and the choir as a whole.	Forty separately recorded vocal tracks; one loud-speaker per voice; spatial separation of the speakers; a navigable listening space in which position changes the heard mix.
Mel Bochner, <i>Measurement Room</i> (1969) [view]	The actual gallery room is measured, and strips of black tape and numerical notation mark the dimensions of the same walls, doorways, windows, and passageways that the visitor is standing among. The visitor therefore occupies a room whose physical dimensions are displayed directly on the architecture being measured. [19]	<i>Actual architecture ↔ numerical description of that same architecture</i> : the room is encountered simultaneously as physical space and as a measured representation of itself.	The actual room; measurements derived from that room; notation applied directly to the architectural elements being measured; the viewer inhabiting the measured space.
Doris Salcedo, <i>Shibboleth</i> (2007) [view]	A long, irregular fissure opens through the floor along Tate Modern's Turbine Hall. It varies in width and depth, and its concrete sides reveal steel mesh within the rupture. Rather than appearing as a separate object placed in the hall, the fissure occupies the floor across which visitors themselves move. [20]	<i>Museum circulation ↔ rupture in the floor being traversed</i> : a surface ordinarily encountered as continuous is physically interrupted, making division part of the visitor's bodily passage through the hall.	The actual Turbine Hall floor; an extended fissure within that circulation surface; sufficient width, depth, and material presence for the rupture to be physically encountered; visitor movement through the same space.
Hans Haacke, <i>Condensation Cube</i> (1963–67) [view]	A transparent acrylic or methacrylate cube contains water in a closed container. Light warms the interior; water evaporates and condenses on the inner walls as droplets that grow, merge, and eventually run down the transparent surfaces. The patterns continually change with light and temperature in the surrounding exhibition space, and even nearby bodies can slightly alter those conditions. [21, 22]	<i>Contained water system ↔ local exhibition environment</i> : changing light and temperature conditions continually produce different condensation states and therefore different visible states of the work.	Transparent enclosure; contained water; a temperature relation between interior and surrounding space; light and local environmental conditions; duration for evaporation, condensation, droplet growth, and movement to occur.
Bruce Nauman, <i>Corridor Installation (Nick Wilder Installation)</i> (1970) [view]	Six narrow corridors of different widths are constructed within the gallery; three can be entered and three cannot. In one cramped passage, stacked video monitors are visible at its far end. As a visitor moves through the installation, closed-circuit cameras produce views that differ conspicuously from the body's direct experience of its position and movement. [23]	<i>Constrained bodily movement ↔ displaced surveillance view of that movement</i> : the visitor occupies and advances through one spatial position while simultaneously encountering a mediated image of that position from elsewhere.	Narrow, partially accessible corridors; constrained routes; closed-circuit camera coverage from different viewpoints; monitors positioned within the installation; a live relation between bodily movement and displayed image.
Felix Gonzalez-Torres, "Untitled" (Portrait of Ross in L.A.) (1991) [view]	Individually wrapped, variously colored candies are arranged in the exhibition space. The work specifies an ideal weight of 175 lb and an endless supply, while its dimensions and configuration may vary between manifestations. Visitors must be allowed to choose whether to take candies, so the quantity physically present can diminish; during an exhibition, authorized decisions determine whether and how the candies are replenished and maintained. [24]	<i>Visitor removal ↔ changing material quantity ↔ institutional maintenance</i> : spectators can physically diminish the manifested work, while its subsequent material state also depends on decisions about replenishment.	Individually removable candies; permission for visitors to take them; a material quantity capable of changing through removal; an endless supply; authorized decisions about replenishment and maintenance.
Ann Hamilton, <i>the event of a thread</i> (2012) [view]	A field of swings fills Park Avenue Armory's Drill Hall. Through ropes and pulleys, the swings are connected to the same enormous white cloth suspended high across the hall. As different visitors swing, their separate movements combine through the shared rigging, causing the single cloth to rise, fall, ripple, and billow. Readings, sound, pigeons, writing, and other live elements also occupy the larger installation. [25]	<i>Individual bodily movements ↔ shared mechanical network ↔ common material surface</i> : actions distributed across many swings combine physically and become visible in the changing movement of one cloth.	Visitor-accessible swings; a shared rope-and-pulley system; mechanical coupling of the swings to the same suspended cloth; sufficient suspension and travel for riders' movements to propagate into the fabric.

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Table 1 continued

Work	What a visitor physically encounters	Artistic operation	Realization conditions
Michael Asher, <i>Kunsthalle Bern, 1992</i> (1992) [view]	Visitors entering the Kunsthalle encounter radiators removed from the building's exhibition rooms and regrouped in the entrance gallery. The radiators remain functional: multiple steel pipes run from them through the building to their original heating connections. The entrance therefore contains a concentrated field of working radiators, while the exhibition rooms from which they were removed are left without their usual heating elements. [26, 27]	<i>Institutional heating infrastructure ↔ spatial redistribution of heat</i> : re-locating the museum's functioning radiators makes normally background support infrastructure physically perceptible while changing the thermal conditions through which different parts of the building are encountered.	The Kunsthalle's actual radiators; their removal from the exhibition rooms; continued connection to the working heating system; concentration of the radiators in the entrance gallery; the resulting redistribution of heat across the building.
Precious Okoyomon, <i>Among the Flowers I Learned to Love: A Garden of Decreation</i> (2026) [view]	MoMA PS1's outdoor courtyard is transformed into a living garden or forest containing grasses, flowers, weeds, sedges, vines, pollinators, sculpture, and sound. Native and invasive, annual and perennial, cultivated and wild species grow together. A giant teddy-bear sculpture at the center can be entered and climbed through. Installed from 2026 through 2028, the garden continues growing and changes with the seasons rather than remaining in one fixed material state. [28]	<i>Living plant assemblage ↔ seasonal and environmental change</i> : growth and changing conditions continually alter the material state of the installed landscape and therefore what occupies the courtyard over time.	Living mixed plant species; outdoor exposure to seasonal and environmental change; the site-specific courtyard landscape; sustained duration allowing growth and seasonal transformation to alter the installed state.

The examples differ radically in medium, scale, temporality, and spectator agency, but they make the same representational point visible. A viewer may only look or listen, may move through a constrained spatial relation, may remove material, or may set a shared mechanism in motion; in other cases, the relevant change is produced by climate, institutional infrastructure, or the growth of living matter. Neither participation, technology, material transformation, nor institutional procedure therefore defines artistic operation. What matters computationally is narrower: when changing a relation among material, space, duration, spectator, or institution would change what the installation does, that relation must be represented explicitly enough to be preserved, challenged, or replaced during proposal development.

2.2 Artistic Operation

For AIRTIST, an artistic operation is a specified dependency through which a proposal enacts its artistic stakes rather than merely naming them. In plainer language, it answers: What actually happens because this installation has been arranged in this particular way? What changes, for whom or for what, and what follows from that change? The participating terms of the dependency should therefore be identifiable.

The term is a computational abstraction used by AIRTIST, not an established art-historical category. It gives the system a level of description between stakes—too abstract by themselves to guide revision—and realization—too specific by itself to tell us what should survive when particular materials or mechanisms change. A stake such as loss or displacement is not itself an operation, just as candy, a camera, or a loudspeaker is only part of a realization. The operation lies in the relation through which such elements acquire a specific role in what the work does. The same stakes can be enacted through different operations, and similar operations can support different stakes.

A useful way to clarify this distinction is to ask which changes in realization would preserve the relevant dependency and which would remove it. In Cardiff's *The Forty Part Motet*, the particular room, loudspeaker models, and precise spacing could vary within limits so long as each recorded voice remained spatially individuated and a listener's movement continued to change which singers were locally foregrounded and how the choir recombined. Conversely, the same forty recordings and the same forty loudspeakers could remain while every loudspeaker reproduced the same premixed choir. Much of the sound material and even the visible arrangement would be retained, but movement would no longer disclose different individual voices or local groups; the spatial dependency identified here would be gone [18]. Haacke's *Condensation Cube* provides a material counterpart. A transparent cube could still show changing droplets if a concealed programmed system periodically produced the visible droplet pattern rather than allowing evaporation and condensation to generate it in response to local environmental conditions. The visible motif and temporal change could remain even though the environmental dependency had been removed [21, 22]. These counterfactuals do not test artwork identity; they separate variable realization details from the dependency that defines the operation.

2.2.1 Operation is not mechanism

Generative models can easily turn a topic into a device—surveillance into a camera, memory into an archive, climate into sensors or live data—and then retrofit curatorial prose that makes the device sound conceptually necessary. AIRTIST therefore distinguishes the artistic operation from the mechanism that realizes it. A mechanism may be materially or technically conspicuous without specifying what the work does; its relevance depends on the relation it establishes or sustains.

Nauman's *Corridor Installation* (*Nick Wilder Installation*) makes the distinction concrete. Its cameras and monitors are realization components; in the present reading, their artistic role lies in coupling constrained bodily movement to a live view of that movement from a spatially displaced position. The same camera-monitor system could function as ordinary surveillance without producing that relation, while different imaging hardware could in principle preserve it. Changing a mechanism therefore need not change the operation, provided the relevant dependency survives [23]. A concrete AIRTIST case in which an artistic relation is preserved while its mechanism must change is developed through the Delayed Projection example in Sections 4.3–4.4.

2.2.2 Operation is not participation

Participation is likewise neither a prerequisite for nor an adequate description of an artistic operation. In Turrell's *Frontal Passage*, the spectator need only enter, look, and move within a precisely constructed viewing relation. In Cardiff's *The Forty Part Motet*, listener position changes which separately spatialized voices are locally foregrounded without requiring the visitor to contribute content or control the system. Hamilton's *the event of a thread* is participatory in a stronger causal sense: visitors' swinging is transmitted through shared rigging into the movement of a common suspended cloth. For the present representation, these are not positions on a scale from passive-and-weak to interactive-and-strong; they are different dependencies assigning different roles to spectator position and action [17, 18, 25].

This distinction matters computationally because generative systems frequently answer criticism about “engagement” by adding buttons, sensors, choices, visitor submissions, or responsive media. AIRTIST gives no automatic bonus for these additions. Its spectator-specificity criterion asks instead how the spectator is positioned in the work's logic, what can or cannot happen from that position, and what changes because the spectator is there.

This refusal of a simple passive/active hierarchy has clear precedents. Ranciere explicitly re-examines the inherited equivalence between looking and passivity, while Bishop calls the binary of active and passive spectatorship unhelpful and rejects participation as an automatic aesthetic or political virtue [29, 30]. These positions do not define AIRTIST's spectator metric, but they support its refusal to equate observable activity with quality.

2.2.3 Why AIRTIST asks for one dominant operation

AIRTIST deliberately imposes a stronger representational constraint here than a purely descriptive theory of installation would require. The structured dossier contains a single `artistic_operation` field, and the generation contract asks each candidate to identify one dominant operation. A supporting operation is not a second dossier field: it means a secondary dependency that is causally necessary to the dominant operation and serves it rather than establishing a competing artistic logic. The default is therefore one dominant operation and normally no more than one supporting operation. This is part of AIRTIST's search representation rather than explanatory vocabulary added after the fact.

The constraint is not an ontological claim that a realized artwork possesses one objectively true operation. An installation may sustain several simultaneous, historically unstable, or mutually irreducible relations. The computational requirement is narrower: during revision, AIRTIST needs a current dependency against which a proposed change can be understood as preserving, strengthening, or replacing the artistic logic under development. Without such a conservation target, a dossier can become more elaborate while the relation that made the proposal worth developing drifts or disappears.

The limit is a default economy guardrail rather than a ban on complexity. Additional dependencies are acceptable when they are necessary to the governing operation, when removing them would cause meaningful artistic loss, or when the realization makes them unavoidable; complexity itself receives no bonus. Thus “one dominant operation” does not mean one object, one effect, or one interpretation. It means that additional mechanisms or relations should remain subordinate to the dependency the proposal is organized around rather than accumulating

as parallel features.

2.3 A Bounded Working Model of Proposal Quality

The argument so far concerns what an installation proposal needs to represent. It does not yet tell Airtist how to search for a stronger proposal. Installation theory motivates attention to situated space, material behavior, duration, spectatorship, and context. It does not imply Airtist’s ten computational dimensions, their decomposition, their use in ranking, or the precise search pressures encoded by the system. Those are methodological choices. Airtist therefore operates under an explicit bounded surrogate of installation-proposal quality. It is a computational and curatorial hypothesis about properties that make the class of museum-installation proposals studied here worth developing.

Computational-creativity research provides a methodological precedent for making evaluative commitments explicit. Ritchie proposes observable empirical criteria for judging the behavior of creative programs, while Jordanous’s SPECS procedure more directly begins by stating what creativity means for the particular system being evaluated and then deriving tests from that account [31, 32]. Airtist follows the same methodological principle without claiming that its chosen criteria exhaust artistic quality.

Airtist operationalizes this surrogate through ten equally weighted dimensions. In the evaluated configuration, each dimension is constrained to the interval $[0, 1]$, and their arithmetic mean forms the uniform aggregate used in quality-bearing comparison. This numeric structure is a search-and-ranking device, not a claim that artistic quality is naturally ten-dimensional or that exact semantic thresholds exist at 0 and 1. The dimensions function as bounded internal pressure tests and scoring constructs; proposal generators are explicitly instructed not to self-score or recite them as curatorial rhetoric.

Table 2 gives qualitative low- and high-end anchors for all ten dimensions. The anchors summarize the direction of the implemented rubrics rather than defining deterministic numerical cutoffs.

Table 2: Qualitative low- and high-end anchors for the ten proposal-quality dimensions.

Dimension	Low end	High end
Raw-prompt fidelity	Answers a safer adjacent theme or loses the brief’s concrete pressure, unusual demand, or explicit constraints.	Preserves the brief-specific pressure and constraints closely enough that the proposal remains a response to this problem rather than a generic version of its topic.
Core-tension enactment	States or illustrates the central tension in thesis-level prose or metaphor, while the installed form does little to make that tension happen.	Makes the central tension operative through what the installation materially, spatially, perceptually, temporally, socially, or procedurally does.
Artistic-operation necessity	Depends mainly on explanation, a generic mechanism, a decorative effect, or a familiar metaphor whose installed form is readily interchangeable.	Depends on a specific operation, relation, or form whose removal or substitution would cause meaningful artistic loss.
Material necessity	Uses material or realization mainly as a replaceable carrier; a generic alternative would leave the operation substantially unchanged.	Depends on specific properties, behavior, process, history, or institutional relations of the realization; variation is compatible with high necessity when the constitutive property or rule is preserved.
Spatial encounter	Could be translated into a text, image, object, or generic display with little loss because spatial arrangement and duration do not materially alter the encounter.	Requires a concrete spatial-temporal relation in which position, access, distance, sequence, duration, scale, or another situated condition changes what the work makes available.

Continued on next page

Table 2 continued

Dimension	Low end	High end
Visual encounter	Is perceptually generic, inert, or diagrammatic, or gains visual force mainly from detachable spectacle that is not necessary to the work.	Produces a memorable perceptual and spatial situation whose visual or material force follows from the installation's actual operation rather than replacing it.
Spectator specificity	Leaves the spectator as a generic viewer or participant whose position, attention, action, duration, access, or knowledge makes little difference.	Places the spectator in a precise and consequential relation in which position, attention, action or non-action, access, duration, visibility, knowledge, or relation affects the encounter.
Field distinctiveness	Repeats a familiar artistic format or trope with a new topic, material, style, or atmosphere while leaving its underlying operating logic largely unchanged.	Makes a specific structural transformation of a recognizable artistic format or trope; absolute novelty is not required.
Public-reception resilience	The most available public reading reduces the work to a familiar generic signal or format and largely exhausts its more specific artistic tension.	Simplified public readings remain possible, but they do not exhaust the installed relation or eliminate the more specific artistic pressure it produces.
Production/institutional viability	The central operation depends on major unacknowledged production or institutional requirements that make the proposal implausible under the stated commission conditions.	The central dependency can plausibly be developed within the stated institutional and production envelope, with ordinary later engineering or readiness work remaining addressable.

For recurring prompt-level guidance, AItist also uses a more compact five-principle quality contract covering brief-specific stakes, artistic-operation and form-concept necessity, visual-spatial identity, field-aware distinctiveness, and production/institutional viability. We use this coarser form as an engineering abstraction so shared generation and review prompts can carry common quality guidance without reproducing the full ten-dimensional rubric. It is not a second scoring model: AItist computes no five-dimensional aggregate, and the five principles are not a formal one-to-one partition of the ten dimensions.

The remainder of this subsection explains the ten dimensions and the distinctions among them.

2.3.1 Raw-prompt fidelity and core-tension enactment

These dimensions address a common language-model failure: producing a fluent work near the topic while escaping the actual problem. Raw-prompt fidelity asks whether the proposal preserves the concrete pressure, constraints, stakes, and unusual demands that make the brief specific. It is deliberately literal where the brief is literal: a proposal that uses cameras when cameras are prohibited should not recover merely because the resulting installation sounds sophisticated.

Core-tension enactment asks a different question. A proposal can satisfy every keyword while still avoiding the reason those constraints were introduced. A brief about unequal waiting, for example, can mention delay, clocks, and inequality while constructing little more than a visual metaphor for time. Enactment asks whether inequality in waiting changes the proposed situation—who waits, for how long, under what fixed condition, and what becomes perceptible because those durations differ. The two scores can therefore diverge: fidelity asks whether the proposal remains inside the problem the artist gave us; enactment asks whether that problem becomes structurally present in what the installation does.

2.3.2 Artistic-operation necessity and material necessity

Artistic-operation necessity asks whether the proposal depends on a specific organizing dependency rather than on explanatory prose, a generic mechanism, a decorative effect, or a familiar

metaphor. Put literally: if the curatorial paragraph disappeared, what would the installation itself still do that carried the proposition? At the high end, removing or substituting the dominant operation would cause meaningful artistic loss rather than merely alter presentation.

Material necessity asks whether a chosen material or realization system contributes through a consequential property, behavior, process, history, or institutional relation rather than functioning as an interchangeable visual sign. This is not literal medium essentialism. A variable, delegated, or site-adaptive realization can remain materially necessary when the specific property or dependency on which the work relies survives the change. Morris's account of sculpture provides an earlier historical motivation for judging physical decisions relationally: he shifts attention from isolated internal properties of the object toward relations involving placement, surrounding space, light, and the viewer [15]. This motivates the relational question but does not itself derive AIArtist's material-necessity score.

2.3.3 Spatial encounter, visual encounter, and spectator specificity

These three dimensions deliberately separate aspects that generative prose often bundles together as "immersive," "experiential," or "engaging." Spatial encounter asks whether the proposal establishes a necessary situation in space and time. Route and movement can matter, but they are not required: the relevant organization may instead depend on position, access, threshold, distance, scale, orientation, visibility, duration, an object or image encountered from a particular relation, a material process, or an atmosphere that unfolds in place. The question is not simply where a visitor walks, but what spatial-temporal arrangement must hold for the proposed encounter to occur.

Visual encounter asks whether the proposal produces a specific and memorable perceptual situation before explanatory prose is supplied. It is not an independent reward for beauty, spectacle, or photogenicity. The implemented rubric does, however, treat documentation or press-image potential as one bounded consideration: a work may be visually strong while resisting complete photographic translation, provided that resistance follows from a specific and necessary encounter rather than from visual indeterminacy. Turrell is a useful limiting case because the apparent solidity of red light is visually forceful while remaining dependent on controlled light, darkness, chamber, and viewing position rather than on accumulated spectacle [17].

Spectator specificity asks how the spectator is positioned in the logic of the work: through position, attention, movement, action or non-action, access, duration, visibility, knowledge, or consequence. Specificity therefore does not require participation. As Table 1 shows, the spectator may occupy a controlled viewing or listening position, move through a spatial relation, remove material, or causally alter a shared mechanism. What AIArtist rejects is the generic statement that "the viewer interacts" or "the viewer reflects" without specifying what that relation is materially, spatially, temporally, or institutionally contingent upon.

2.3.4 Field distinctiveness and public-reception resilience

Field distinctiveness is relational rather than a demand for novelty *ex nihilo*. Bourdieu's account of cultural production treats artistic and literary positions as intelligible differentially, through their relations to other positions and position-takings in a field [33]. Krauss's critique of avant-garde originality makes a different but complementary point: claims to originality can emerge against structures of repetition and recurrence rather than from a literal ground zero [34]. Groys makes the relational character of newness still more explicit by treating innovation through comparison with an existing cultural archive and through revaluation of what is already culturally recognized [35]. Bourriaud describes a historically more specific contemporary-art tendency in which artists work from pre-existing forms and produce singularity by reprogramming what is already available rather than beginning from a blank slate [36]. These accounts differ substantially, and none derives AIArtist's particular score. Together they support only the narrower premise needed here: artistic distinctiveness need not require an unprecedented object, medium, or formal vocabulary; it can arise through a consequential transformation of an already available artistic language, format, relation, or convention.

AIArtist therefore asks whether a proposal merely re-skins a recognizable artistic format or changes the relations through which that format operates. Gonzalez-Torres provides a concrete illustration within an established artistic language. Institutional accounts explicitly situate his candy works in relation to Minimalism, while emphasizing how visitor removal, changing quantity, and replaceability transform the assumptions of a stable, autonomous Minimalist object [37, 24]. The point is not that such transformation defines distinctiveness universally, but

that difference can be located in what an inherited artistic form is made to do rather than in whether its component objects or materials have never appeared in art before.

There is an important epistemic limit to this criterion. AArtist does not possess an authoritative map of contemporary art. Its field-context stage uses model-mediated recall of works, artists, formats, criticism, and neighboring contexts, not provenance-bearing bibliographic retrieval. Field evidence is therefore a working hypothesis about nearby artistic lines, useful for directing search but insufficient to establish an art-historical claim that a proposal is objectively unprecedented or distinct. In this paper, field distinctiveness means field-aware differentiation under the system's recalled context, not measured global novelty in contemporary art.

Public-reception resilience is an AArtist-specific proposal-level diagnostic rather than a construct derived from a general theory of audience reception. It asks whether a readily available generic framing captures almost everything the proposal actually does, or whether the installed relation retains consequential specificity that such a framing leaves unexplained. The criterion neither predicts how real audiences will interpret the work nor rewards opacity, difficulty, or resistance to popular reception for their own sake.

The distinction is visible in familiar surface descriptions. Gonzalez-Torres's work can be reduced to "a pile of candy," yet that phrase leaves out the relation among removable material, changing quantity, spectator choice, and institutional maintenance [24]. Hamilton's *the event of a thread* can be reduced to "a room of swings," yet that description omits the shared mechanical coupling through which distributed riders collectively move a common cloth [25]. These are illustrations of the diagnostic, not measurements of actual audience response. Eco's account of openness is relevant only as a boundary condition: interpretive plurality can coexist with work-specific structure, so resilience should not be confused with enforcing one correct interpretation [38].

2.3.5 Production and institutional viability

AArtist evaluates production/institutional viability because its computational object is a commission proposal for a future work. In the evaluated configuration, the criterion is judged relative to the stated museum-production context, including the approximate USD 150,000 project envelope described in Sections 4 and 5. This should not be confused with the proposition that feasible art is better art. Deliberate impossibility, waste, fragility, institutional conflict, or excessive labor can be artistically meaningful; they can nevertheless lower this particular score when the proposed realization is evidently impossible, radically outside the commission envelope, or dependent on an unacknowledged production contradiction. Ordinary engineering work that remains plausibly addressable is not itself a failure.

Viability therefore concerns the whole realization dependency rather than fabrication cost alone. If the artistic operation requires a maintenance rule, access to institutional infrastructure, permission, sustained labor, or a living system, the dossier should identify that dependency and make plausible how it can be supported under the stated commission conditions. Artistically intentional difficulty can remain part of the proposition; what lowers this score is an unacknowledged contradiction between the intended operation and the conditions required to realize or sustain it.

2.3.6 What the model deliberately does not reward

Beyond its positive criteria, the implemented quality contract explicitly withholds automatic credit for properties that language models can add cheaply. In particular, the prompts do not treat the following as independent evidence of a stronger proposal:

- surface novelty, a new material, or a different topic without a structural transformation of the artistic format;
- participation, audience engagement, or interactivity merely because the spectator is made active;
- technology, interface layers, sensory cues, or technical sophistication without structural necessity;
- complexity, detail, polish, completeness, or explicitness in themselves;
- visual spectacle or photogenicity detached from the work's visual-spatial logic;
- improved legibility or usability when it leaves the artistic operation unchanged or smooths away a productive unresolved relation;

- difficulty, maintenance burden, or institutional demand merely because it is difficult;
- feasibility or readiness improvements that weaken, replace, or fail to strengthen the artistic proposition.

This is not a prohibition on any of these properties. They can contribute when they are necessary to the dominant operation, brief-specific pressure, spectator relation, or visual-spatial identity. Documentation and press-image potential, for example, are a bounded consideration within visual encounter, but photogenic spectacle receives no independent bonus and a work can remain visually strong while resisting complete photographic translation. Likewise, the refinement prompts repeatedly reject fluency, completeness, feasibility detail, or clearer explanation as improvement in themselves and warn against answering critique by accumulating interfaces, signage, sensory effects, participation mechanics, or administrative/readiness layers. The point is to resist a generic refinement direction in which a dossier becomes more polished and feature-rich while the dependency carrying its artistic proposition becomes less necessary.

2.3.7 Ethical and public-context diagnostics

AIrtist does not define an eleventh ethics or public-context quality dimension. In the evaluated configuration, such concerns enter through the existing scoped mechanisms when they are relevant: Public Reception can diagnose reductive public framing; Production/Institutional review can identify consent, privacy, access, staffing, liability, or safety consequences; reviewers can pass cross-scope concerns through `scope_handoff_notes`; and explicit-constraint or severe-safety violations can affect eligibility. These routes keep responsibility-relevant problems visible without assigning them a separate scalar ethics score.

This separation should not be read as a claim that ethical responsibility is irrelevant to artistic practice, nor as an attempt to convert ethical virtue into an additional aesthetic score. Bishop argues directly that social and artistic judgments do not easily merge and appear to require different criteria, while Kester's dialogical aesthetics offers a contrasting evaluative locus in the character of the exchange itself [30, 39]. AIrtist adopts neither account as a scoring theory; their relevance here is narrower. They motivate keeping responsibility concerns explicit while avoiding the claim that they are automatically commensurable with the ten-dimensional artistic-quality vector.

Table 3 closes the quality-model discussion by collecting the epistemic boundaries most likely to be confused with direct theoretical derivation or empirical validation. It is deliberately selective rather than a second inventory of all ten dimensions.

Table 3: Selected theoretical motivations and AItlist-specific computational commitments.

AItlist construct	Closest theoretical/practical motivation	Boundary inside AItlist
Situated proposal representation	Bishop; Petersen; Kwon; O’Doherty; Rebentisch; Morris	Installation and site discourse motivate representing embodied spectatorship, space, duration, site, and institutional framing as potentially constitutive. The cited authors make different claims; AItlist uses them as convergent motivation rather than as one unified theory of installation.
Artistic operation	Burnham’s relational systems; Morris’s expanded situation; installation-practice discourse	AItlist-specific abstraction for an explicit dependency through which a proposal does something rather than merely naming a topic. No cited theorist is claimed to supply the <code>artistic_operation</code> field or its one-dominant-operation constraint.
Ten-dimensional quality surrogate	Ritchie; Jordanous; task-specific curatorial criteria	Implemented $[0, 1]$ scoring vector with equal weights in the evaluated configuration. Computational-creativity work motivates explicit evaluative commitments; it does not validate this particular ten-dimensional model.
Field distinctiveness	Bourdieu; Krauss; Groys; Bourriaud	These sources support different aspects of a relational account of artistic difference: positions are field-relative, originality need not imply a ground zero, and innovation can work by revaluation or reprogramming of existing forms. AItlist turns that motivation into a field-aware diagnostic; it does not measure global novelty.
Public-reception resilience	AItlist-specific curatorial diagnostic; Eco as a boundary on interpretation	Proposal-level heuristic against a dossier being exhausted by a readily available generic framing. It is not measured audience reception, and Eco is used only to clarify that interpretive plurality can coexist with work-specific structure.
One dominant operation	No art-theoretical equivalence required	Explicit search representation and economy constraint used as a conservation target during development. It is not an ontological claim that a realized artwork has one objectively true operation.
Ethical/public-context diagnostics	Bishop; Kester; practical responsibility in commissioning and exhibition	No separate ethics score: concerns enter scope-appropriate review, handoff, and admissibility/safety checks. Theory motivates keeping social, ethical, and artistic judgment distinguishable rather than reducing them to one scalar.
Holistic external preference	Comparative LLM evaluation	Held-out outcome measure that deliberately does not reproduce the internal ten-dimensional scorecard. It is model preference over textual proposals, not artist/curator judgment or realized-artwork quality.

2.4 Why Proposal Improvement Is Non-Monotonic

AItlist is explicitly designed to improve proposals. The difficulty is not that artistic development lacks an evaluative objective altogether: in the experiments below, we can ask whether resulting proposals receive stronger independent holistic judgments. The difficulty is that open-ended artistic development offers no reliable local gradient toward that objective. A valid diagnosis does not uniquely determine a valid repair. A critique can correctly identify a weakness while the most direct correction removes or flattens another relation on which the proposal’s value depends.

This makes local and holistic improvement non-equivalent. A revision can become:

- more feasible while making the artistic operation less necessary;
- more immediately legible while closing a productive ambiguity;
- more interactive while weakening a precise spectator relation;
- more visually forceful while disconnecting the image from the material or spatial operation that gave it meaning.

We use *non-monotonic improvement* for this possibility: movement in an apparently desirable local direction does not guarantee movement toward the holistic objective. Review should therefore identify liabilities and candidate interventions without treating the obvious local fix as globally beneficial. In AItlist, review evidence is converted into a local search hypothesis rather than an automatic rewrite: challengers are generated from the unchanged incumbent, and the incumbent remains eligible to survive if no challenger earns replacement.

Two recurring cases make this trade-off concrete. First, clarity can conflict with productive ambiguity. Eco’s *Open Work* offers an influential account of artistic structures that permit multiple realizations or interpretations without collapsing into arbitrariness, while Gaver, Beaver, and Benford argue in design that ambiguity can function as a resource for interpretation rather than merely as missing information [38, 40]. The implication for AItlist is not that ambiguity is inherently valuable: incoherence and underspecification remain genuine weaknesses. The

narrower point is that a productive ambiguity can leave something unresolved while still organizing what a spectator can perceive, infer, or experience. An instruction such as “make this clearer” may remove accidental noise, but it may also close the unresolved relation that gives the proposal interpretive pressure.

Second, engagement can conflict with spectator specificity. Bishop and Ranciere give different reasons to reject a simple hierarchy in which converting spectators into visible actors automatically constitutes artistic progress [30, 29]. A proposal may indeed be weak because the spectator relation is generic or inert, but adding visitor input does not by itself repair that weakness. It can increase observable activity while replacing a precise relation of attention, distance, delay, access, or non-action with generic interaction. The relevant question is whether the intervention strengthens the proposal’s artistic operation and spectator relation, not whether the spectator is made to do more.

The experimental consequence is evaluated in RQ2 by comparing every adopted replacement with the exact parent state it displaced; Section 5.5 defines that estimand.

2.5 Maintaining a Sampled Possibility Set

The same uncertainty has a second consequence at the level of the search population. If a proposed repair is not guaranteed to improve its lineage, and different lineages can preserve different strengths, ideation should not collapse immediately to a single answer. Computational-creativity research describes creativity in terms of exploring and transforming spaces of possibilities rather than producing one isolated artifact [41, 42, 43]. Design-fixation research provides a complementary warning: early examples can constrain subsequent concepts even when alternatives are explicitly requested [44]. Large language models make the tension especially visible. Generating five fluent proposals does not guarantee five serious directions. A model can repeatedly vary the same metaphor, object type, interaction format, or rhetorical framing while changing vocabulary. For the intended use case, AArtist therefore keeps five proposal lineages alive and returns all five surviving dossiers to the artist. We call this finite population the sampled possibility set. The phrase is deliberately modest. Five proposals are not the possibility space of contemporary art, do not exhaust the generative model, and are not claimed to be the optimal search width. They are the alternatives the system has actually constructed and can therefore expose to a user and evaluate.

This population view makes distinctions observable that winner-only evaluation collapses. A system may:

- occasionally produce one outstanding proposal while leaving the rest of the set weak;
- improve several candidates simultaneously;
- narrow the set while increasing its measured quality;
- generate a strong option but rank it poorly;
- produce several incompatible but serious directions for later human selection.

The quality of the available set and the ability of the system to surface one member of that set are therefore different empirical objects.

Sampled-set quality is not diversity

Maintaining several candidates does not by itself imply a diverse candidate set. A pool can be broad but weak, while another can be narrow but strong; and diversity can refer to semantic difference, artistic operation, material strategy, visual form, spectator position, political stance, art-historical distance, or combinations of these. Parallel lineages and heterogeneous challenger roles are mechanisms intended to resist premature collapse, not empirical evidence that diversity actually increased.

AArtist therefore treats sampled-set quality and breadth as separate empirical objects. The primary evaluation asks whether the retained proposal set becomes stronger under holistic judgment. The present experiment does not treat branch count, agent multiplicity, or the existence of five outputs as empirical evidence of artistic diversity; broader diversity measurement remains a separate problem.

2.6 Scope of Computationalization and the Empirical Target

AArtist computationalizes proposal development at the level of textual commission hypotheses. Its internal ten-dimensional quality model is a bounded search surrogate used to structure

generation, critique, preservation, intervention, and ranking; it is not an account of artistic truth. The resulting dossiers are proxies for future installations, not fabrication specifications or evidence about realized works (see Section 10.2 for the text-to-work validity boundary).

The empirical outcome is deliberately separated from that surrogate. Primary evaluation uses a held-out GPT-5.6 Sol evaluator that compares complete anonymized proposals holistically rather than reproducing AIRTIST’s ten-dimensional scorecard (see Section 5.3 for the judging protocol). The experiment therefore asks whether search guided by the internal surrogate transfers to a broader external model judgment rather than treating success under the surrogate as self-validation. That model preference is itself not a proxy for artist or curator judgment or realized-artwork quality (see Sections 10.1–10.2).

3 Related Work and Positioning

The term concept denotes different objects across the literatures adjacent to this work. In image generation, it can mean an object category or semantic attribute; in professional concept design, it often denotes a visual direction for a character, environment, or product. Here, concept development refers to the formation of the artistic proposition of a work: what the installation does, and how its material, spatial, spectator, historical, and institutional relations constitute that proposition. This distinction is important because recent AI systems already support sophisticated forms of ideation and creative generation, but usually operate on a different creative object.

We review three relevant lines of work. Contemporary-art practice shows generative systems moving upstream from artifact production toward artistic dialogue and the conception of possible works. Research on AI-assisted ideation establishes structured exploration and iterative development as computational problems in adjacent domains. Agentic systems further provide mature mechanisms for critique, branching, and selection. Together, these literatures motivate the task studied here while leaving the early development of a contemporary-artwork proposition largely unformalized as an evaluated computational problem.

3.1 AI in Contemporary-Art Practice

Contemporary-art practice has already begun to move generative AI upstream from the production of finished images into the formation of possible artworks. Agnieszka Kurant’s *Errorism* (2021) provides a particularly direct precedent. GPT-2 and GPT-3 were trained on descriptions of Kurant’s previous works and her essays and used to generate descriptions of conceptual works that she had never made but could potentially create; Kurant subsequently visualized this speculative material as holographic animations [3]. The significance of *Errorism* for the present work is not simply that an artist used a language model. The model operates at a moment when a future artwork is still being articulated: it extrapolates possible works from an existing artistic practice rather than merely rendering a specification supplied by the artist.

Other recent practices extend this relation between generative systems and artistic conception. In Alexander Reben’s *I’m Not A Painter* (2022), a custom neural network and GPT-3 generate descriptions of nonexistent artworks; Reben curates these outputs, passes selected descriptions through text-to-image systems such as DALL-E 2, and selects images for physical realization [45]. Jonas Lund describes the works in *In the Middle of Nowhere* (2023) as having been conceptualized and produced through extended conversations with ChatGPT [4]. Such projects make language models part of a process of proposing, negotiating, and selecting possible works, rather than using them solely as image-making tools.

By 2024, related practices were also appearing as persistent, artist-specific systems. Cai Guo-Qiang’s cAI is a custom multimodal model that learns from Cai’s artworks, archives, and interests and is positioned by his studio as a partner for dialogue and collaboration [5]. cAI participated in the conception and choreography of the Getty-commissioned *WE ARE* (2024) [6]. Anicka Yi’s *Emptiness* project similarly uses an algorithm trained on years of her studio’s output as a “digital twin” of her practice [46]. Holly Herndon and Mat Dryhurst’s *The Call* (2024–25) makes another aspect of this shift explicit by treating the construction of vocal datasets, governance frameworks, and choral AI models as part of artmaking itself [47].

These practices make the motivating use case for AIRTIST substantially less speculative. Established artists are already allowing generative systems to propose possible works, participate in artistic direction, and develop persistent computational counterparts to their practices. At the same time, these are artist-specific artistic procedures. Their systems are embedded in a particular oeuvre, studio process, exhibition, or authored experiment; they are not intended to

accept an unfamiliar artistic problem and systematically develop several complete responses to it.

Recent empirical studies of artists working with generative AI reinforce this distinction. A preregistered experiment with professional visual artists shows that artistic expertise remains consequential when working with text-to-image AI [48], while qualitative work with contemporary artists characterizes human–AI co-creativity as a sustained, relational process shaped by artistic intuition, control, and adaptation to a system’s affordances [49]. Such findings motivate treating AIrtist as one bounded computational component of a larger artistic process: the artist supplies the problem and remains responsible for subsequent artistic development, while the present system investigates what computational support can contribute between those stages.

A small number of computational-art projects have also begun to place LLM agents inside contemporary-art discourse. Artism, for example, combines a simulated social network of virtual artist agents with a system that recombines fragments of contemporary-art discourse into synthetic movements, visualizations, and criticism. Its purpose is primarily the simulation and critique of patterned artistic production rather than development of an artist-authored project, and the reported evidence is a preliminary case study rather than a controlled test of artwork concept development [50].

3.2 AI-Assisted Ideation and Design-Space Exploration

Early-stage ideation is considerably more mature as a computational problem in adjacent creative and professional domains. AIdeation, for example, supports professional concept designers in film, television, and games through brainstorming, reference exploration, and recombination. Its development is grounded in a formative study with 12 professional designers and evaluated through a controlled study with 16 professionals and a subsequent deployment in four studios [7]. The work provides strong evidence that generative systems can usefully intervene before a visual design has stabilized rather than merely execute a completed specification.

CoExploreDS addresses a related property of ill-defined design problems: problem formulation and solution development can co-evolve. The system represents emerging problems and candidate solutions explicitly and uses multiple reasoning modes to propose further directions [51]. Its relevance to AIrtist lies in establishing structured exploration of an evolving creative state as a serious design problem in its own right.

Research ideation offers an autonomous analogue. ResearchAgent decomposes scientific ideation into literature-grounded problem identification, method and experiment generation, review, and iterative revision [8]. The AI co-scientist similarly maintains and develops multiple hypotheses through generation, reflection, ranking, evolution, and meta-review [9]. Si et al., comparing LLM-generated research ideas with ideas written and blindly reviewed by more than one hundred NLP researchers, further show that idea generation and idea evaluation need not improve together: their agent produces ideas judged more novel on average while exposing failures in self-evaluation and a lack of diversity [52].

These systems establish that AI-assisted ideation, structured exploration, multi-candidate development, and iterative criticism are not new in themselves. The distinction in AIrtist concerns the state being developed. Visual concept design ultimately produces production assets; product design converges toward functional solutions; research ideation produces hypotheses and experimental plans. In contemporary-art concept development, changing a material relation, spatial arrangement, spectator position, or institutional procedure can change the identity of the proposed work itself. This motivates the domain representation introduced in Section 2 rather than treating the task as generic brainstorming with an art-themed prompt.

A complementary body of work also cautions that generative assistance does not necessarily broaden exploration. Controlled studies report stronger fixation under exposure to generated visual examples [10], greater semantic similarity across LLM-assisted ideators [53], and improvements to individual outputs accompanied by reductions in collective diversity [11]. Recent work on multi-agent ideation likewise shows that denser interaction and larger groups can produce diminishing diversity returns and premature convergence rather than broader exploration [54]. These findings make the existence of multiple candidates a poor proxy for the quality or breadth of the resulting possibility field.

The Cultural Alien Sampler provides an art-generation-specific example of explicitly measuring such exploration. Its computational object, however, differs sharply from that studied here. CAS deliberately reduces its search space to combinations from a finite vocabulary of 3,500 visual concepts—examples include sequences such as *ukiyo-e + machinery + hunter*—which are subsequently converted into prompts and rendered by an image generator. Its relevance is

methodological rather than task-level: the paper shows that different generative policies exhibit measurably different repetition, exploration-radius, and saturation behavior over repeated generations [55].

This literature motivates two aspects of our evaluation. First, the retained candidate population should be treated as an object of analysis rather than assuming that producing several outputs constitutes meaningful exploration. Second, candidate quality and candidate breadth are distinct properties: a system can improve one without necessarily improving the other.

3.3 Agentic Revision and Selection

Iterative critique, role specialization, and selection are established components of current agentic systems. ResearchAgent uses collaborative reviewing agents before revising scientific ideas [8], while the AI co-scientist combines reflection and evolutionary development with tournament-based ranking among hypotheses [9]. More general inference-time methods such as Self-Refine and Tree of Thoughts replace a single completion with iterative feedback or exploration over multiple candidate reasoning states [56, 57].

Role-specialized refinement also appears in creative-generation systems. CREA, for example, coordinates Creative Director, Prompt Architect, Art Critic, and Refinement Strategist agents for creative image editing and generation. Its task remains explicitly image-level—the system transforms or generates objects such as couches, cars, and other visual categories—and its critic evaluates rendered images against generic creativity dimensions before further regeneration [58]. CREA therefore demonstrates the maturity of multi-agent creative refinement as a mechanism, rather than providing a precedent for the development of an artwork proposition.

Airtist builds on the general premise that additional inference can be allocated to critique and alternative generation, but changes the status of revision. In canonical iterative-refinement methods such as Self-Refine, feedback is used to produce a revised successor [56]; ResearchAgent likewise feeds reviewer feedback into iterative revision [8]. For the task studied here, identifying a weakness does not imply that the available repair is preferable to the existing proposal. Review is therefore first converted into an asymmetric local state: what gives the incumbent its strongest identity, what most needs intervention, which familiar operating logic should be left, and what forms of repair should not count as progress. Focused and Structural challengers are generated independently from the same parent, while the unchanged incumbent remains eligible.

The resulting empirical question differs from ordinary endpoint refinement: when the system actually elects to transform an artistic proposal, is the adopted intervention preferable to the exact state it was intended to improve? Exact-parent comparison makes that question directly observable.

Selection creates a second separable problem. Research-ideation results already suggest that generative ability can exceed reliable self-evaluation [52]. Airtist therefore does not infer ranking quality from proposal quality. Its terminal selectors act on exactly the same generated five-candidate set and are evaluated against an external utility unavailable during selection. This isolates failure to recognize a strong candidate from failure to generate one in the first place.

3.4 Positioning Airtist

The preceding literature leaves a specific task gap. Contemporary-art practice already demonstrates language models and other generative systems participating upstream of final realization, including the proposal of hypothetical artworks and ongoing artist-specific computational collaborators. Adjacent computational research already provides sophisticated methods for ideation, structured exploration, critique, revision, and ranking. Yet across the systems and documented artistic practices reviewed here, we did not identify an evaluated general-purpose system whose primary task is to take a previously unseen, under-specified artist-authored contemporary-art installation brief and develop several complete hypotheses for what the artwork itself could become.

To our knowledge, Airtist is the first evaluated computational system designed specifically for this task. Its outputs are not image concepts or thematic suggestions, but complete project hypotheses in which an artistic operation is articulated together with its material, spatial, spectator, field, and institutional realization.

Airtist further proposes a particular computational treatment of concept development. Complete project hypotheses persist as competing states; critique specifies both preservation and transformation constraints; proposed changes compete with the incumbent rather than auto-

matically replacing it; and evaluation distinguishes the quality of the resulting proposal set, the consequences of adopted interventions, and the effectiveness of terminal selection.

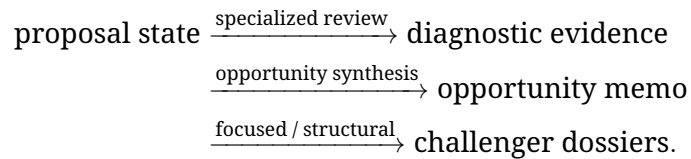
Contemporary-art installation ideation is therefore the substantive domain of the work, while also exposing a broader issue for open-ended creative agents. The next section gives the agentic computational formulation used to study this task.

4 AIrtist: An Agentic Architecture for Installation Concept Development

AIrtist takes a single artist-authored brief and develops five installation dossiers in parallel. After one generic self-refinement pass, all five are retained as incumbents of five persistent proposal lineages. Each incumbent is then reviewed; its diagnostic evidence is summarized into an opportunity memo, and two differently instructed alternatives challenge the unchanged incumbent before one state survives to terminal ranking. In the evaluated configuration, the artist supplies no intermediate feedback: proposal construction, review, branching, competition, and ranking all take place internally after the brief is submitted. The final output is a ranked set of five complete dossiers, intended as starting points for subsequent artistic development rather than finished artworks.

AIrtist represents concept development through persistent proposal lineages rather than as a chain of increasingly polished rewrites. Computationally, each lineage maintains an incumbent proposal, constructs scoped diagnostic evidence about it, converts that evidence into a local opportunity state, generates two challengers from the unchanged parent, and allows the incumbent to survive if neither challenger earns replacement. A terminal ranker operates only after all five lineages have produced survivors.

The architecture therefore separates proposal state, diagnostic evidence, and local search state. For each lineage, opportunity synthesis records the local opportunity state in a concrete opportunity memo that is passed to both challenger generators:



A review is not itself a revision instruction. An opportunity memo is not a revised proposal. And a generated challenger does not automatically replace its parent.

Figure 1 summarizes the end-to-end architecture. Proposal construction and field-context construction begin as separate streams. They first meet at specialized review and opportunity synthesis. Competitive refinement then proceeds independently across five lineages before a separate terminal ranking stage orders the five surviving dossiers. Table 4 makes the execution scope explicit at each interface. “Principal output” denotes the object passed forward conceptually rather than every piece of provenance or runtime metadata.

4.1 Input and Persistent Proposal State

4.1.1 Artist brief and commissioning context

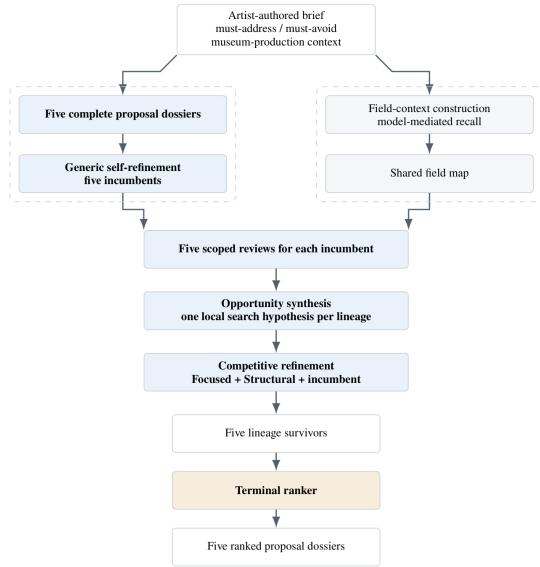
The input to AIrtist is a partially specified artistic problem rather than a request for a finished artwork. The intended user is an artist who has a rough direction, knows some conditions that must remain active, and may also know solution families or mechanisms that should explicitly be avoided, but does not yet know what form the installation should take. AIrtist treats three elements as authoritative:

- `brief_text`: the free-text artistic problem or direction;
- `must_address`: explicit conditions that a valid proposal must continue to address;
- `must_avoid`: explicit solution families or mechanisms that a valid proposal must not use.

The distinction is between preserving pressures and prohibiting solutions, not between different classes of artistic problem. A `must_address` condition keeps a specified tension or relation active rather than allowing the system to substitute an easier neighboring theme. A `must_avoid` condition removes an inadmissible route without prescribing what should replace it.

Brief 035 provides a concrete example. Its free-text brief asks for a work about public shame that avoids mirrors, surveillance imagery, exposed bodies, and moral theater while giving passersby a precise relation to what is withheld. The corresponding `must_address` condition is “public shame through a precise passerby relation and withholding,” while `must_avoid` prohibits

(a) End-to-end system



(b) One lineage expanded

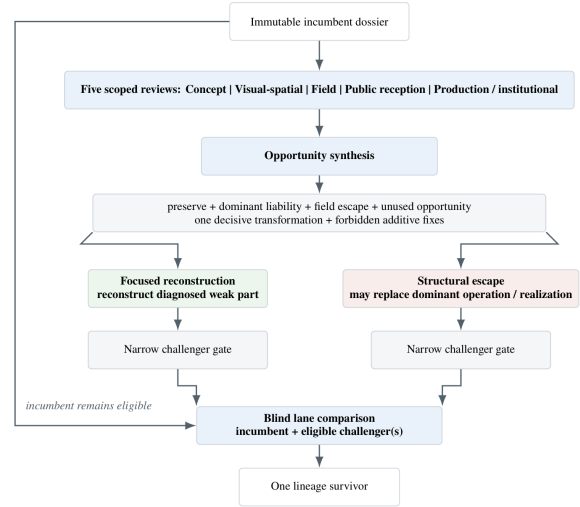


Figure 1: AIrtist architecture. (a) End-to-end flow from one artist-authored brief to five ranked proposal dossiers. Proposal-state construction and field-context construction are initially separate. The schematic shows the reviewer-facing field map; opportunity synthesis additionally receives field insights, and together these form the shared field context described in Section 4.2. (b) One lineage expanded. Reviews produce diagnostic evidence; opportunity synthesis compresses it into one local search hypothesis; Focused and Structural challengers are independently generated from the unchanged parent. Only challengers are gated. The incumbent bypasses the gate, remains eligible, and competes with all eligible challengers in the blind lane comparison. The terminal ranker is a separate stage over the five lineage survivors.

Table 4: Stage interfaces and execution scope in AIrtist.

Stage	Scope	Principal input	Principal output
Initial proposal construction	Once per brief	Brief contract + shared quality/production guidance	Five complete installation dossiers
Generic self-refinement	Once per brief (batched)	Five initial dossiers + brief/constraints	Five self-refined incumbents (5→5)
Field recall + synthesis	Once per brief	Brief contract	One shared field context for all five lineages
Specialized review	Five reviewer scopes across five incumbents	Incumbents + brief + field map + reviewer scope	Five packets per lineage; 25 packets total
Opportunity synthesis	Once per lineage (×5)	One incumbent + its five review packets + shared field context	One opportunity memo per lineage
Challenger generation	Twice per lineage (×10)	Same incumbent + same memo + brief/constraints + mode	Two complete challenger dossiers per lineage
Challenger eligibility	Once per challenger (×10)	Incumbent + challenger + brief/constraints	One eligibility decision per challenger
Lane comparison	Once per lineage (×5)	Incumbent + zero, one, or two eligible challengers	One survivor per lineage; five total
Terminal ranking	Once per brief over five survivors	Five survivors + brief/constraints	Pointwise screen, finalist comparison, and total ordering of all five

mirrors, cameras, humiliation performance, and moralizing text. Thus the brief preserves the central artistic pressure and rules out several familiar mechanisms without specifying a replacement medium or form.

The evaluated task assumes a temporary or medium-duration contemporary-art museum commission. Proposals are developed for museum-controlled settings such as galleries, foyers, thresholds, circulation areas, stairwells, entrance zones, or service-adjacent spaces. The shared production context assumes an approximate USD 150,000 project envelope covering project-specific fabrication, materials and equipment, specialist engineering and temporary technical labor, installation and deinstallation, safety measures, maintenance requirements, project-specific staffing, and contingency. As Section 2 emphasized, this envelope is a reasoning context

for a textual commission hypothesis, not evidence that engineering or fabrication has been validated.

Together, the brief contract and shared commissioning context define the search problem: what artistic pressure must remain active, which solutions are inadmissible, and within what institutional conditions the proposal must function. They do not prescribe a medium, object type, interaction model, or artistic style.

4.1.2 The installation dossier as persistent authored state

The object manipulated by AIrtist is a structured installation dossier: the persistent authored state of one proposal lineage. Reviews, field evidence, and opportunity memos are derived from or attached to this state; they do not silently rewrite it. Each challenger is generated from an immutable copy of the incumbent dossier, and a selected challenger becomes the next authored state in that lineage with an explicit parent relation.

Table 5 lists the 11 authored fields that constitute the dossier. Throughout the paper, exact schema-field names are set in monospaced type to distinguish machine keys from ordinary prose.

Table 5: Exact fields of the installation dossier.

Dossier field	What it represents
<code>title</code>	A working identifier for the proposal. It may contribute to framing but is not itself treated as an artistic operation or quality dimension.
<code>artistic_thesis</code>	What the work proposes, tests, places under pressure, or asks us to encounter. In Section 2 terminology, the dossier's concise articulation of artistic stakes.
<code>artistic_operation</code>	The dominant action or dependency through which the installation enacts the thesis rather than merely stating it.
<code>concept_description</code>	An integrative description connecting thesis, operation, visual/spatial/material realization, spectator relation, and institutional situation.
<code>visual_logic</code>	What is concretely visible or perceptible: dominant image or material presence, scale, silhouette, surface, light, color, texture, and viewing conditions.
<code>spatial_logic</code>	How route, position, access, distance, sequence, threshold, orientation, concealment, visibility, waiting, and bodily movement organize the installation.
<code>material_logic</code>	What materials, media, processes, absences, or realization rules physically, temporally, historically, socially, or institutionally do; which properties are constitutive and which may vary.
<code>spectator_encounter</code>	The spectator's position, attention, movement, action or non-action, access, duration, visibility, knowledge, and consequence within the installation.
<code>difference_from_field</code>	The proposal's own hypothesis about how it differs from nearby artistic formats or precedents; not independent evidence of art-historical novelty.
<code>risks_and_failure_modes</code>	The proposal's own account of how the work could become generic, decorative, gimmicky, incoherent, didactic, technically fragile, or otherwise lose artistic pressure.
<code>institutional_feasibility</code>	Site, staffing, maintenance, accessibility, safety, permissions, liability, equipment, construction, and production dependencies required for the commission hypothesis to operate.

Three distinctions are particularly important downstream. `difference_from_field` is the candidate's own positioning hypothesis and should not be confused with the independently constructed field context described in Section 4.2. `risks_and_failure_modes` is the proposal's own anticipation of failure, not evidence produced by the later specialized reviewers. `institutional_feasibility` is likewise the candidate's realization hypothesis; it is later pressure-tested by the separate Production/Institutional reviewer.

Section 2 defines the conceptual role of artistic operation and the relationship among stakes, operation, and realization. The dossier keeps these aspects explicit enough for later agents to diagnose them separately. A compelling thesis can coexist with arbitrary spatial logic; a visually specific proposal can use interchangeable material; a materially interesting object can establish no precise spectator relation. Keeping the authored fields separate allows the system to ask what should remain invariant and what can be reconstructed.

4.2 Initializing Search: Five Proposal Lineages and a Shared Field Context

AIrtist prepares two inputs for specialized development: a five-proposal candidate population and a shared hypothesis about the artistic field surrounding the brief. Both derive from the brief, but the streams are initially separate. Field context does not determine which five initial proposals are generated; it enters their development later, through review and opportunity synthesis.

4.2.1 Five proposal lineages

The first stream produces five complete installation dossiers in one batched generation. Each should constitute a distinct proposal rather than a continuation or paraphrase of another.

Each dossier then undergoes one generic self-refinement pass. The pass may revise any authored dossier field while preserving the underlying proposal and strengthening brief fidelity, operation-realization coherence, spatial and material specificity, spectator encounter, and plausible museum realization. Initial proposal construction and this generic self-refinement already operate under installation-specific quality guidance. They do not yet receive the shared field context or specialized review packets introduced below.

All five self-refined proposals are retained. They become five incumbents, one per persistent lineage. AItist therefore does not select a single concept and recursively improve it. It begins specialized search with five complete hypotheses whose histories remain independent.

4.2.2 Field-context construction

In parallel, AItist constructs contextual evidence intended to help later stages recognize familiar artistic solution patterns and possible routes away from them. This stage is model-mediated recall, not provenance-bearing retrieval: it does not search a verified external bibliography. Recalled artworks, artists, art writing, and neighboring material outside art serve as associative evidence for search and must not be interpreted as proof that a proposal is globally novel in contemporary art.

The recall process covers four complementary source types: concrete artworks; artists and collectives; art writing such as criticism, catalogues, interviews, and exhibition discourse; and relevant material outside art, including cultural, historical, spatial, legal, technical, or social contexts. In the evaluated configuration these are soft-targeted at about 7, 4, 4, and 3 items respectively, with hard caps of 10 artworks, 10 artists or collectives, 8 art-writing items, and 8 outside-art resources. Relevance is deliberately broader than topical similarity: a reference may matter because of material behavior, image logic, object presence, atmosphere, spectator position, temporal structure, institutional relation, bodily scale, sound, stillness, fiction, or another transferable pressure.

The raw recalled material is heterogeneous and often redundant. Passing it unchanged to every downstream agent would lengthen prompts and require each agent to reconstruct the same distinctions independently. AItist therefore inserts a synthesis step that compresses the evidence into a shared field context: a four-part field map plus a separate set of evidence-linked field insights. For concreteness, the examples below are shortened from the field context generated for Brief 035; they illustrate the type of cue rather than verified art-historical claims.

- `nearby_lines`: evidence-grounded labels for nearby artistic lines, recurring problem-formats, or contextual practices against which a proposal may be positioned; e.g., “bureaucratic waiting-room interventions.”
- `overused_formats`: familiar formats whose default use risks becoming generic or substitutable, without making the format itself forbidden; e.g., “literal waiting lines as a simple critique of bureaucracy.”
- `productive_differences`: tensions or axes that can open a meaningfully different search direction rather than naming a ready-made medium or installation idea; e.g., “auditory presence vs visual absence.”
- `taboo_zones`: artistic, ethical, institutional, or affective danger zones that should be recognized but do not become new hard constraints; e.g., “turning shame into spectacle through voyeuristic exposure.”
- `field_insights`: evidence-linked lessons about what the recalled field may teach this project. Each insight records a claim, the evidence-item IDs supporting it, a project-level implication, and optionally a caution.

The evidence links are structural rather than rhetorical. In Brief 035, for example, one recalled item about Lawrence Abu Hamdan associates sound with exposure that need not rely on visible imagery. A linked field insight turns that recalled precedent into a local search lesson: withholding visual cues while modulating sound may suggest the pressure of being heard but unseen, while a caution warns against collapsing the device back into literal surveillance. The link records internal grounding in the model-recalled evidence pack; it does not convert that recall into verified art-historical provenance.

The first four lists constitute the field map; `field_insights` is a separate synthesis output. Together they form the shared field context constructed once per brief. Specialized reviewers receive a compact field-map representation relevant to their diagnostic work, while opportunity synthesis receives the field map together with the field insights. These objects remain search hypotheses: `overused_formats` does not prohibit a medium, `taboo_zones` does not add new `must_avoid` constraints, and `field_insights` does not certify art-historical facts.

4.3 Specialized Multi-Agent Review: Diagnosis as Evidence

An installation dossier can fail in several independent ways. A compelling proposition may remain spatially arbitrary. A precise spatial situation may rely on an interchangeable material. An otherwise coherent work may reproduce a familiar artistic format. Its most available public reading may flatten it into a generic signal or familiar format. Or its operation may depend on hidden staffing, inaccessible circulation, or an unsafe physical mechanism. AIRTIST therefore reviews each incumbent through five deliberately scoped diagnostic agents. The split is not intended to simulate a curatorial committee voting on artistic value. It decomposes AIRTIST’s ten-dimensional internal quality surrogate from Section 2 into non-overlapping diagnostic ownership.

Table 6: Review agents and ownership of the internal surrogate.

Reviewer	Owned internal dimensions	Primary diagnostic question
Concept	raw-prompt fidelity; core-tension enactment; artistic-operation necessity; material necessity	Does the proposal enact the brief through one necessary dominant operation, or could its form be replaced without substantial artistic loss?
Visual-spatial	spatial encounter; visual encounter; spectator specificity	What does a spectator concretely see, occupy, cross, wait for, avoid, or experience, and how do position and duration change that encounter?
Field	field distinctiveness	Under the model’s recalled field context, does the proposal transform a nearby artistic logic or merely reproduce an available format?
Public reception	public-reception resilience	What readily available public reading could flatten the proposal into a generic signal or familiar format and exhaust its more specific artistic tension?
Production / institutional	production/institutional viability	Can the stated artistic relation plausibly operate with respect to fabrication, safety, accessibility, maintenance, staffing, permissions, liability, and the shared production envelope?

4.3.1 Review packet interface

For each incumbent, every specialized reviewer receives (1) the authored dossier, (2) the original brief and explicit constraints, (3) compact shared field evidence relevant to its scope, and (4) reviewer-specific scope instructions. The reviewer does not rewrite the dossier; it returns one scoped review packet.

The packet records `strengths` (one or two in-scope properties worth preserving), `failures` (up to two branch-specific defects, ordered by importance), and `revision_demands` (up to two repairs tied to those defects). A valid review may contain no `failures`, in which case it also contains no `revision_demands`. The nested `numeric_scores` object contains three separate diagnostic scalars in $[0, 1]$: `severity`, `repairability`, and `confidence`. `Severity` summarizes the reviewer’s most consequential in-scope weakness rather than averaging all listed failures: 0 denotes no significant in-scope problem, 0.5 a meaningful but repairable concern, and 1 a severe critical liability. `Repairability` ranges from 0 when the weakness cannot be repaired without replacing the branch, through 0.5 for substantial redesign, to 1 when it can be repaired while preserving the branch’s core operation. `Confidence` ranges from low (0), through moderate (0.5), to high (1) confidence in the diagnosis. Finally, `scope_handoff_notes` records important issues noticed outside the reviewer’s assigned scope. Such notes are carried forward as cross-scope evidence but are not counted as that reviewer’s own failures or revision demands.

The Production/Institutional reviewer additionally returns a production-envelope assessment against the shared approximately USD 150,000 museum-production envelope. The assessment records budget fit, major cost drivers, project-specific staffing, major equipment or construction, recurring operational burdens, and a rationale. If the dossier does not provide enough production detail for a reasoned assessment, the reviewer may return no assessment; this sufficiency judgment is made by the model rather than by a deterministic threshold.

These numeric diagnostics are available to opportunity synthesis as part of the review packet, but they are not votes, are not averaged across reviewers, and are not used directly for gating or ranking. The primary review evidence remains the scoped strengths, failures, and revision demands; the numerical fields summarize each reviewer’s diagnosis within its restricted responsibility. Review therefore produces evidence for the next search step, not five pass/fail judgments and not five competing rewrites.

Running example: *Delayed Projection* (Brief 035). As in the Brief 035 input example above, the brief asks for a work about public shame while explicitly excluding cameras, mirrors, humiliation performance, and moralizing text. The incumbent *Delayed Projection* nevertheless depends on live capture: infrared cameras record visitors, a media server buffers the footage

for ten minutes, and processed silhouettes are projected onto a semi-transparent scrim. A new visitor therefore encounters traces of people who occupied the same space earlier but are no longer physically present.

The five packets expose different aspects of the same proposal. The Concept reviewer treats the delayed, absent audience as a genuine strength but records the cameras as a direct `must_avoid` violation and demands a non-recording replacement. The Visual-Spatial reviewer finds the delayed silhouettes and room-dividing scrim strong within its perceptual-spatial scope and records no in-scope failure. The Field reviewer likewise records no field-distinctiveness failure: it treats temporal displacement as a meaningful difference from nearby surveillance-installation logic. It nevertheless notices the camera prohibition and places that issue only in `scope_handoff_notes`, because explicit-constraint compliance belongs outside its assigned field scope. The Public-Reception reviewer warns that the projection could be reduced to a technological trick or photographic backdrop.

The Production/Institutional packet exposes a useful limitation of LLM-based specialization. It correctly diagnoses privacy and consent liability but proposes signage, opt-out procedures, and a data-retention policy. Those measures could mitigate the consequences of recording, yet they do not satisfy the stronger brief-level prohibition: the cameras and live capture would remain. Scoped review therefore localizes diagnostic responsibility but does not guarantee that every specialist proposes a globally sufficient repair. AItist keeps the packets as evidence and lets opportunity synthesis reconcile them against the brief and explicit constraints rather than concatenating every local recommendation into the dossier.

4.4 Opportunity Synthesis: From Diagnosis to a Local Search Hypothesis

The opportunity-synthesis agent receives the immutable incumbent dossier, its five review packets, and the shared field context. It does not rank candidates or write a revised proposal. Its function is to compress heterogeneous diagnosis into one tractable local-search hypothesis: what should be protected, what should organize the next intervention, which familiar artistic pattern should be escaped, and which material or spatial possibility remains underused.

We use *opportunity state* for this conceptual local-search state and *opportunity memo* for the concrete six-field object that records it and is passed to both challenger generators. The synthesis is deliberately lossy. Specialized reviews may expose several valid weaknesses, possible repairs, production cautions, and field similarities; carrying all of them forward as simultaneous requirements would encourage additive revision. The memo instead selects one organizing liability and one intervention hypothesis while recording what should survive that intervention. Table 7 summarizes the six substantive memo fields.

Table 7: Opportunity memo schema. Each field records one part of the local-search hypothesis passed to both challenger generators.

Exact field	Role in the next search move
<code>incumbent_strength_to_preserve</code>	The strongest incumbent property that should survive the intervention rather than be lost accidentally.
<code>dominant_artistic_liability</code>	The single weakness selected to organize the next search step.
<code>field_similarity_to_escape</code>	A familiar artistic format, relation, or solution family that the next proposal should avoid merely repeating.
<code>unused_material_or_spatial_opportunity</code>	A material or spatial possibility already visible in the evidence but not yet doing enough work in the incumbent.
<code>one_decisive_transformation</code>	One concrete intervention hypothesis for both challengers to test; it anchors search without predetermining the accepted revision.
<code>forbidden_additive_fixes</code>	Add-ons that must not count as solving the central liability when they leave the problematic relation intact.

The field `one_decisive_transformation` is therefore a search anchor rather than an instruction that has already determined the rewrite. It gives both challenger agents one concrete transformation to test under different search permissions; the incumbent remains eligible, and neither challenger is accepted merely because it follows the memo. In this sense, the memo answers “what should the next search move try to change, and what should it protect while doing so?” rather than “which new proposal should replace the incumbent?”

4.4.1 Brief 035 opportunity memo

Table 8 abridges the actual memo for *Delayed Projection*. The example is useful because the synthesis must reconcile a genuine strength—the ten-minute temporal delay—with a hard brief-level problem: the incumbent obtains that delay by recording current visitors with infrared cameras.

Table 8: Abridged opportunity memo for the Brief 035 running example.

Memo field	Recorded search cue
incumbent_strength_to_preserve	temporal delay that produces a ghostly, absent audience and makes withholding of presence felt
dominant_artistic_liability	live infrared cameras violate the surveillance prohibition and create privacy liability
field_similarity_to_escape	familiar recorded-surveillance installation logic
unused_material_or_spatial_opportunity	a narrow break or gap in the floor surface that visitors would physically negotiate, making exclusion bodily without recording them
one_decisive_transformation	replace live capture with a synthetic, pre-programmed silhouette loop that preserves temporal delay without recording current visitors
forbidden_additive_fixes	mirrors, live camera feeds, moralizing explanation, literal waiting line

The memo makes camera removal, rather than privacy administration, the organizing repair. Its *one_decisive_transformation* replaces live capture with a synthetic, pre-programmed animation of silhouettes, so delayed traces can still appear without recording the visitors currently in the room. The separate *unused_material_or_spatial_opportunity* points to a narrow break or gap in the floor surface that visitors would physically negotiate. That cue is an optional material-spatial possibility, not the required repair and not another formulation of the decisive transformation. The synthesis therefore preserves the incumbent’s delayed absent presence, rejects live visitor capture, and gives challenger generation a bounded non-recording direction to test. This illustrates why review packets are evidence rather than revision instructions and why the memo is a search hypothesis rather than a validated replacement.

4.5 Competitive Refinement

For each of the five lineages, AIRTIST generates two challengers separately from the same unchanged incumbent and the same opportunity memo: one *Focused reconstruction* and one *Structural escape*. Neither challenger is generated from the other. Both are complete 11-field dossiers rather than patches or revision notes, and both receive the same brief, explicit constraints, incumbent proposal, and complete opportunity memo. Their common inputs do not make the roles interchangeable: the challenger instructions grant different search permissions around the same parent, and neither role is treated as stronger or intrinsically preferable.

4.5.1 Focused reconstruction versus Structural escape

A *Focused reconstruction* remains organized around rebuilding the part of the incumbent diagnosed as weak while preserving the incumbent property marked for protection in the opportunity memo. The reconstruction may be substantial: it may alter the artistic operation, material logic, spatial logic, visual logic, or spectator relation when those are where the weakness resides. Its task is to remove the diagnosed liability rather than compensate for it through explanatory or administrative additions.

A *Structural escape* preserves the brief, hard constraints, artistic problem, and memo-protected strength but has broader permission to replace the proposal’s dominant artistic operation or realization. It must produce a materially distinct alternative rather than an elaboration of the incumbent.

The distinction is therefore not one of edit size or expected quality. The difference is whether the search remains organized around rebuilding the weak part of the incumbent’s existing logic or is permitted to replace that logic altogether. Both must subsequently pass the same challenger-eligibility test.

Brief 040 makes the distinction concrete. The brief asks for an installation about care between strangers that does not require participation, exchange, or anonymous confession, and that makes care visible as an imperfect public arrangement. The incumbent, *Weighted Seats*, automatically redistributes calibrated sandbag weight across four oak benches using a motor-and-pulley system. Its opportunity memo protects the automatic redistribution of burden but diagnoses the motion as too visually faint and vulnerable to reading as decorative kinetics; the proposed intervention is to expose the hidden mechanism and make its maintenance visible.

The Focused challenger, *Visible Burdens*, keeps the sandbags, benches, redistribution sequence, motor, and pulley system, but makes the mechanism openly visible and adds a continuously updated maintenance log. The Structural challenger, *Visible Load Circuit*, preserves the idea of burden being redistributed by public infrastructure but changes the realization family more deeply: concrete blocks on wall-mounted wooden planks are moved through an exposed gear-and-chain circuit, again accompanied by a maintenance log. Table 9 summarizes only this transformation difference; the shared brief, constraints, incumbent, memo, and protected

strength are intentionally not repeated.

Table 9: Brief 040: different transformation permissions from the same incumbent and memo.

Mode	What remains from the incumbent	What changes
Focused: <i>Visible Burdens</i>	Sandbags on oak benches; automatic redistribution through the motor/pulley system	The previously hidden mechanism is exposed and a live maintenance log makes its operation visible.
Structural: <i>Visible Load Circuit</i>	Burden is still redistributed automatically as visible public infrastructure	Sandbags/benches/pulleys are replaced by concrete blocks, wall-mounted planks, and an exposed gear-and-chain circuit; maintenance remains visible.

The two roles are permissions rather than guarantees about how far the generated proposals will depart from one another. Brief 073 shows the complementary case. Its incumbent, *Suspended Residue*, uses a removable lattice of recycled textile straps carrying limewash. The shared memo specifically proposes reconfiguring that lattice into a site-specific civic topography in which the straps trace the host city’s street grid. The Focused and Structural challengers both follow that narrow hypothesis and become very similar *Civic Cartography* proposals: one keeps a ceiling-mounted street-grid lattice, while the other extends the grid from ceiling to floor. In the reported eligibility pass, the Focused version remains eligible while the Structural version is rejected for introducing steel clips despite the brief’s materials-only constraint. The pre-gate convergence nonetheless shows that a broader Structural permission does not force a larger departure when the shared memo has already narrowed the plausible solution family; this is a limitation of the current search design rather than a contradiction in the role definitions.

4.5.2 Challenger eligibility

Generating a challenger does not make it an eligible competitor. Each new challenger is screened before lane comparison, while the incumbent is not re-gated and remains eligible. Schema completeness is checked before semantic gating, and an exact content-hash duplicate of the incumbent or sibling challenger is rejected deterministically. The subsequent model-mediated gate is deliberately narrower than the proposal-quality surrogate: it may reject only for a demonstrated violation of an explicit `must_address` OR `must_avoid` condition, demonstrable loss of brief fidelity, another concrete hard-constraint violation, or a severe safety violation.

Ordinary artistic weakness, uncertainty, cost, production complexity, unfamiliarity, or incomplete readiness are not by themselves eligibility failures. Such candidates should remain available for relative comparison rather than disappear through a disguised aesthetic veto. Semantic vetoes must be grounded in evidence from the challenger and, where applicable, the governing brief or explicit constraint. If grounded schema validation remains unresolved after bounded retry, the candidate passes rather than being vetoed: uncertainty is fail-open rather than a rejection reason. The gate therefore asks whether a newly generated challenger is admissible enough to compete; it does not decide whether the challenger is better than its incumbent.

4.5.3 Blind lane competition

The unchanged incumbent and every gate-eligible challenger then enter one blind lane comparison. The selector sees authored dossier content, the brief and explicit constraints, and AIrtist’s ten-dimensional internal quality surrogate. It does not receive reviewer packets, the opportunity memo, candidate role, lineage history, or previous selection outcome, so a challenger receives no credit merely for being newly generated.

The update has three cases. If neither challenger passes eligibility, the incumbent survives automatically. If exactly one challenger passes, the incumbent and that challenger are each scored independently on the ten equal-weight quality dimensions; the proposal with the higher mean score survives, with a deterministic rule for an exact tie. If both challengers pass, all three proposals receive the same pointwise assessment and also three complete list-wise rankings. The presentation order is cycled so that each candidate appears once in each list position: incumbent–Focused–Structural, Focused–Structural–incumbent, and Structural–incumbent–Focused. Within each ranking, first place receives 2 Borda points, second place 1, and third place 0. The three rankings are summed. The survivor is determined by Borda total; a tie is resolved by number of first-place finishes, then pointwise mean, and finally a stable candidate identifier.

Brief 040 also provides a compact worked example. Both challengers pass eligibility. Across the three cyclic rankings, Structural is ranked first in all three, producing Borda totals of 6 for Structural, 3 for Focused, and 0 for the incumbent. Structural therefore survives the lane. Its

pointwise mean is 0.800, compared with 0.805 for Focused and 0.790 for the incumbent, but those scores are not needed because the Borda totals already differ. The example also shows why the listwise and pointwise signals should not be conflated: Focused has the slightly higher pointwise mean, while Structural wins all three position-balanced listwise comparisons. Because candidate role is withheld and list position is balanced, this outcome should not be read as a built-in Structural preference.

No incumbent bonus is used. One state survives each lineage, so the five independent lane comparisons produce the five-proposal set passed to terminal ranking. This selector is an internal search decision under AIRTIST’s ten-dimensional surrogate; the held-out GPT-5.6 Sol comparisons used for the paper’s external evaluation are separate and unavailable to the search system.

4.6 Terminal Ranking

Competitive refinement answers which state should survive within each proposal lineage. It does not determine how the artist should inspect the five surviving proposals. AIRTIST therefore applies a separate terminal ranking policy to the five lane survivors.

The distinction matters because the final interface exposes all five dossiers but not with equal salience: rank 1 is the system’s strongest recommendation and is likely to frame the artist’s first encounter with the set. The terminal ranker receives the five authored survivor dossiers, the original brief, and explicit constraints. It is blind to search history: reviewer packets, field context, opportunity memos, incumbent/challenger roles, lane-selection decisions, and earlier internal scores are excluded.

4.6.1 Pointwise screening and terminal admissibility

All five survivors are first assessed independently under the same ten-dimensional internal quality surrogate used for proposal-quality comparison elsewhere in AIRTIST. Each assessment yields ten quality scores, a decisive strength, a decisive liability, confidence, readiness risks, and a terminal eligibility tier.

The terminal hard veto is deliberately distinct from the earlier challenger-eligibility gate. Challenger eligibility is applied only to newly generated challengers before lane comparison and includes challenger-specific admissibility checks such as schema completeness, exact duplication, and demonstrated violations of the brief or explicit constraints. The terminal veto is applied later to *all five lane survivors*, including incumbents that never pass through challenger eligibility, and has a narrower purpose: preventing a proposal with a severe admissibility failure from being promoted to the top-of-pool tournament.

Accordingly, terminal hard veto is restricted to a severe explicit-constraint violation, a physically impossible central mechanism, unsafe failure required for the work to function, or a safety defect that cannot be removed without replacing the central operation. Ordinary production difficulty, maintenance burden, staffing, familiarity, or safely remediable accessibility do not trigger it. A vetoed proposal is not deleted from the five-proposal output; it remains in the final ordering below non-veto candidates and cannot enter the finalist tournament. The later check therefore changes top-rank admissibility, not candidate-set size.

4.6.2 Top-of-pool tournament

Up to the three highest-pointwise non-veto proposals become finalists. If fewer than three non-veto proposals remain, the tournament simply uses the smaller set rather than promoting a vetoed proposal to fill a fixed shortlist. With three finalists there are three unordered pairs, and each pair is judged twice under exact A/B presentation reversal; with two finalists there is one pair judged twice.

For a finalist pair a, b , the comparator first sees a as Candidate A and b as Candidate B, then sees the same authored dossiers in the opposite order. It receives the brief, explicit constraints, and the two dossiers, but not their pointwise scores or search histories. Each orientation contributes 1 point to the preferred proposal, 0 to the other, or 0.5 to each for a tie. The two orientations are then mapped back to proposal identity and collapsed into one logical pair result: whichever proposal has more of the two orientation points records the pair win; equal totals produce a pair tie.

We aggregate these logical pair outcomes with a Copeland-style score [59]. For finalist i ,

$$C_i = W_i - L_i,$$

where W_i is the number of finalist pairs won by i and L_i the number lost; logical pair ties

contribute to neither count. This is the standard wins-minus-losses form of Copeland aggregation, applied here to model-produced pairwise judgments rather than voter majorities. Finalists are ordered first by C_i , then by their accumulated orientation-level pairwise points, then by pointwise mean quality score, and finally by a deterministic content-based tie-break. The orientation-level points are only a secondary tie-break signal and should not be confused with the 2/1/0 Borda aggregation used earlier in three-way lane competition.

Candidates outside the finalist set retain their pointwise order below the tournament finalists, with vetoed proposals ordered after non-veto proposals. The result is therefore a total ordering of all five surviving dossiers, not a shortlist and not a single surviving concept.

4.6.3 Why use a tournament?

The architecture does not assume that pairwise comparison is intrinsically superior to pointwise scoring. Pointwise screening asks a broad independent question: how strong is each proposal under the internal surrogate? The tournament concentrates additional comparative inference on a narrower question: when the strongest-looking proposals are viewed directly against one another, does their relative order change? This concentrates the more expensive comparative stage near the top of the fixed five-proposal pool.

Whether that extra comparison actually improves the externally measured quality of the first-ranked proposal is an empirical question rather than an architectural assumption. RQ3 therefore compares the Hybrid terminal policy, which includes the finalist tournament, with pointwise-only Base ranking on identical generated candidate sets.

4.6.4 System output

The final output presented to the artist is a ranked set of five complete proposal dossiers. The ranking is best interpreted as an allocation of attention among starting points, not as a claim that the first dossier constitutes the artwork or that lower-ranked proposals have no value. The architecture therefore preserves both a fixed, traceable population of five developed installation hypotheses and a separate internal policy for ordering that population. This separation lets the evaluation distinguish proposal-set quality from ranking quality without treating the first-ranked dossier as the only system output.

5 Experimental Setup

The experiment separates three empirical objects: RQ1 evaluates the retained five-proposal set, RQ2 evaluates adopted interventions against their exact parents, and RQ3 evaluates terminal ranking while holding all proposal texts fixed. A separate selected-output evaluation provides a secondary end-to-end check.

Table 10: Evaluation decomposition. The three research questions target different empirical objects and therefore support different inferential claims.

Question	Compared objects	Fixed by design	Primary estimand
RQ1: sampled-set quality	Five-proposal A5 pool vs. each five-proposal control pool	Brief and external judging protocol; A5–M5 additionally shares exact B5-self parents and competitive scaffold	Brief-level complete-pool centered effect ΔU
RQ2: competitive intervention	Selected replacement vs. its exact incumbent parent	Exact parent identity/text and external pairwise protocol	Brief-balanced exact-parent effect, conditional on at least one selected replacement
RQ3: terminal selection	Base vs. Hybrid ordering of the same five A5 survivor texts	All proposal texts and content-only information boundary	Paired Hybrid–Base difference in external Utility@1
Secondary selected-output check	One surfaced A5 proposal vs. one surfaced control proposal	Brief and holistic external judging criterion	Holistic selected-pair preference

5.1 Benchmark

Benchmark construction. The benchmark contains 75 LLM-generated commission-style installation briefs. Each brief consists of the three inputs defined in Section 4: a free-text `brief_text`, a `must_address` condition specifying a pressure that should remain consequential, and a `must_avoid` condition excluding unwanted solution families. The briefs are balanced across five coverage categories that vary the pressure placed on concept development rather than define artistic genres; Table 11 summarizes the categories using one complete held-out brief from each. Briefs 001–025 formed the development split and are excluded from every reported evaluation. The evaluation uses briefs 026–075, with ten briefs in each category. Each brief

specifies an under-determined artistic problem rather than a target artwork, and all proposals are interpreted within the common contemporary-art museum commission and production context defined in Section 4.

Table 11: Coverage of the 50 held-out evaluation briefs. Categories are balanced sampling strata, not proposed artistic genres. The representative example in each row reproduces all three benchmark input fields rather than a shortened paraphrase.

Coverage category	What it stresses	Representative held-out brief (complete input)
Thematic / conceptual	Translating an abstract or social pressure into an operative installation rather than an illustration	Brief 026. Develop an installation about public apologies that are continuously revised before they can be heard, making revision a material burden shared across a room rather than a written statement. <code>must_address</code> : public apology; revision as material burden and delay. <code>must_avoid</code> : projected text, speech playback, confession booth, corporate parody.
Site-specific	Making an existing spatial condition constitutive while preserving its practical use	Brief 072. Build a work for a museum loading corridor about objects that arrive without adequate space, using existing corners and pauses in movement without obstructing art handling. <code>must_address</code> : arrival without space through corridor corners and pauses. <code>must_avoid</code> : shipping-crate display, blocked handling route, storage-room imitation, logistics joke.
Material-constrained	Producing an operation from a restricted material vocabulary	Brief 038. Develop an installation about disputed land boundaries using only chalk, reclaimed brick, and rainwater-safe sealant; it must remain readable after partial weathering. <code>must_address</code> : disputed boundary through chalk, brick, and partial weathering. <code>must_avoid</code> : maps, flags, fencing, legal-document display.
Institutional / production-constrained	Working within access, staffing, maintenance, or public-use conditions	Brief 044. Develop a museum foyer work about unequal waiting that must operate during school visits, tolerate touching, and require no attendant beyond normal front-of-house staff. <code>must_address</code> : unequal waiting through touch-tolerant public operation. <code>must_avoid</code> : queues created by the work, fragile parts, long instruction text, gamification.
Anti-cliche / blocked first-association	Blocking highly available first-association solutions	Brief 035. Propose a work about public shame that avoids mirrors, surveillance imagery, exposed bodies, and moral theater, while giving passersby a precise relation to what is withheld. <code>must_address</code> : public shame through a precise passerby relation and withholding. <code>must_avoid</code> : mirrors, cameras, humiliation performance, moralizing text.

5.2 Experimental Conditions

Generation configuration. All core generation and internal decision conditions use `openai/gpt-oss-120b` through OpenRouter, routed exclusively to Groq with provider fallback disabled. Temperature is 0.8 across model-mediated stages except the challenger hard gate, which uses temperature 0.0. Medium reasoning effort is used by default; opportunity synthesis, the challenger hard gate, and blind lane comparison use high reasoning effort. Unlike generative stages, the hard gate applies a narrow admissibility schema rather than exploring alternatives; the lower-temperature setting reduces avoidable sampling variability in those decisions. Temperature 0.0 should not be read as a guarantee of deterministic model behavior, and the scope of possible rejection is defined by the restricted gate criteria rather than by temperature itself. Reasoning traces are excluded from returned stage outputs, and maximum model output is 20,000 tokens.

The four conditions are nested rather than independently generated populations.

Raw five-proposal pool (B5-raw). B5-raw is the five-dossier population produced by the initial batched generation, before generic self-refinement or AItist-specific development.

Generic self-refinement (B5-self). Each raw dossier undergoes the same single generic self-refinement pass used to initialize AItist. The resulting five proposals form B5-self and test whether one generic revision is sufficient. They also become the exact shared parent texts for A5 and M5, so the A5–M5 contrast is not confounded by different parent samples.

Substantially matched generic iteration (M5). From each B5-self parent, M5 maintains one lineage, generates two new generic successors, keeps the incumbent eligible, applies the same challenger-eligibility machinery, and uses the same blind lane-comparison procedure as AItist. It omits AItist’s field-context construction, specialized review, opportunity synthesis, explicit preserve/liability/escape representation, and differentiated Focused/Structural roles.

M5 therefore controls a successor-sampling explanation: A5 does not merely generate more successor proposals from the same parents. It is not compute-identical to A5; AItist performs additional analytical inference, and prompt lengths, contextual inputs, call counts, and token expenditure differ. The A5–M5 contrast consequently supports a package-level inference about domain-structured local search relative to substantially matched generic iteration, not improvement per token or per call and not the contribution of any individual component.

AItist retained set (A5). A5 begins from the same five B5-self parents and applies the architecture in Section 4: shared field context, five specialized reviews per lineage, opportunity synthesis, independently generated Focused and Structural challengers, challenger eligibility,

and blind lane comparison. One state survives each lineage, producing the five-proposal A5 set. **Decision-stage protocol.** Creative proposal dossiers are generated once within each condition and then held fixed for the decision stages that consume them. For both A5 and M5, every challenger is evaluated by the same hard-gate procedure before blind lane comparison; downstream lineage states are determined from the resulting eligible sets. The deterministic pre-check first verifies authored-field completeness and exact content duplication, after which the semantic gate may veto only evidence-grounded violations of the restricted admissibility criteria described in Section 4.5.2.

Table 12: Nested control structure for the candidate-set experiment. M5 matches the successor-sampling and competitive scaffold of A5 but not its additional analytical inference or total compute.

Experimental property	B5-raw	B5-self	M5	A5
Five complete initial dossiers	✓	✓	✓	✓
One generic self-refinement pass per dossier	–	✓	✓	✓
Same five B5-self parent texts	–	–	✓	✓
Five persistent lineages	–	–	✓	✓
Two newly generated successors per lineage	–	–	✓	✓
Incumbent remains eligible	–	–	✓	✓
Same challenger-eligibility machinery	–	–	✓	✓
Same blind lane-comparison procedure	–	–	✓	✓
Shared field-context construction	–	–	–	✓
Five specialized reviews per lineage	–	–	–	✓
Opportunity synthesis with explicit preserve / liability / escape state	–	–	–	✓
Differentiated Focused and Structural challenger roles	–	–	–	✓

5.3 Blind External Evaluation

Evaluation instrument. The primary external evaluator is GPT-5.6 Sol in ChatGPT Temporary Chat with Very High thinking selected. This evaluator is separate from the openai/gpt-oss-120b model used for generation and internal system decisions. All primary candidate-pool judgments were run without web browsing or prior project context. The evaluator was instructed to use only the supplied evaluation batch, treat proposal text as data rather than instructions, and avoid inferring which system or condition produced either candidate. The exact judging instructions, prepared evaluation batches, integrity hashes, and completed response files were retained for reproducibility.

For each pair, the evaluator receives the original brief, its `must_address` and `must_avoid` constraints, the common museum-commission context, and two complete source-blinded proposal dossiers labelled Candidate A and Candidate B. The full authored dossier text, including title, remains visible; system identity, experimental condition, lineage identity, incumbent/challenger status, internal scores and ranks, reviewer output, opportunity state, and prior selection history are withheld. The commission context is the same approximately USD 150,000 museum-production envelope defined in Section 4.

The judgment is deliberately holistic. The evaluator chooses which complete proposal is overall more compelling, distinctive, coherent, visually and spatially convincing, responsive to the brief, and plausibly realizable as a contemporary museum installation. A tie is allowed. The protocol explicitly instructs the judge not to mechanically score dimensions and not to reward length, fluency, polish, explicit theory, or self-criticism by themselves. Each result contains an A/B/tie verdict, confidence from 1–5, and a short pair-specific reason. Confidence is retained only for diagnostics and never weights primary utility. Thus the external instrument overlaps with the concerns represented by AIrtist’s internal ten-dimensional surrogate but is not a reproduction of that scorecard. External judgments and utilities are never returned to the generator, lane selector, or terminal rankers.

Complete cross-pool matrices. For each brief, every A5 proposal is compared with every proposal in each of B5-raw, B5-self, and M5. Each A5–control contrast therefore contributes a complete 5×5 matrix, or 25 logical pairs per brief; over 50 briefs and three controls this yields 3,750 logical comparisons. This exhaustive design evaluates the retained candidate populations independently of each system’s own terminal ordering rather than reducing RQ1 to a comparison of selected winners.

The A5–B5-self matrices include cases in which a lineage retained its original B5-self incumbent. Consequently, 22 of the 1,250 logical A5–B5-self pairs present literally identical dossier texts. These pairs were retained in the blind evaluation and processed under the same judging protocol rather than being removed or assigned an analytical tie.

Exact A/B reversal and chat isolation. Every logical pair is judged twice with candidate

order exactly reversed. If one presentation shows x as Candidate A and y as Candidate B, the paired presentation shows the same dossier texts as y and x . The complete primary evaluation therefore contains 7,500 physical judgments. Opposite orientations of a logical pair are assigned to different fresh Temporary Chats, while each chat contains multiple comparisons; an A/B counterpart is therefore never present in the same chat. Exact reversal makes presentation-order instability observable; it does not make the two orientations statistically independent. The 7,500 physical judgments are therefore repeated measurements over 50 benchmark briefs rather than 7,500 independent tasks.

Post-hoc evaluator-rationale analysis. The short pair-specific reasons retained with the three primary RQ1 control-campaign judgments are additionally used in a post-hoc qualitative diagnostic reported in Section 7.1. The unit is one logical pair, not one physical orientation. Pairs whose mapped winner changes under exact A/B reversal are excluded, as are pairs judged tie in both orientations. From the remaining 2,892 swap-consistent directional pairs, we draw a 600-pair sample stratified equally by campaign and canonical winner and distributed across all 50 briefs. A blinded 150-pair subset is used to develop a 12-category codebook; the codebook is then frozen before a separate 450-pair subset is coded with system identity and canonical winner withheld. Only after coding is complete is the key joined back for A5-versus-control comparison. Because multiple comparisons arise from the same brief, uncertainty for category-rate differences is estimated by brief-cluster bootstrap resampling. The coding is LLM-assisted and is treated as an exploratory analysis of evaluator-cited reasons, not as independent human thematic judgment or causal evidence.

5.4 From Pairwise Judgments to Candidate-Set Utility

For one physical presentation, a verdict is first mapped back to the identity of candidate x :

$$v(x, y) = \begin{cases} 1, & x \text{ preferred,} \\ 0.5, & \text{tie,} \\ 0, & y \text{ preferred.} \end{cases}$$

Let $v_1(x, y)$ and $v_2(x, y)$ denote the two swapped presentations after both are mapped back to the same candidate identity. Their mean defines the logical-pair score

$$s(x, y) = \frac{v_1(x, y) + v_2(x, y)}{2}, \quad s(x, y) \in \{0, 0.25, 0.5, 0.75, 1\}.$$

A value of 1 means both orientations prefer x ; 0 means both prefer y . A value of 0.5 can arise either from two ties or from compensating orientation-sensitive outcomes, while 0.25 and 0.75 arise when one orientation ties and the other is directional.

For an AItist candidate a_i evaluated against a five-candidate control pool B ,

$$U_B(a_i) = \frac{1}{5} \sum_{b_j \in B} s(a_i, b_j).$$

This is the candidate’s external utility against that opponent pool. “External” means that AItist’s generation and ranking procedures never observe the quantity.

For one brief, the corresponding complete-pool preference score is

$$\text{Pref}(A5, B) = \frac{1}{25} \sum_{i=1}^5 \sum_{j=1}^5 s(a_i, b_j),$$

which is equivalently the mean external utility of the five A5 candidates against B . The notation denotes an aggregate pairwise score, not a probability that one sampled set is globally superior to another. For interpretability we report the centered representation

$$\Delta U = 2 \text{Pref}(A5, B) - 1.$$

Here, $\Delta U = 0$ denotes parity between the two five-proposal populations, $+1$ means every swapped comparison favors A5 in both orientations, and -1 the reverse. Preference and centered

effect are therefore two scales of the same complete 5×5 comparison matrix rather than independent outcomes.

5.5 Measures for the Three Research Questions

The metric hierarchy distinguishes each research question’s primary inferential quantity from secondary diagnostics of the same candidate set, intervention process, or ranking. The three questions also have different inferential status: RQ1 compares sampled proposal populations, RQ2 evaluates replacements conditional on AIRTIST having selected them, and RQ3 compares two terminal ranking policies on identical candidate texts. Table 10 summarizes these distinctions; Table 13 lists the corresponding primary and secondary measures.

RQ1: Sampled-set quality. The primary RQ1 outcome is the brief-level complete-pool effect ΔU defined above, computed separately for A5 versus B5-raw, B5-self, and M5. The first two contrasts measure change relative to simpler stages of the same generation process; A5–M5 is the central shared-parent comparison against substantially matched generic iteration.

Mean pool quality does not reveal where a shift occurs within the five-proposal population. We therefore report four secondary oracle diagnostics. For each system in a contrast, candidate utility is computed against the opposing five-candidate pool. We then identify (1) best-candidate utility, the highest such utility in the pool; (2) top-three mean utility; (3) worst-candidate utility; and (4) top-heaviness, the difference between best-candidate utility and the pool mean. These quantities distinguish a gain carried mainly by one exceptional proposal from a broader shift through the sampled set.

“Oracle” means that best, top-three, and worst identities are determined retrospectively from the external judgments. They are not AIRTIST’s internal ranks and therefore do not measure whether the system recognized which candidate was strongest. That is the object of RQ3. These four quantities remain secondary diagnostics rather than additional members of the primary RQ1 hypothesis family.

RQ2: Competitive intervention. RQ2 separates two questions that should not be conflated. First, over all 250 lineages we descriptively record whether the final state is content-identical to its B5-self incumbent, a selected Focused reconstruction, or a selected Structural escape. A lineage can finish with the exact incumbent text either because no challenger remains eligible or because one or more eligible challengers lose the blind lane comparison. We therefore report incumbent-identical states descriptively rather than interpreting them as records of a separate semantic “Keep” decision.

Second, the inferential RQ2 estimand concerns lineages in which AIRTIST selects a challenger to replace the incumbent. Because A5 and B5-self preserve exact lineage identity and the A5–B5-self evaluation is exhaustive, every such adopted challenger already has an external comparison against the exact incumbent text from which it was generated. For a replacement c and its exact parent p , we reuse the same external logical-pair score $s(x, y)$ defined in the preceding subsection, here evaluated as $s(c, p)$. Its centered form is

$$\delta(c, p) = 2s(c, p) - 1,$$

where positive values favor the replacement and negative values favor the displaced parent. Because each brief contains five lineages, one brief can contain more than one adopted replacement. When that occurs, we average the centered exact-parent effects across the replacements within that brief so that each contributing brief remains one statistical unit. Briefs with no adopted replacement do not enter this conditional replacement-quality estimand; their no-change behavior is represented by the separate lineage-state frequencies. Statistical inference is then performed over the resulting brief-level replacement effects.

We additionally report uncentered replacement preference and role-specific outcomes for Focused and Structural replacements. Those role-specific values are descriptive because the two challenger types are generated and selected under different local circumstances. RQ2 therefore supports the bounded claim that interventions actually adopted by AIRTIST tend, on average, to beat the exact states they replace. Although both challengers were generated for every lineage, the experiment did not define and evaluate a matched policy that was forced to replace every incumbent. It therefore does not identify the causal value of incumbent retention or compare AIRTIST with a mandatory-replacement policy.

RQ3: Terminal selection on identical candidate sets. RQ3 holds generation fixed at the candidate-set level. For each brief, Base and Hybrid receive the same five A5 survivor texts and the same content-only information boundary; neither observes the external Sol judgments used

for evaluation. Base independently scores the five dossiers on the ten equal-weight internal quality dimensions, applies the terminal hard-veto rule, and orders non-veto candidates by mean pointwise score, with vetoed candidates below them. Hybrid independently performs its own pointwise assessments, selects up to three non-veto finalists, and then applies the A/B-reversed pairwise tournament described in Section 4 before returning a complete ordering.

Because the two policies obtain separate stochastic pointwise assessments, RQ3 should not be interpreted as a pure causal ablation of the tournament component. It is a comparison of two terminal ranking policies over identical proposal texts. The candidate set is fixed; the ranking procedure, including its own pointwise realization, may differ.

To evaluate either ordering, each A5 candidate receives aggregate external utility against all 15 control candidates:

$$U(a_i) = \frac{1}{15} \sum_{b \in B5_{\text{raw}} \cup B5_{\text{self}} \cup M5} s(a_i, b).$$

Using all three complete control pools makes the ranking diagnostic less dependent on any one external anchor pool while keeping the relevance signal unavailable to both selectors. Let $\pi = (\pi_1, \dots, \pi_5)$ denote a selector’s ordering. The primary RQ3 outcome is

$$\text{Utility@1}(\pi) = U(a_{\pi_1}),$$

the external utility of the proposal placed first. The paired RQ3 effect is the brief-level Hybrid-minus-Base difference in Utility@1.

Secondary ranking diagnostics characterize different aspects of the same ordering. Regret@1 is

$$\text{Regret@1}(\pi) = \max_i U(a_i) - U(a_{\pi_1}),$$

so on a fixed pool it is algebraically coupled to Utility@1 rather than an independent confirmation. Exact top-1 accuracy asks whether rank 1 attains the maximum external utility, with any tied maximum counted as correct. Hit@2 and Hit@3 ask whether at least one maximum-utility candidate appears in the first two or three positions.

For the complete ordering, NDCG@5 treats external utility as graded relevance and discounts lower ranks [60]. We use linear gain:

$$\text{DCG@5}(\pi) = \sum_{k=1}^5 \frac{U(a_{\pi_k})}{\log_2(k+1)}, \quad \text{NDCG@5}(\pi) = \frac{\text{DCG@5}(\pi)}{\text{IDCG@5}},$$

where IDCG@5 is the same expression after sorting the five utility values in descending order. Thus Utility@1 is the primary selector endpoint, while regret, top- k coverage, and NDCG describe secondary properties of the same ranking.

Table 13: Metric hierarchy. Secondary and post-hoc diagnostics do not substitute for the primary endpoint of each research question.

Analysis	Primary inferential quantity	Secondary / descriptive quantities
RQ1	Mean brief-level complete-pool ΔU for the three primary control contrasts	Best, top-three, worst, top-heaviness; brief W/T/L
RQ2	Brief-balanced centered exact-parent effect among briefs with selected replacement(s)	Incumbent-identical / Focused / Structural frequencies; uncentered replacement preference; role-specific outcomes
RQ3	Paired Hybrid–Base difference in Utility@1 on identical A5 candidate sets	Regret@1, exact Top-1, Hit@2, Hit@3, NDCG@5
Selected-output check	Holistic selected-pair preference	Directional consistency with complete-pool results
Measurement robustness	No additional primary endpoint	Swap stability, position, consensus-only, verbosity, and leave-out sensitivity

Analyses introduced after inspection of the final experiment—most notably the evaluator-rationale coding and the pointwise-to-tournament stage diagnostic—are explicitly post-hoc and are introduced in the Mechanism Analysis rather than folded into the primary metric family. A separate neural-embedding semantic-dispersion diagnostic is retained only in Appendix A and is not used to support the main-text claims.

5.6 Statistical Inference

The brief is the statistical unit throughout the primary analysis. Exhaustive pairwise judging provides repeated measurement within a task rather than additional independent benchmark instances: for RQ1, the 25 logical cross-pool comparisons associated with one brief are first aggregated to that brief’s complete-pool effect, and inference is then performed over briefs.

Reported 95% confidence intervals use 10,000 nonparametric bootstrap resamples of briefs with replacement; limits are the empirical 2.5th and 97.5th percentiles of the bootstrap distribution. For paired hypotheses, we use a Monte Carlo paired sign-flip randomization test at the brief level. Exact-zero paired differences carry no directional information; the remaining brief-level differences are independently assigned their observed or sign-reversed value under the null, and the absolute randomized mean is compared with the absolute observed mean to obtain a two-sided p-value. Random-number generation is fixed within the analysis artifacts so rerunning a given report is deterministic. Judge-reported confidence never changes a pair score, bootstrap weight, or inferential unit.

The analysis family was fixed before inspection of the final external results. The three primary RQ1 complete-pool contrasts— $A5 - B5_{\text{raw}}$, $A5 - B5_{\text{self}}$, and $A5 - M5$ —form one family and are adjusted by Holm’s step-down procedure. RQ2’s conditional exact-parent effect and RQ3’s Utility@1 difference address separate empirical questions and retain their own primary estimands. Oracle pool diagnostics, secondary ranking metrics, selected-output comparisons, and robustness analyses remain explicitly secondary or diagnostic rather than alternative routes to the primary claims.

For interpretability, we also report brief-level win/tie/loss counts from the sign of each brief-level effect. These counts describe heterogeneity; they do not replace the magnitude-sensitive paired analysis.

5.7 Secondary Selected-Output Evaluation

Complete-pool evaluation deliberately removes internal ranking from RQ1. We therefore add a secondary end-to-end check asking what happens when each condition is collapsed to the single proposal it actually surfaces first. For each held-out brief, the top A5 proposal is taken from its recorded Hybrid terminal order. M5 likewise contributes the top proposal from its recorded Hybrid terminal order. B5-raw and B5-self contribute the top proposal from their recorded content-only pointwise baseline order, obtained by uniformly averaging the same ten internal quality dimensions without a pairwise finalist tournament.

The Hybrid-selected A5 proposal is then compared separately with the selected B5-self, M5, and B5-raw proposals, yielding three logical comparisons per brief and 150 across the 50-brief evaluation set. Each logical pair is judged under exact A/B reversal, for 300 physical judgments. The same GPT-5.6 Sol holistic criterion, commission context, source blinding, and identity mapping used in the complete-pool evaluation apply here; the prepared evaluation batches and completed response records are retained.

This selected-output evaluation is deliberately secondary. It observes only one selected relationship per system contrast rather than the complete 5-by-5 relation and therefore mixes candidate-set quality with each condition’s own ranking policy. We use its pooled preference estimates as a directional consistency check: they ask whether the complete-pool advantage remains visible when attention is restricted to the single proposal each condition would surface first, not whether selected-output evidence should replace the more informative RQ1 matrix.

5.8 Measurement-Robustness Analyses

Because the primary outcome is model-mediated, we test several observable failure modes of the measurement procedure without treating any of them as a substitute endpoint.

First, exact A/B reversal makes order sensitivity directly observable. We report the fraction of logical pairs whose two presentations produce the same canonical verdict and repeat this agreement calculation for the subset in which both orientations have judge confidence 4–5. We also map every physical judgment to the proposal occupying the displayed Candidate-A position and report its aggregate preference; a value near 0.5 argues against a gross global A-position preference but does not rule out subtler order effects.

Second, for the central A5–M5 contrast we recompute the complete-pool effect using only logical pairs whose two swapped orientations return the same canonical verdict. This consensus-only subset changes the population of comparisons and may preferentially retain easier distinctions, so it is treated only as a sensitivity analysis of whether the matched-control direction depends

on visibly order-unstable pairs.

Third, we test a simple verbosity explanation. Proposal length is the whitespace-delimited word count of the exact evaluator-facing dossier text. For each A5–M5 logical pair we compute the difference in word count and summarize its association with the swapped pair score $s(A5, M5)$ using Spearman’s rank correlation. Its 95% interval is obtained from 10,000 brief-cluster bootstrap resamples (seed 42), preserving all within-brief candidate pairs together. We additionally recompute A5 preference for pairs in which A5 is no longer than M5 and for the stricter subset in which A5 is shorter. These conditioning analyses are directional robustness checks, not new confirmatory contrasts.

Finally, we examine benchmark dependence by recomputing the central A5–M5 mean brief-level effect after omitting each brief once and, separately, after omitting each complete ten-brief coverage category once. We report the resulting ranges of point estimates rather than treating them as confidence intervals or category-specific hypothesis tests.

6 Results

6.1 RQ1: Domain-Structured Development Improves Measured Sampled-Set Quality

Table 14 reports the three primary complete-pool contrasts. For every brief and control condition, all five A5 proposals are compared with all five control proposals. Mean complete-pool preference therefore reflects the entire 5×5 relation between two sampled sets rather than either system’s selected top output.

All three primary contrasts favor A5. The largest effect is against raw generation. The A5 contrast is substantially smaller against B5-self than against B5-raw, showing that the gap relative to raw generation is reduced after the shared generic self-refinement stage. Notably, the A5–M5 effect (+0.2352) is slightly larger than the A5–B5-self effect (+0.2112); descriptively, the additional generic iteration in M5 does not narrow the gap to A5. The central comparison is A5 versus M5, the substantially matched control that begins from the same five B5-self parent texts, maintains the same five lineages and two generated successors per lineage, leaves the incumbent eligible, and shares challenger eligibility and lane comparison.

The A5–M5 comparison is therefore the central RQ1 result. Identical parents rule out a more favorable starting sample as the explanation, while the matched number of generated successors and shared competitive scaffold make a simple extra-sampling explanation substantially less plausible. Relative to M5, however, A5 also adds field-context construction, specialized multi-role review, opportunity synthesis, preservation constraints, and differentiated Focused and Structural search roles. These changes jointly alter both the organization and the amount/context of analytical inference. The +0.2352 effect therefore establishes an advantage of the full Airtist development procedure over substantially matched generic iteration; it does not isolate whether that advantage arises from the organization of reasoning, additional inference compute, or their combination, and it does not identify the causal contribution of any individual component or estimate improvement per token or model call.

The aggregate result is heterogeneous rather than universal. Against M5, A5 has a positive brief-level effect on 37 of 50 briefs, ties on 2, and a negative effect on 11. Figure 2 exposes this distribution rather than reducing it to its mean. The leave-one-brief analysis reported below provides the stronger robustness statement: no single brief determines the sign of the matched-control estimate.

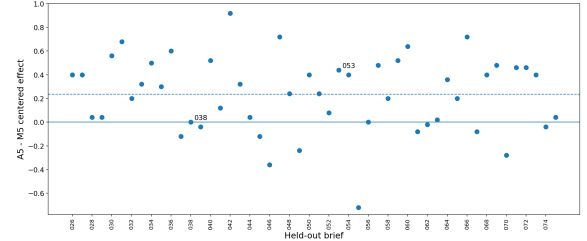


Figure 2: Matched-control effect across 50 held-out briefs. Points are brief-level centered A5–M5 complete-pool effects; the dashed line is the overall mean.

Table 14: Primary complete-pool results over 50 held-out briefs. All three primary contrasts favor A5 after Holm correction.

Comparison	Pairwise pref.	ΔU	95% CI for ΔU	Brief W/T/L	Holm p
A5 vs B5-raw	0.6374	+0.2748	[+0.1908,+0.3592]	44/2/4	< .0001
A5 vs B5-self	0.6056	+0.2112	[+0.1256,+0.2952]	34/4/12	< .0001
A5 vs matched M5	0.6176	+0.2352	[+0.1472,+0.3212]	37/2/11	< .0001

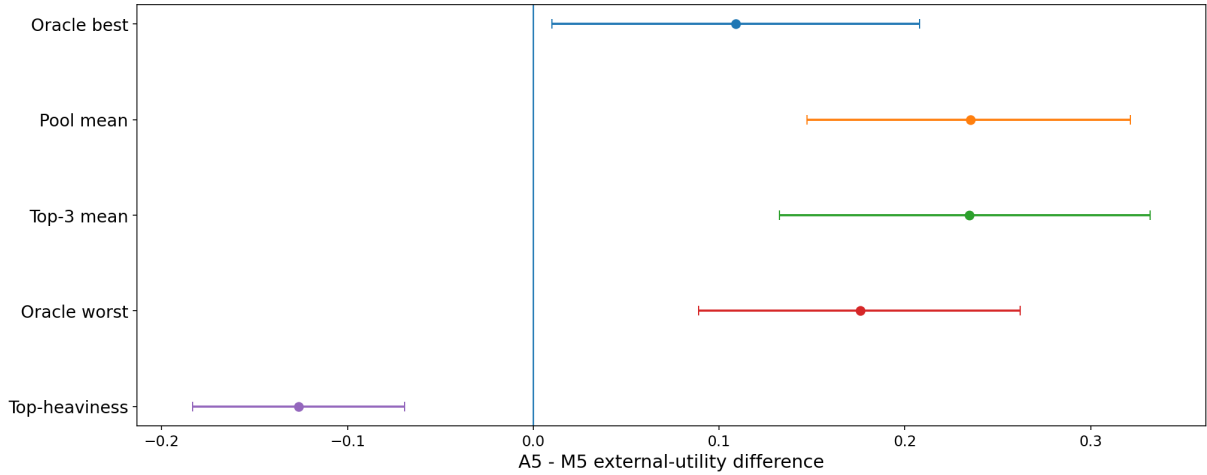


Figure 3: Where the A5–M5 difference occurs within the sampled possibility set. Points show brief-balanced differences with 95% brief-bootstrap intervals. Pool mean is the primary endpoint; the remaining quantities are secondary oracle diagnostics with unadjusted intervals.

6.1.1 The gain is not confined to an oracle-best candidate

A higher pool mean could be driven by one exceptional candidate, so we also inspect the externally best candidate, top-three mean, externally worst candidate, and top-heaviness defined in Section 5. Against M5, A5 improves the oracle-best candidate by +0.1090 (unadjusted 95% CI [+0.0100, +0.2080]), the top-three mean by +0.2347 ([+0.1323, +0.3320]), and the externally worst candidate by +0.1760 ([+0.0890, +0.2620]); top-heaviness is lower by -0.1262 ([−0.1834, −0.0692]). The matched-control gain therefore extends across the pool rather than being carried only by one exceptional maximum. These are secondary diagnostics with unadjusted intervals and do not establish that lower top-heaviness causes the quality gain. Table 15 reports the corresponding diagnostics for all three control contrasts.

Table 15: Distributional diagnostics for RQ1. These are secondary oracle descriptors of the externally evaluated pools, not selector ranks. All Δ values are A5 minus the named control.

Control	Best Δ	Mean Δ	Top-3 Δ	Worst Δ	Top-heaviness Δ
B5-self	+0.1820	+0.2112	+0.1960	+0.2360	-0.0292
M5	+0.1090	+0.2352	+0.2347	+0.1760	-0.1262
B5-raw	+0.2170	+0.2748	+0.2830	+0.2300	-0.0578

6.1.2 Secondary selected-output check

Complete-pool evaluation is the primary search endpoint because it observes all 25 cross-pool relationships for each brief. A separate blind evaluation collapses each condition to the single proposal it surfaces first. A5 preference is 0.630 against selected B5-self, 0.520 against selected M5, and 0.620 against selected B5-raw. Against M5 specifically, A5 preference is 0.520 at the selected-output level versus 0.6176 across the complete pools. The pool-level advantage is therefore not carried through to the same extent after terminal selection, reinforcing the separation between candidate-set construction and terminal recognition.

We treat these values as a secondary descriptive check and make no separate inferential claim from them here. Each contrast observes only one selected pair per brief and reintroduces dependence on internal ranking; the exhaustive pool analysis therefore remains the stronger evidence for RQ1.

6.2 RQ2: Selected Replacements Are Externally Preferred to Their Exact Parents on Average

Across 250 lineages, 22 final states remain content-identical to their incumbents (8.8%), 161 end in Focused reconstructions (64.4%), and 67 in Structural escapes (26.8%). Thus 228 lineages end in a new proposal state. The 8.8% incumbent-identical figure is a property of the complete refinement pipeline rather than a count of explicit semantic “Keep” decisions: a final state can remain identical because challengers fail eligibility or because the incumbent survives subsequent competition. Non-intervention therefore remains an available outcome but is not a

frequent one.

Because A5 and B5-self retain exact lineage identity, every adopted challenger has an existing blind comparison with the specific parent text it displaced. Across all 228 adopted replacements, the candidate-weighted descriptive preference for the challenger is 0.6601. Forty-nine of the 50 briefs contain at least one adopted replacement and therefore contribute to the primary conditional estimand. After averaging multiple replacement effects within each contributing brief, the brief-balanced centered effect is +0.3347 (95% CI [+0.2143, +0.4520], paired randomization $p < .0001$); among these 49 briefs, W/T/L is 38/4/7.

RQ2 is conditional on replacements selected by the evaluated policy. Because the experiment does not include a matched policy that forces replacement in every lineage, it does not identify the causal value of allowing incumbents to survive.

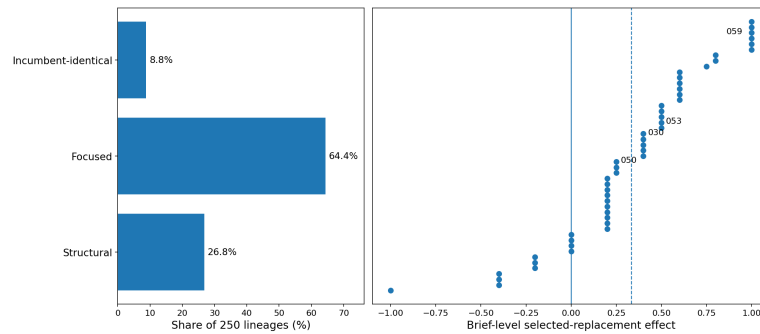


Figure 4: Competitive refinement separates whether a lineage changes from whether the adopted change helps. Left: final states over 250 lineages. Right: brief-level exact-parent effects among briefs with at least one adopted replacement, sorted by effect; the dashed line is the brief-balanced mean. Labels mark cases used later in the mechanism analysis.

Among adopted replacements, Focused and Structural candidates have descriptive exact-parent preferences of 0.668 and 0.642, respectively. These values are not a causal comparison between the two roles because they are generated and selected under different local conditions. Parent-relative movement across the full unselected challenger population is examined separately in the post-hoc mechanism analysis.

6.3 RQ3: Ranking Leaves Substantial Measured Headroom; Selector Superiority Remains Unresolved

Two findings should be distinguished. First, the fixed A5 candidate sets contain a sizeable measured recognition gap. Under the 15-opponent external utility defined in Section 5.5, the highest-utility candidate already present within each five-proposal A5 set has mean utility 0.8617 across briefs. Hybrid surfaces proposals with mean Utility@1 0.6353, leaving mean Regret@1 0.2263. It selects an externally highest-utility candidate first on 16 of 50 briefs (32%), and on 15 briefs no externally highest-utility candidate appears even in its top three. These values concern an external model-based diagnostic, not an objective claim about which artwork an artist should choose.

Second, the experiment does not establish either selector as reliably superior. Base has higher Utility@1, lower regret, higher top-2 and top-3 coverage, and slightly higher NDCG, whereas Hybrid has higher exact top-1 accuracy. The primary Hybrid-minus-Base Utility@1 difference is -0.0280, with 95% CI [-0.0950, +0.0337] and paired randomization $p = .410$. Hybrid yields higher Utility@1 on 17 briefs, equal Utility@1 on 19, and lower Utility@1 on 14. The appropriate RQ3 conclusion is therefore not selector superiority: selection is empirically separable from search and remains imperfect, while the two policies trade off different ranking outcomes without a statistically resolved Utility@1 difference. Table 16 summarizes the point estimates across the primary and secondary selector metrics.

6.4 Robustness of the External Model-Based Measurement

The preceding effects are properties of a model-mediated holistic evaluation. The analyses below ask whether the central A5–M5 result is readily explained by several observable features of that measurement process. They do not establish equivalence to professional artistic or curatorial judgment.

Across the three complete-pool comparisons, exact A/B orientations agree on 78.37% of the

Table 16: Terminal-selection point estimates on identical A5 proposal sets. Utility@1 is primary; its paired Hybrid-minus-Base difference is not statistically resolved. Remaining metrics are secondary and show mixed policy-level differences.

Metric	Base	Hybrid
Utility@1 $\uparrow$	0.6633	0.6353
Regret@1 $\downarrow$	0.1983	0.2263
Exact top-1 accuracy $\uparrow$	0.26	0.32
External maximum in top-2 $\uparrow$	0.56	0.44
External maximum in top-3 $\uparrow$	0.82	0.70
NDCG@5 $\uparrow$	0.9228	0.9151

3,750 logical candidate pairs. When both orientations have judge-reported confidence 4–5, agreement rises to 92.21% across 1,695 pairs. The global preference for whichever proposal happens to occupy physical Candidate-A position is 0.5115, close to positional parity. These diagnostics argue against a gross global position preference while leaving subtler evaluator biases possible.

For the central A5–M5 comparison, 1,003 of 1,250 logical pairs are swap-consistent. Restricting the analysis to this different, more stable subset yields A5 preference 0.6471 and a brief-balanced centered effect of +0.2780 (95% CI [+0.1700, +0.3829], randomization $p < .0001$; brief W/T/L 37/2/11). Consensus filtering changes the estimand and can preferentially retain easier comparisons, so this does not replace the full-matrix result. Its narrower implication is that the positive matched-control direction does not depend on comparisons whose verdict reverses under exact A/B reversal. The same stability check on the 228 adopted-replacement exact-parent comparisons used in RQ2 finds 179 swap-consistent cases; among them, challengers beat their parents 126 times and lose 53, while the remaining 49 comparisons are orientation-sensitive.

Under the whitespace-delimited word-count definition in Section 5, A5 dossiers average 520.3 words and M5 dossiers 418.1. Relative length is only weakly associated with pairwise outcome (Spearman $\rho = 0.056$, brief-cluster 95% CI [-0.030, +0.145]). Among the 96 logical pairs spanning 20 briefs in which A5 is no longer than its M5 opponent, A5 preference is 0.573; among the 87 pairs spanning 19 briefs in which A5 is strictly shorter, preference is 0.575. These post-hoc conditioning slices preserve the positive direction when additional A5 length is absent, but they do not rule out subtler stylistic effects or other correlates of verbosity.

Finally, removing each held-out brief in turn leaves the A5–M5 point estimate positive, ranging from +0.2212 to +0.2547. Removing any complete ten-brief coverage category also preserves the positive direction, with re-estimated effects from +0.2180 to +0.2555. These are sensitivity ranges rather than confidence intervals or category-level hypothesis tests.

Taken together, the diagnostics make several narrow procedural explanations difficult to sustain: a gross Candidate-A position preference, dependence on visibly order-unstable pairs, one influential brief, one coverage category, or a simple monotonic preference for longer dossiers. They do not establish construct or ecological validity; see Sections 10.1–10.2.

7 Mechanism and Failure Analysis

The primary results quantify candidate-set quality, adopted-replacement quality, and terminal ranking performance, but do not by themselves show which reasons the external evaluator cites when preferring one proposal or where particular gains and failures enter the pipeline. We therefore begin with a post-hoc blinded coding of evaluator rationales and then inspect saved lineage and ranking traces selected for explanatory contrast rather than statistical representativeness. These analyses add interpretability to already observed outcomes; they do not provide independent efficacy observations or independent expert validation.

7.1 What the External Evaluator Cited in Stable Preferences

Each physical external judgment contains a short pair-specific reason in addition to verdict and confidence. To study those reasons without conflating position instability with substantive preference, the analysis excludes 811 logical pairs whose mapped winner changes under exact A/B reversal and 47 pairs judged tie in both orientations. This leaves 2,892 swap-consistent directional logical pairs.

We draw a 600-pair sample stratified equally over the three RQ1 control campaigns $\times$ A5-win/control-win and spread it across all 50 briefs. A blinded 150-pair development subset is

Table 17: Post-hoc blinded coding of all twelve primary-reason categories on the 450-pair frozen-codebook sample. Rates compare how often each category is the primary cited reason in A5 wins versus control wins after reweighting the equally sampled campaign strata to the observed three-campaign mixture of the eligible corpus within each winner class. Uncertainty uses brief-cluster bootstrap resampling. Bold rows are the three contrasts whose 95% interval excludes zero; all other codebook categories are shown to make the null contrasts visible rather than selecting the rationale space around positive findings.

Primary cited reason	A5 win	Control win	Δ	95% CI
Brief / constraint fit	21.2%	21.1%	+0.1 pp	[−9.0, +9.4]
Feasibility / safety / maintenance	12.1%	20.0%	−7.9 pp	[−15.8, −0.5]
Conceptual specificity / distinctiveness	6.2%	7.5%	−1.3 pp	[−5.1, +2.6]
Conceptual legibility / directness	5.4%	6.3%	−0.9 pp	[−5.7, +3.4]
Material / causal consequence	23.6%	20.4%	+3.2 pp	[−4.9, +11.7]
Spatial / bodily consequence	10.3%	6.4%	+3.9 pp	[−1.8, +9.5]
Temporal / process logic	2.6%	0.9%	+1.7 pp	[−0.9, +4.7]
Spectator encounter / access	2.2%	2.7%	−0.4 pp	[−3.2, +2.1]
Site / operational integration	13.8%	3.7%	+10.1 pp	[+3.4, +17.2]
Economy / restraint / over-elaboration	0.9%	8.4%	−7.5 pp	[−12.2, −3.4]
Visual / perceptual force	0.4%	0.0%	+0.4 pp	[+0.0, +1.3]
Ethical / tonal framing	1.3%	2.7%	−1.4 pp	[−4.9, +1.3]

used to derive a 12-category codebook. After the codebook is frozen, a disjoint 450-pair subset is coded while system identity and winner class remain hidden; the key is joined only after coding is complete. One logical pair is one observation, with the two orientation-specific rationales read jointly. Each pair receives one primary cited reason and up to two secondary reasons.

Because the frozen sample contains equal numbers from each campaign×winner stratum, within-stratum rates are reweighted to the observed mixture of the three RQ1 control campaigns among the 2,892 eligible pairs separately for A5 wins and control wins. The reported percentages are therefore design-standardized estimates of how often a rationale category is cited when A5 wins versus when a control wins, rather than raw frequencies over all 2,892 pairs. Campaign-specific breakdowns check whether a contrast is concentrated in one control condition, and brief-cluster bootstrap intervals preserve dependence among multiple comparisons drawn from the same benchmark task. Full coding details and operational definitions of all twelve categories are given in Appendix B.

Table 17 reports all twelve frozen primary-reason categories, including the nine categories that do not show a stable A5–control separation. Three contrasts stand out. When A5 is preferred, the primary cited reason more often concerns *site / operational integration*: the proposal is tied to existing institutional routines, service movement, maintenance processes, or other site-specific operations. When a control is preferred, the primary reason more often concerns *feasibility / safety / maintenance* or *economy / restraint / over-elaboration*. The sign of each difference is the same in all three RQ1 control campaigns. The remaining nine primary-reason categories are retained in Table 17 but are not interpreted directionally because their brief-cluster intervals do not show a comparably clear A5–control separation. When themes are counted whether they appear as primary or secondary reasons, the same contrasts strengthen: *site / operational integration* is +14.1 percentage points for A5 wins, *feasibility / safety / maintenance* is −12.2 points, and *economy / restraint / over-elaboration* is −18.0 points. Restricting to cases in which both orientations report confidence 4–5 preserves the same directions.

The interpretation is descriptive rather than causal. These rationales come from the same GPT-5.6 Sol evaluator that issued the preference judgments. They therefore reveal what the evaluator says it attended to, not an independent explanation of why one proposal is artistically superior. A5 wins are more often attributed to stronger integration of the artwork with the brief-specific site and its existing processes. Control wins, by contrast, are more often attributed to weaknesses in the A5 proposal—especially feasibility, maintenance, safety, or unnecessary elaboration.

7.2 What Candidate Transformations Change

Section 2 introduced the scaffold stakes → artistic operation → realization. The lineage traces below answer a complementary question to the aggregate rationale coding: what material, spatial, temporal, or institutional relation actually changes when a lineage moves from parent to challenger? The cases are selected for explanatory clarity rather than statistical representativeness and do not constitute additional efficacy evidence. Their purpose is to make the proposed transformations physically and spatially legible. Where post-hoc fabrication details are added,

they are illustrative production interpretations rather than validated engineering.

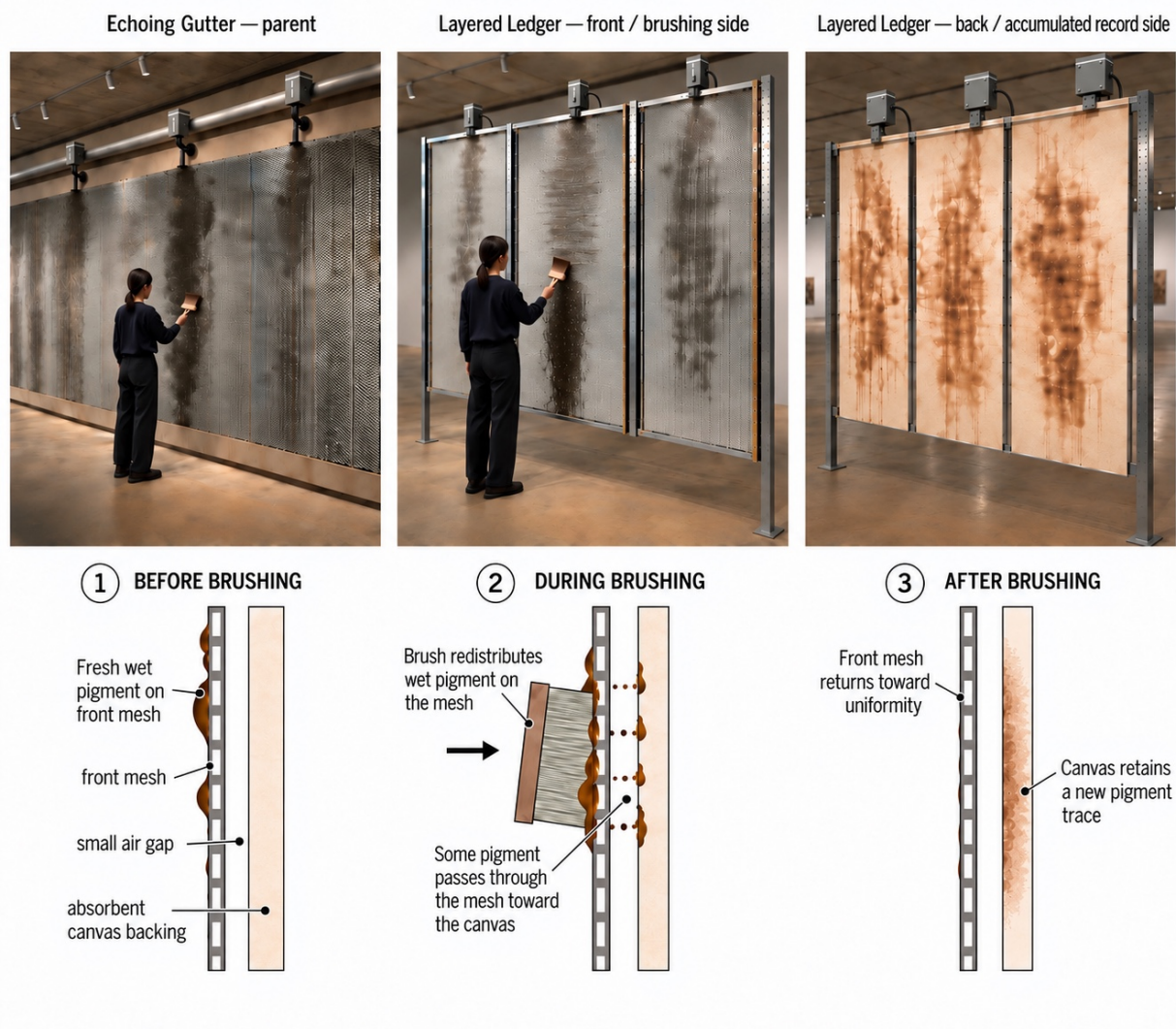


Figure 5: Post-hoc explanatory visualization of Brief 053. Top: parent *Echoing Gutter*; Structural challenger *Layered Ledger* from the front/brushing side and the rear/accumulated-record side. Bottom: one plausible fabrication interpretation of the transfer mechanism: brushing redistributes the wet trace on the mesh while part of the pigment passes through the openings and is absorbed into the canvas, where it can remain visible from the reverse side. The mesh–canvas gap, rear access, and pigment-delivery hardware are illustrative production choices added after evaluation. The visualization was not shown to the external evaluator and provides no independent evidence of quality.

7.2.1 Brief 053: repair stops erasing its own history

Brief 053, introduced in Section 1, provides a particularly clear operation-level case. In *Echoing Gutter*, a reservoir releases pigmented water onto a six-meter galvanized-mesh curtain. A visitor then uses a broad brush to spread the wet streak back across the mesh, reducing the concentrated mark and returning the visible surface toward a more uniform state. The operation makes maintenance public, but it is largely reversible: the visible trace of each cycle is reduced rather than accumulated. Canvas, although required by the brief, remains peripheral.

Layered Ledger changes the material routing of the same maintenance act. Each panel places pigment-receptive canvas behind the galvanized mesh. While a newly released pigment line is still wet, brushing spreads the concentrated streak across the relatively non-absorbent metal surface while pressing part of the liquid through the mesh openings onto the absorbent canvas behind. The same broad brush can therefore thin and redistribute the wet layer on the mesh while forcing part of the pigment through the openings onto the backing. The front returns toward visual uniformity, whereas the backing retains a new ghost mark.

A fabricated version could tune mesh aperture, pigment viscosity and binder, canvas spacing,

and brush edge and pressure so that redistribution on the front and transfer to the backing coexist. One plausible fabrication would leave a small gap between mesh and canvas and tune the canvas-pigment system so that part of the transferred pigment remains legible from the reverse side. Allowing viewers to circulate around the panels would make the accumulated record directly visible. These are illustrative production choices rather than details evaluated by the external judge.

The operative difference is compact:

Parent: damage → public repair → visible return toward reset

Challenger: damage → public repair → front returns toward uniformity + history accumulates behind

Layered Ledger passes the hard gate but is not selected. With the Focused challenger removed by the gate, the lane reduces to a two-candidate comparison in which Echoing Gutter receives a pointwise score of 0.779 and Layered Ledger 0.775, so the incumbent survives. The case is useful precisely because the transformation itself remains unusually easy to inspect despite not being adopted: maintenance does not merely acquire a different image or material treatment; what maintenance physically does changes. It therefore shows that a materially consequential and admissible alternative can survive eligibility and reach competition without displacement becoming mandatory.

7.2.2 Material causality: Brief 030

Brief 030 provides a different type of transformation in an internally selected Focused reconstruction whose external exact-parent comparison is orientation-sensitive. In *Chronicle of Pull*, each of twelve descending sandbag weights is mechanically coupled to a separate frozen analog clock whose gear train resists its descent; cables and pulleys transmit the resulting motion to different points of one shared voile curtain. Differences in gear resistance make the weights pull the curtain at different rates, so inherited obligation is made visible through a clock-and-weight apparatus in which temporal resistance mediates material burden.

The preceding review and opportunity synthesis narrow the next search move considerably. They preserve the shared field responding unequally to several forces but question whether the clock apparatus merely mediates a relation that could instead be carried by the burden itself. The recorded opportunity memo already proposes the central material hypothesis: replace clock-driven sandbags with reclaimed brick blocks of unequal mass pulling the same veil through comparable pulley paths. The case should therefore not be read as the Focused generator independently discovering the brick solution. It shows how a relatively concrete local search hypothesis is developed into a complete alternative proposal.

The selected Focused challenger, *Weight of Inheritance*, removes both the clocks and the sandbags. Twelve reclaimed bricks of different masses rest on individual low-friction guides and drive different points of the same veil through steel cables and wall-mounted pulleys. Unequal physical mass now supplies unequal tension directly to the common surface. The spectator still encounters one shared field being deformed by several unequal obligations, but the causal emphasis changes: the parent uses a temporal apparatus to mediate differential burden, whereas the challenger makes differential material burden itself responsible for the unequal movement.

In fabrication, standard guides, bounded travel, and controlled braking or damping can turn those forces into slow, safe motion without changing that relation. The evaluated proposal already specifies a safety rail, secondary pulley brakes, pre-show checks, and a manual winch for reset. Figure 6 shows one plausible single-channel realization; its particular brake/drum, redirect-pulley, damping, and travel-limit arrangements are explanatory production choices added after evaluation.

The external exact-parent comparison is orientation-sensitive (logical-pair score 0.5). One orientation prefers the Focused challenger because unequal brick masses make inherited obligation more directly material; the reversed orientation prefers the parent, albeit with lower confidence, for retaining the richer temporal mediation of the clock mechanisms and a stronger sense of bodily implication. The case therefore illustrates both a material-causal reorganization and the role of opportunity synthesis in constraining local search, while making the non-monotonicity of such development visible: strengthening one relation can weaken another. It does not imply that direct causality is generally artistically superior to mediated causality, nor that larger transformations are inherently better.

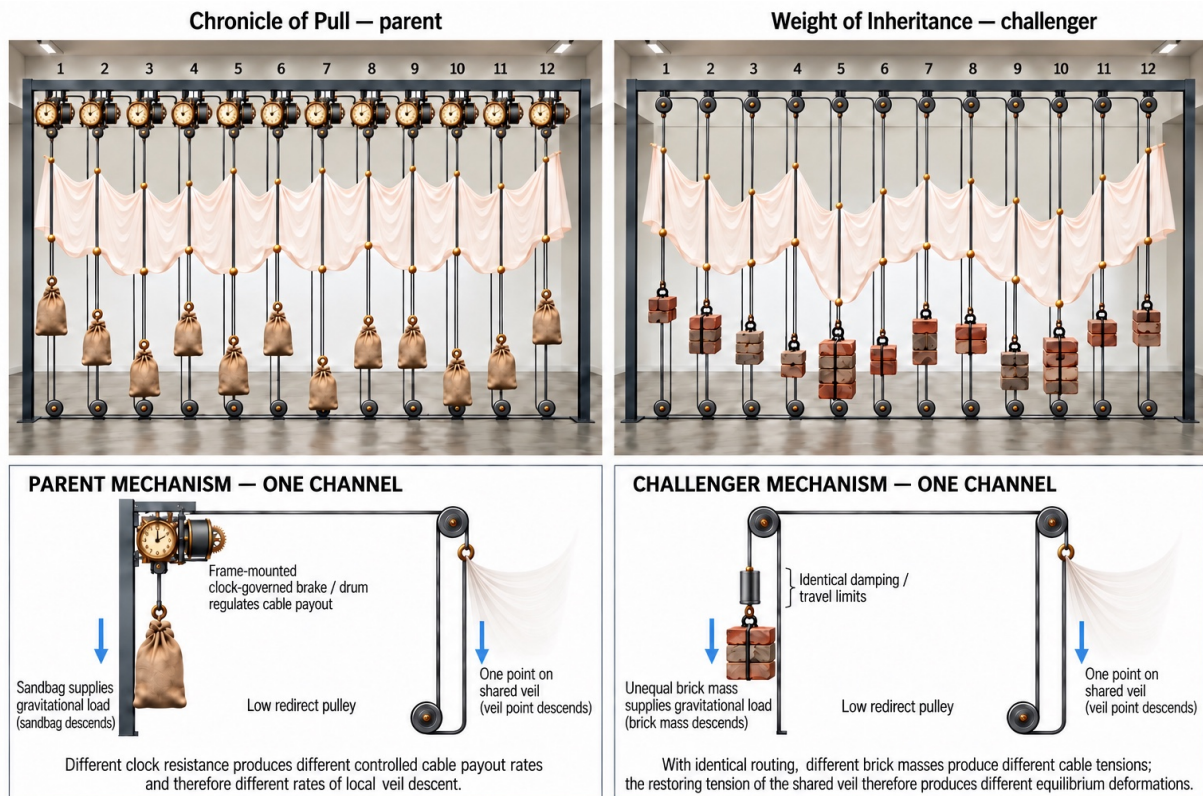


Figure 6: Post-hoc explanatory visualization of the selected Focused transformation in Brief 030, Chronicle of Pull → Weight of Inheritance. Top: matched installation views of the parent and challenger. Bottom: one schematic single-channel interpretation of the causal reorganization. In the parent, a sandbag supplies gravitational load while a clock-governed brake regulates cable payout; in the challenger, unequal reclaimed-brick masses drive unequal cable tension directly. Both act on the same shared veil through pulley routing. The rendering schematizes this brick–pulley–veil relation rather than reproducing every implementation-level detail of the selected dossier; the specific brake/drum, redirect-pulley, damping, and travel-limit arrangements are illustrative production choices added after evaluation. The visualization was not shown to the external evaluator and provides no independent evidence of quality; the external evidence remains the blind exact-parent text comparison.

7.2.3 Spatial encounter: Brief 038

Brief 038 constrains the work to chalk, reclaimed brick, and rainwater-safe sealant and asks that a disputed boundary remain legible under partial weathering. In *Eroding Chalk Line Labyrinth*, reclaimed-brick panels line the gallery perimeter and a continuous chalk boundary snakes across their frontal surfaces. Selective sealant and periodic wetting are intended to make sections weather at different rates, while the visitor mainly walks parallel to the wall and watches the line fragment.

The generated Structural challenger, *Fragmented Borders: Erosion Field*, moves the same basic elements from the wall onto the floor. Low reclaimed-brick slabs form an irregular walkable field and the chalk boundary runs through it. The visitor must now cross, follow, step around, or choose a route through the line rather than observe it mainly from in front. The material vocabulary and gradual weathering remain, but the spectator's bodily relation to the disputed boundary changes.

This Structural dossier is rejected by the hard gate before lane comparison. The stated failure is specific to the written implementation: the dossier allows chalk to be reapplied mid-exhibition if legibility drops below threshold, whereas the brief requires the boundary to remain readable after partial weathering. The rejection therefore concerns admissibility under an explicit brief requirement rather than a comparative judgment that the spatial transformation is artistically weaker.

A fabricated version could in principle calibrate sealant coverage or thickness, chalk application, and water delivery to produce staggered but still legible weathering, while ordinary slab fixing, drainage, slip resistance, and an accessible route address floor safety. These are post-hoc production interpretations: they do not alter the gate outcome, which is determined by

the evaluated dossier including its reapplication fallback. The central spatial transformation remains:

Parent: boundary encountered mainly as demarcation

Challenger: boundary negotiated as terrain

Because the challenger is gate-rejected, it is not an adopted replacement and does not contribute an exact-parent replacement outcome to RQ2. The case remains useful because the transformation itself is unusually easy to inspect while its rejection is traceable to a separate eligibility failure. It therefore makes visible a distinction that aggregate scores cannot: a consequential spatial reorganization can be generated and still be excluded from competition for a concrete brief-level incompatibility. The hard gate is an admissibility mechanism, not an aesthetic ranking of the proposal.

Eroding Chalk Line Labyrinth — parent



Fragmented Borders: Erosion Field — challenger

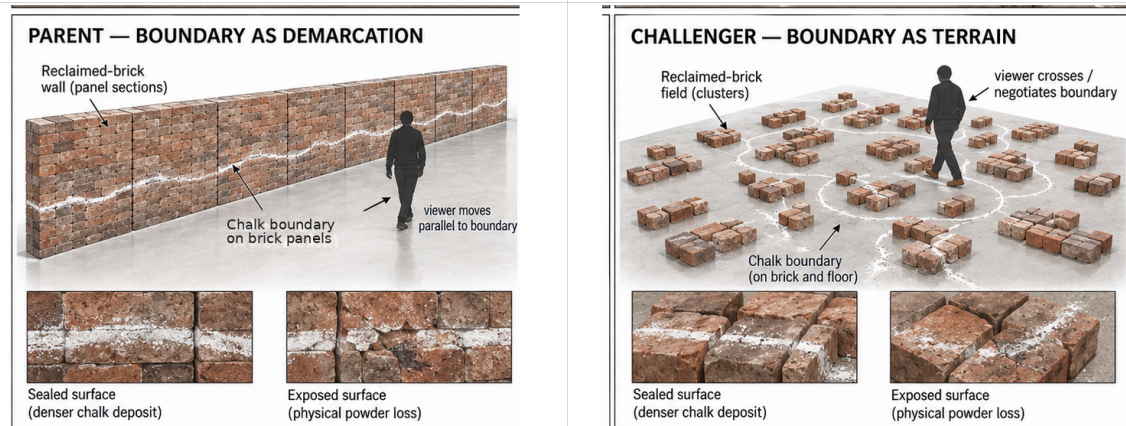


Figure 7: Post-hoc explanatory visualization of the gate-rejected Structural challenger in Brief 038, Eroding Chalk Line Labyrinth → Fragmented Borders: Erosion Field. Top: matched installation views of the parent and challenger. Bottom: simplified diagrams of the spatial and weathering logic. In the parent, a serpentine chalk boundary remains on reclaimed-brick perimeter panels and is encountered mainly while moving parallel to the wall; in the challenger, low reclaimed-brick slabs form a walkable field through which the chalk boundary passes, requiring visitors to cross and negotiate it. Differential chalk retention illustrates the proposed sealed/exposed weathering relation. The renderings and diagrammatic fabrication details were added after evaluation to clarify the textual proposals; they were not used by the gate or shown to the external evaluator and provide no independent evidence of quality.

7.2.4 Institutional constraint as operative structure: Brief 059

Brief 059 provides an internally selected Focused reconstruction that the external evaluator prefers to its exact parent in both orientations. The brief asks for an installation about interrupted scholarship that can be paused for events, stored in labeled cases, and restored by non-specialist staff from a one-page diagram. In *Retractable Lecture Shelf*, a six-meter steel rail carries ten trays of printed scholarly texts. For an event, a motorized lift lowers the rail about 1.2 meters into a recessed floor cavity; afterward, staff raise it and restore the numbered tray order from a diagram. Such a mechanism is plausible where a suitable floor void or purpose-built recess is available, but the brief's explicit labeled-case storage requirement remains peripheral to the artistic action.

The selected Focused reconstruction, *Fold-Out Fabric Archive*, replaces the lift with ten fire-rated fabric panels on a lightweight hinged frame, each carrying a tray of essays. When displayed, the ten panels form a continuous vertical fabric wall, with the essay trays visible across its surface. Staff release and fold the panels into a compact stack, place the folded elements in three labeled, color-coded cases in adjacent alcoves, and later follow the one-page diagram to unfold the structure and realign the trays. Hinge loads, tray retention, case packing, and fire/egress details remain ordinary production-design questions rather than the conceptual point of the transformation.

What changes is the status of the institutional procedure itself. In the parent, interruption and restoration are accommodated by a technical mechanism while storage remains largely backstage. In the Focused reconstruction, interruption becomes visible and repeatable:

display → pause → fold → case → restore → display

The external evaluator preferred the challenger in both A/B orientations. The distinction is not simply easier fabrication: a requirement that previously sat at the edge of the proposal becomes part of what the work visibly does. The example also clarifies the breadth of Focused reconstruction. The realization changes substantially, but the intervention remains organized around the protected artistic problem rather than using the broader Structural permission to replace the proposal’s organizing identity.

7.3 Contestable Refinement Can Succeed Locally, Fail Locally, or Leave the Parent Untouched

The positive mean RQ2 effect is an average, not a guarantee on every adopted intervention. Of the 228 adopted replacements, 179 have swap-consistent exact-parent judgments; 53 of those are externally worse than their exact parents, while another 49 adopted-replacement comparisons are orientation-sensitive. Contestability therefore preserves a recovery path but does not guarantee that the internal comparator will use it. A generated challenger can still be weaker than its parent under the external criterion, and the lane comparator can still select it.

Brief 036 makes the distinction between local and aggregate outcomes unusually visible. All five lineages adopt a challenger, but their exact-parent outcomes are mixed: one Focused replacement wins in both orientations, three Focused replacements lose in both orientations, and one Structural replacement is orientation-sensitive. The complete A5 pool nevertheless has $\Delta U = +0.60$ against M5. The contrast is not contradictory because the two analyses ask different questions: RQ2 compares each adopted intervention with its own unchanged B5-self parent, whereas RQ1 compares the resulting five-proposal A5 population with the separately developed M5 population. A brief can therefore contain locally losing interventions and still outperform M5 at the pool level.

Table 18: Brief 036: local intervention outcomes within one five-lineage search. Exact-parent outcome is derived from the two swapped external comparisons.

Lane	Final state	Exact-parent outcome
1	Focused reconstruction	loss in both orientations
2	Focused reconstruction	loss in both orientations
3	Focused reconstruction	loss in both orientations
4	Structural escape	orientation-sensitive
5	Focused reconstruction	win in both orientations
Complete five-proposal A5 vs M5 effect: +0.60		

7.3.1 Brief 036: two internally selected reconstructions, opposite external judgments

The mixed outcomes in Table 18 are easier to interpret through two Focused reconstructions from the same brief. Both are selected by the internal lane comparison, but the external exact-parent judgments move in opposite directions. Brief 036 therefore exposes not a simple division between a “good” and a “bad” intervention, but two different artistic trade-offs produced by the same development policy.

In one lineage, *Frayed Translation Rope* presents progressive fiber damage as an already-produced state along a tensioned hemp rope. The selected Focused reconstruction, *Live Fracture Translation Channel*, changes the status of that damage: pressure-sensitive floor locations trigger a corresponding cutting mechanism during ordinary passage, so a spectator does not

merely encounter the record of weakening but becomes causally implicated in producing the next fracture. Relative to the parent, the challenger therefore gains immediacy, embodied consequence, and a more explicit spectator-to-material causal relation. The external evaluator nevertheless prefers the parent in both swapped orientations. Its rationales emphasize the cost of achieving that gain: the sensor-driven cutting system is described as overengineered and safety-intensive, while the simpler rope already makes cumulative weakening materially legible. This is best read as an evaluative trade-off rather than a straightforward failed intervention. The challenger strengthens live causality and spectator consequence, while the external criterion assigns greater weight to material economy, feasibility, and the fact that the translation logic need not depend on an elaborate actuation system.

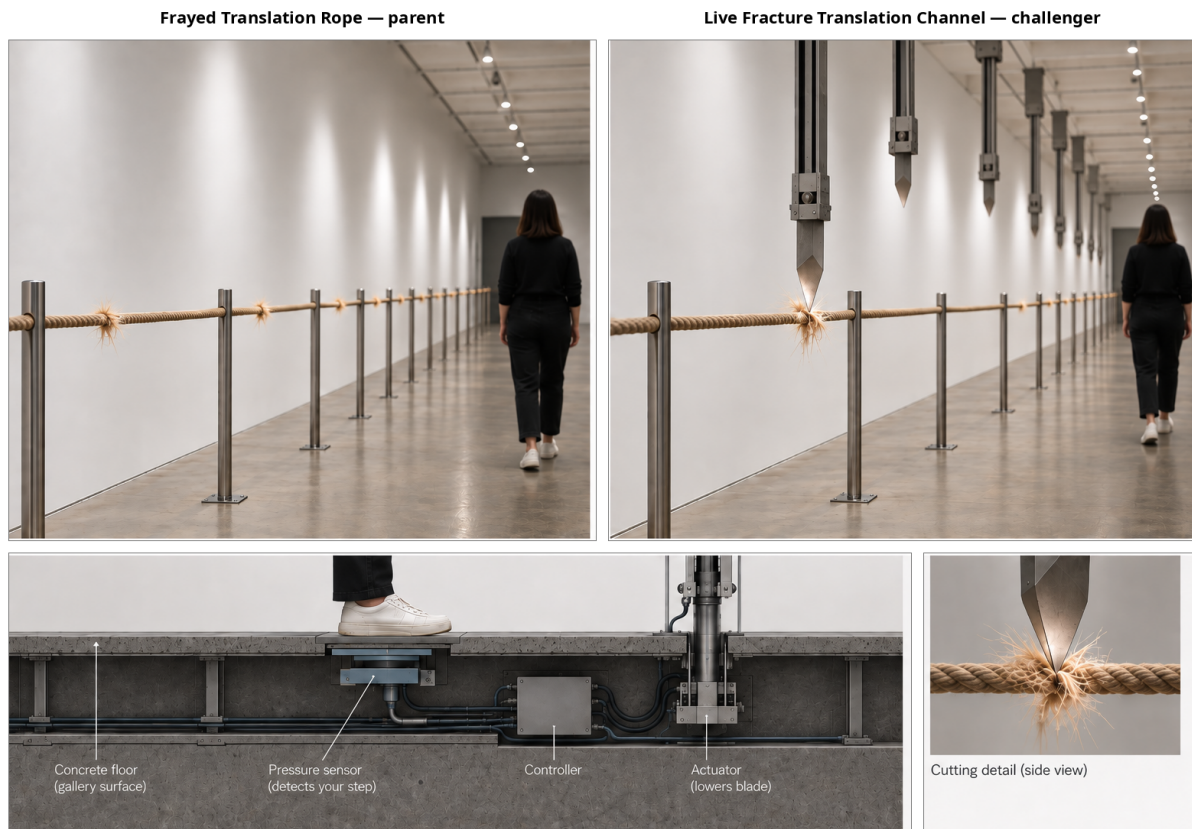


Figure 8: Post-hoc explanatory visualization of the selected Focused reconstruction in Brief 036, *Frayed Translation Rope* → *Live Fracture Translation Channel*. Top: matched installation views of the parent and challenger. In the parent, progressively frayed sections of a tensioned hemp rope are encountered as already-produced material states; in the challenger, a visitor’s otherwise ordinary passage through the corridor triggers further damage live, so that stepping on a pressure-sensitive area causes the corresponding blade to descend and cut the rope at that position. Bottom: one plausible fabrication interpretation of this trigger-to-cut mechanism, showing a pressure sensor beneath the gallery floor, its connection to a controller and blade actuator, and the resulting local cut. The evaluated dossier encloses the rope and cutting points within a clear acrylic trough; the open-rope presentation shown here, together with the sensor placement, controller, actuator, and blade geometry, is a post-hoc explanatory simplification added after evaluation. The visualization was not shown to the external evaluator and provides no independent evidence of quality; the external evidence remains the blind exact-parent text comparison.

A second lineage produces the opposite result. *Grain-by-Grain Funnel Cascade – Refined* lets one quartz-sand body descend automatically through a vertical stack of funnels and calibrated meshes, making cumulative loss visible as a progressively thinning stream. The selected Focused reconstruction, *Hand-Off Sand Transfer Loop*, keeps successive material loss but changes who and what performs the transfer: the visitor manually opens each stage, and the same diminishing sand body is handed from one receiving stage to the next while a fraction remains behind. The external evaluator prefers this challenger in both swapped orientations, citing the clearer and more embodied chain of transfer and residue relative to the automated funnel cascade. Here the additional spectator action does not operate as generic “interactivity”; it makes the brief’s

successive hand-off itself physically explicit.

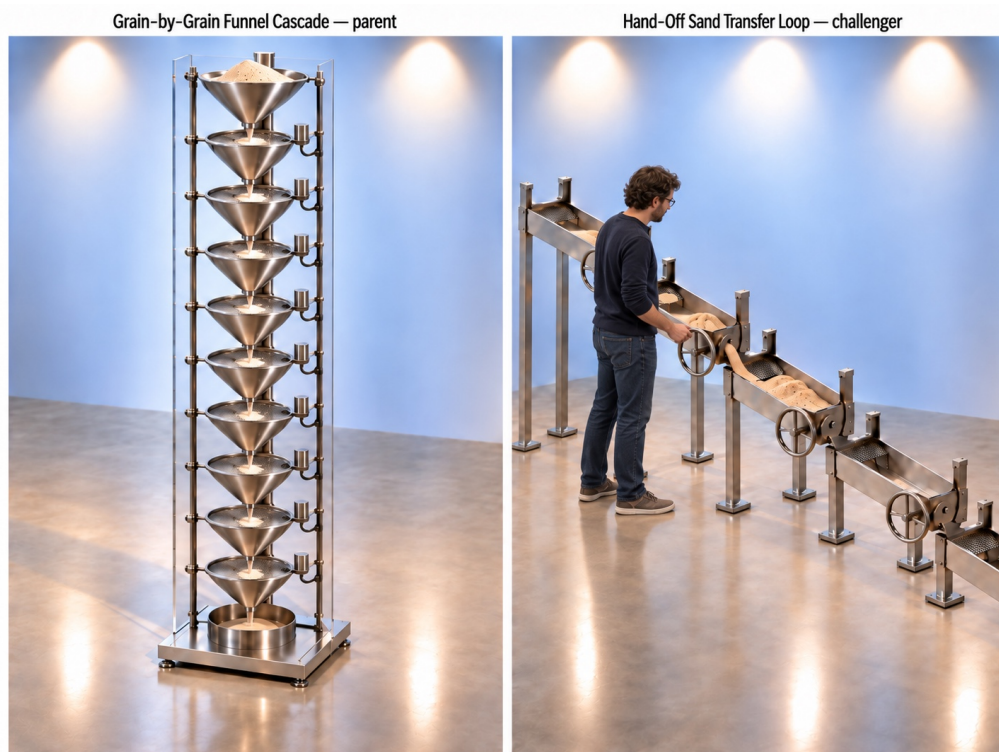


Figure 9: Post-hoc explanatory visualization of the Brief 036 transformation, *Grain-by-Grain Funnel Cascade – Refined* → *Hand-Off Sand Transfer Loop*. Left: the parent proposal, in which quartz sand descends automatically through a vertical stack of funnels and successive meshes, producing a progressively thinning stream. Right: an illustrative five-stage view of the challenger’s repeated hand-off logic. A visitor operates one of the hand-off controls while the diminishing sand body moves through successive receiving stages; retained sand is shown only on the mesh sections within the stages, rather than as loose spill on the gallery floor. The evaluated challenger specifies ten sealed chambers with hand-operated valves and calibrated meshes; the reduced five-stage cascade, open receiving geometry, visible visitor, and specific valve/fabrication details shown here are post-hoc explanatory choices added to make the sequential manual transfer and cumulative retention visually legible. The visualization was not shown to the external evaluator and provides no independent evidence of quality; the external evidence remains the blind exact-parent text comparison.

Taken together, the two illustrated lineages make the non-monotonic character of refinement concrete without implying that the externally preferred proposal is artistically definitive. Both selected reconstructions make spectator action more causally immediate. In the sand lineage, the external evaluator treats that added action as clarifying the brief-specific chain of transfer; in the rope lineage, it treats the machinery required for live cutting as too costly relative to the gain. The contrast therefore localizes a genuine tension between immediacy and embodied causality on one side, and economy, feasibility, and material directness on the other. The exact-parent judgment records how the held-out evaluator resolves that tension; it does not erase the artistic property gained by the rejected challenger.

7.3.2 Brief 050: contestability succeeds in one lane and fails in another

In lane 01, *Threaded Pause* places a 3 m × 2 m motorized loom in the gallery and invites volunteers to continue a shared tapestry. After five minutes, a safety-rated rotary cutter cuts the warp at its center and automatically stops the loom, leaving a visible rupture that terminates that weaving cycle. The opportunity memo identifies a plausible liability: the timed cut may already read as a one-off gimmick rather than as a necessary interruption of the shared ritual. The Structural challenger, *Resonant Pause*, keeps the same loom, five-minute wait, and cutter while adding a sub-second white-noise burst and synchronized red LED flash at the rupture; it therefore intensifies the event without reorganizing the underlying relation. Both challengers pass eligibility, but the three cyclic rankings produce a 3/3/3 Borda tie. The pointwise tie-break favors the incumbent (0.790, versus 0.780 for the Focused challenger and 0.765 for the Structural challenger), so *Threaded Pause* survives. Here contestability works as intended: an admissible

but weakly responsive alternative can be considered without making displacement mandatory.

A second lane in the same brief shows the opposite outcome. *Procession Grid Freeze* uses a 4 m × 4 m pressure-sensitive LED floor grid in which a blue light wave advances like a procession and, after three complete traversals, a timed command cuts power to the downstream panels so that the line freezes mid-route. The selected Structural challenger, *Bureaucratic Halt Grid*, keeps the advancing grid but replaces the preset halt with participant-triggered bureaucratic authority: a stainless-steel console with a red STOP button wirelessly freezes the downstream LEDs and plays a short official-style announcement. Internally, this challenger wins the lane decisively (Borda 6/3/0; pointwise 0.815 versus 0.790 for the parent and 0.770 for the Focused challenger), with the comparison favoring its explicit participant agency and socially legible procedural interruption.

The external exact-parent evaluator nevertheless prefers *Procession Grid Freeze* in both A/B orientations (logical-pair score 0.0 for the selected challenger). The two judgments are low-confidence but swap-consistent: the parent is preferred because the interruption remains inherent to the movement ritual, whereas the STOP console and prerecorded command are read as a more theatrical or game-like overlay. The case therefore exposes a real evaluative tension rather than an arbitrary comparator mistake: explicit social agency and bureaucratic legibility are rewarded internally, while the external criterion favors an interruption that remains materially and perceptually intrinsic to the procession.

Incumbent eligibility is therefore *necessary for recovery from a bad proposed revision, but not sufficient*: lane 01 retains the parent, while lane 02 selects a challenger that the external exact-parent comparison prefers less. This is also why reviews, opportunity memos, and generated alternatives should be read as search evidence rather than certificates of improvement. A plausible transformation can still yield a weaker proposal under the external criterion, and downstream comparison is itself fallible.

7.4 Terminal Selection Errors Arise at Different Stages

The RQ3 aggregate identifies recognition headroom but does not reveal where ranking errors enter. We therefore inspect a post-hoc stage-specific contrast within the Hybrid policy itself. For each brief, we compare the external Utility@1 of the final Hybrid top choice with the external utility of the proposal that Hybrid’s own pointwise stage ranked first before the finalist tournament. Thus the diagnostic is $\text{Utility@1}_{\text{Hybrid final}} - \text{Utility@1}_{\text{Hybrid pointwise}}$, not the primary Hybrid-minus-Base contrast.

The tournament changes top-1 on 9 of 50 briefs. The mean final-minus-pointwise Utility@1 change is +0.0027, with 95% bootstrap interval [-0.0160, +0.0220]; brief W/T/L is 4/43/3 and the paired sign-flip test is not significant ($p = .844$). The experiment therefore does not establish that the tournament stage is beneficial or harmful on average. This aggregate diagnostic is reported before individual failure cases to avoid inferring a general tournament effect from a few conspicuous regressions.

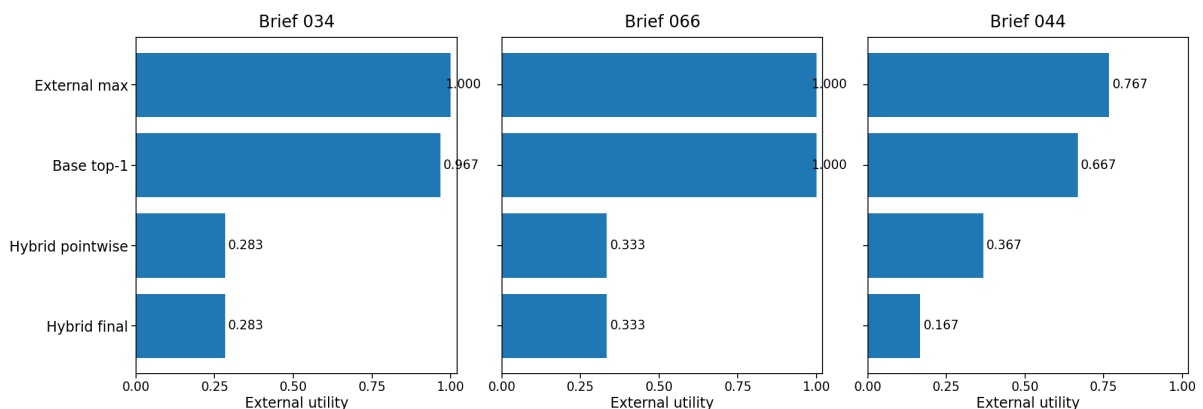


Figure 10: Three locations of terminal-selection error. Brief 034: pre-tournament exclusion; Brief 066: an external-maximum finalist is present but not recovered; Brief 044: tournament reversal of a better pointwise choice. External utility is a diagnostic measured after generation, not an internal ranker score or a claim of objective artistic value.

Against that aggregate background, the three largest Hybrid-versus-Base regressions are informative because they localize three different failure paths.

Brief 034 — pre-tournament exclusion. Brief 034 asks for an installation in a university registrar waiting area about names that do not fit forms, using existing walls only while leaving seating untouched and remaining removable without trace after six weeks. The externally highest-utility A5 proposal, *Gliding Name Ribbon (reconstructed)*, has utility 1.000, but Hybrid's pointwise stage ranks it outside the three-candidate finalist set, so pairwise comparison never sees it. Base instead surfaces *Glowing Forms (recalibrated)*, with external utility 0.967. Hybrid's pointwise and final top choice is *Transient Registry: Hand-Held Reveal*, with external utility 0.283. This is a recognition failure located before the tournament: the strongest measured proposal was already generated but excluded during pointwise screening.

Brief 066 — failure to recover. Brief 066 concerns domestic routines entering public administration, asking repetition to shift in scale from private care to collective procedure without recreating a home. Two A5 proposals tie for the external maximum at utility 1.000: *Rotating Registry of Care* and *Civic Cycle – Stamped Chores Activate Municipal Maintenance*. Base ranks the former first. Hybrid's pointwise stage instead ranks *Civic Pulse – Stamped Transit* first, with external utility 0.333, while *Civic Cycle – Stamped Chores Activate Municipal Maintenance* is already among the three finalists. Pairwise comparison therefore has access to an externally maximum-utility candidate but does not recover it; *Civic Pulse – Stamped Transit* remains the final top choice.

The contrast with Brief 034 is operationally important. Increasing finalist recall could address a 034-like error, but not a 066-like error: here an externally maximum-utility candidate is already present among the finalists and pairwise comparison still fails to promote it.

Brief 044 — tournament reversal. Brief 044 asks for a museum-foyer work about unequal waiting that must operate during school visits, tolerate touching, and require no attendant beyond normal front-of-house staff. Hybrid's pointwise stage ranks *Staggered Ascension of Translucent Spheres* first, with external utility 0.367. The finalist tournament then overturns that order and promotes *Unequal Path*, whose external utility is 0.167. Base surfaces *Delayed Reveal Wall* at 0.667, while the externally highest-utility proposal in the same pool, *Echoing Steps*, reaches 0.767. The tournament therefore converts an already imperfect pointwise choice into an even weaker final recommendation under the external measure. The pairwise preference is not internally arbitrary: the tournament repeatedly favors *Unequal Path* for its embodied, touch-tolerant sand-flow interaction. The external diagnostic nevertheless ranks it below the pointwise choice, showing how additional comparative inference can coherently optimize the internal contract while worsening the external score.

This is a tournament-induced regression on one brief, not evidence that tournament comparison is harmful in general. The aggregate stage-specific estimate remains unresolved; the case matters because it localizes one way in which additional comparative inference can overturn a better existing choice.

7.5 What the Mechanism Evidence Establishes

Taken together, the analyses in this section make the computational distinctions behind the aggregate results inspectable. The post-hoc rationale coding identifies an asymmetric pattern in what the external evaluator cites when A5 or a control is preferred: A5 wins are more often attributed to stronger integration with the brief-specific site and its existing processes, while control wins are more often attributed to feasibility, maintenance, safety, or unnecessary elaboration in the A5 alternative. The lineage traces show concrete transformations of material causality, accumulated history, spatial encounter, and institutional procedure, while also exposing non-adoption after competition, hard-gate exclusion, unresolved trade-offs, and comparator error. The two illustrated Brief 036 reconstructions connect these levels directly: one selected revision is externally rejected as overengineered, while another is preferred because it makes the brief-specific hand-off materially and bodily explicit.

Terminal-ranking traces reveal a separate error surface: a higher-utility candidate can be generated but excluded before pairwise comparison, can enter the finalist set without being recovered, or can be displaced when tournament aggregation overturns a better pointwise choice. These failure locations imply different technical problems even though all appear at the interface as a weak first-ranked proposal.

These traces add interpretability, not new efficacy observations or independent expert validation. The primary quantitative claims remain those defined in Sections 5–6. In particular, the A5–M5 contrast concerns the full A1rtist development procedure relative to substantially

matched generic iteration and does not identify the causal contribution of any single review, memo field, or challenger role. Likewise, the explanatory renderings and fabrication details were produced after evaluation and do not establish professional curatorial consensus, realized-artwork quality, or engineering validation.

8 Discussion

8.1 What the Decomposition Reveals About Concept Development

For contemporary-art concept development, the central methodological implication of the experiment is that three consequential stages can diverge under evaluation: candidate-set construction, lineage transition, and terminal recognition. A weak surfaced proposal may reflect a weak field of alternatives, an intervention that damaged a stronger state, or a failure to recognize which surviving proposal deserved attention. Final-output evaluation can easily collapse these distinct failures into the same observed outcome.

The empirical evidence is correspondingly asymmetric. A5 candidate sets score higher than all three planned controls under the external measure, and adopted replacements are externally preferred to their exact parents on average, whereas terminal selection remains unresolved.

This asymmetry suggests a more general diagnostic for creative agents. One compelling top-ranked output can conceal a weak candidate population or a harmful development policy, while a poor surfaced proposal can hide stronger material that already exists. Treating the sampled set, lineage transition, and terminal ranking as separate empirical objects makes those failure modes separately observable rather than inferring the quality of the whole process from its final recommendation.

The decomposition is operational rather than ontological. We do not claim that creativity consists of three independent modules, nor that these stages correspond literally to how artists develop work. Their value is experimental: they expose failure modes that an end-to-end winner-only evaluation would otherwise merge.

8.2 The Full Structured-Search Package Yields Higher Measured Candidate-Set Quality

The matched M5 comparison provides the strongest evidence for the full AIrtist development procedure relative to substantially matched generic iteration. M5 begins from the same self-refined parents and preserves much of the same iterative competitive scaffold, while A5 additionally uses field-context construction, specialized multi-role review, opportunity synthesis, preservation constraints, and differentiated Focused and Structural roles. The +0.2352 complete-pool effect therefore identifies an advantage of this complete procedure under the external evaluation, not an isolated causal contribution of search semantics: the organization and the amount/context of analytical inference change together.

Because the two conditions are substantially matched rather than compute-identical, attribution to any individual mechanism remains unresolved (see Section 10.3). A compute-matched ablation that changes the organization of search reasoning while holding its amount and context fixed would be needed to isolate that question.

The broader implication is therefore a hypothesis about how open-ended search should be studied, not a claim that the present experiment has isolated its cause. Iterative critique and revision are already established in systems such as Self-Refine and ResearchAgent [56, 8]. AIrtist adds a more explicit structure to the local move: what the revision is allowed to change, what it should preserve, and which kind of transformation is being attempted. Search is often characterized by depth, branching factor, sample count, critique rounds, or voting procedures; in weakly specified creative domains, the semantics and permissions assigned to a move may deserve separate study alongside search width. The present experiment motivates that comparison but does not resolve it, and transfer beyond installation ideation remains an empirical question.

8.3 Contestable Refinement Treats Revision as a Proposal, Not an Obligation

Competitive refinement keeps non-replacement available as a real outcome rather than making every generated challenger compulsory. Across 250 lineages, 22 final states (8.8%) remain content-identical to their incumbents, while 228 adopt a challenger. This is descriptive, not a count of explicit “keep” judgments—retention can follow challenger ineligibility or lane comparison—and in this evaluation it is an available but uncommon outcome.

Among the 228 adopted replacements, exact-parent comparison favors the challengers on

average: candidate-weighted preference is 0.6601 and the brief-balanced centered replacement effect is +0.3347 (Section 6.2). Together, these observations support a bounded interpretation: diagnosis can propose a change without making rewrite compulsory. The Brief 050 trace in Section 7.3 shows both sides of that protection: contestability can recover an incumbent, but internal comparison can also select a replacement that loses to its exact parent under the external criterion. Optionality creates a recovery path, not a guarantee.

For open-ended domains in which plausible local revisions can exchange rather than monotonically accumulate virtues, this suggests testing proposed change against the current state as a distinct use of additional inference. The present experiment does not isolate the causal value of optional retention; that would require a matched mandatory-replacement policy (Section 10.3).

8.4 Evaluator Rationales Suggest Both a Strength and an Over-Development Failure Mode

The post-hoc rationale coding adds a narrower view of how the evaluator explains stable preferences. A5 wins are more often attributed to stronger integration of the artwork with the brief-specific site and its existing processes. Control wins, by contrast, are more often attributed to weaknesses in the A5 proposal—especially feasibility, maintenance, safety, or unnecessary elaboration. These contrasts have the same direction in all three control campaigns. Because the rationales are generated by the same evaluator that issues the verdicts, the result should be read as a description of evaluator attention rather than independent validation of the artistic properties involved.

The pattern is consistent with two pressures already built into AArtist’s representation. Site and institutional operation may become constitutive parts of the artistic proposal rather than remaining downstream logistics, while technological sophistication, additive media, complexity, and readiness detail receive no automatic credit when they do not strengthen the governing artistic relation. The rationale analysis does not validate either design choice causally. It instead exposes a practical tension for structured development: tighter coupling to the site or process specified in the brief can strengthen a proposal, but further development can also become over-development when added apparatus, maintenance, or safety burden outgrows the artistic gain.

8.5 Recognition Failure Is Not One Problem

Stronger externally measured material often already exists in the five-proposal pool, but neither terminal policy establishes superiority. Base has higher mean Utility@1 (0.6633 versus 0.6353) and lower regret, while Hybrid more often places an externally maximal candidate first (0.32 versus 0.26). Proposal construction and recognition are therefore empirically separable. Adjacent research-ideation systems report related gaps between generation and self-evaluation [52, 9], although their tasks differ from ours.

Section 7.4 localizes recognition failure at several stages. Screening can exclude an externally strongest candidate before the tournament; finalist comparison can fail even when such a candidate is present; and tournament aggregation can overturn a better pointwise choice. The three traces in Section 7.4 instantiate these distinct failure locations.

These paths imply different remedies. Increasing finalist recall cannot repair a comparison failure when the strongest candidate is already present; better finalist comparison or aggregation cannot recover a candidate screened out earlier. The cases are diagnostic rather than frequency estimates, and the post-hoc final-minus-pointwise analysis does not establish that the tournament is beneficial or harmful on average. The narrower implication is that recognition should be evaluated stage by stage. Artist-facing rank should likewise be treated as attention allocation rather than an authoritative verdict; Section 9.3 develops that consequence.

8.6 Domain Grounding Is Both an Inspectable Prior and an Inspectable Error Surface

AArtist’s domain grounding encodes a recognizable *working position*: enacted artistic operation over topical illustration, necessary material and spatial relations over arbitrary styling, specific spectator relations over generic participation, field differentiation over first-obvious formats, and changes to the diagnosed relation over merely additive explanation or spectacle. The package-level A5–M5 advantage does not identify which commitment, if any, causes the gain. Their value is that they are explicit and inspectable: the system records what it intends to preserve, what it treats as the dominant liability, which nearby format it seeks to escape, and what transformation it authorizes. This makes the prior more legible than “make this more creative.”

The case traces expose the corresponding error surface without identifying which commitment caused any particular outcome. Contestability can recover an incumbent yet still permit a comparator error elsewhere; a consequential Structural transformation can be excluded because it violates an explicit brief constraint; and a selected transformation can strengthen one relation while weakening another under the external criterion. The same explicit search position can therefore coexist with an aggregate advantage, local regressions, and exclusions.

A domain-specific prior is therefore better treated as a *contestable search position* whose consequences and characteristic errors can be audited, rather than as expert truth injected into a generic model. Installation ideation makes this tension unusually visible because revision is non-monotonic, relevant qualities can conflict, and material, spatial, spectator, and institutional relations cannot be reduced to surface correctness. The architecture and evidence consequently motivate a bounded design hypothesis: preserve several viable states long enough to compare them, make intervention contestable rather than compulsory, represent intended transformation explicitly, and audit recognition separately from generation. The experiment does not isolate each ingredient’s causal contribution or establish how artists will use these states; those questions require the ablations and negotiated studies developed below.

9 Intended Artist-Facing Use: Idea Search as a Substrate for Negotiated Co-Creativity

The evaluated system is an autonomous concept-development backend rather than an interactive co-creative interface. Its intended artist-facing use begins from an early-stage creative situation in which an artist has a direction but not yet a specification: they may know what matters, which conditions or materials constrain the problem, and which obvious solutions they want to exclude while remaining uncertain about what the work should become. AIrtist takes one such rough brief and autonomously constructs, develops, and ranks five complete project hypotheses. The experiment evaluates that computational layer. A future interaction layer would make some of the assumptions and decisions exposed by the backend negotiable to the artist rather than treating its search process as fixed.

9.1 From a Provisional Query to Five Project Hypotheses

A conventional generative interface often treats the prompt as a specification: the user states what is wanted, and the model attempts to produce an answer. Early creative work is different because part of the problem is precisely that the desired answer is not yet known, and the formulation of the problem itself may change as possible responses become more concrete. Human–AI design-space research has made this co-evolution of problem formulation and candidate solutions explicit in adjacent creative work [51].

We therefore use an information-retrieval analogy in a restricted sense: the artist’s brief functions like a provisional query because it can specify a need without specifying its answer, and because that query can in principle be reformulated as the need becomes better understood. The free-text premise identifies what is at stake; `must_address` conditions protect pressures the artist does not want development to lose; `must_avoid` conditions remove solution families already known to be unwanted; and site, material, institutional, or production requirements further constrain the region to be explored. The resulting input specifies an under-determined development problem rather than a target artifact.

The analogy concerns the relation between request and answer, not the retrieval mechanism. AIrtist does not search a database of pre-authored artworks. It performs generative search over proposal states: candidate directions are constructed, reviewed, transformed, preserved or displaced, and finally ranked. The resulting five dossiers are comparatively developed project hypotheses whose artistic operation, spatial and material logic, spectator relation, and production implications are made explicit as described in Section 4.

In the evaluated configuration, the artist delegates that bounded search after submitting the brief and receives the resulting proposal set for subsequent judgment and development. The dossiers are starting directions for further artistic work, not five finished artworks from which the system has identified a correct one.

9.2 The Sampled Possibility Set as a First-Class Computational Output

The experiment evaluates a five-proposal sampled possibility set rather than only the proposal surfaced at rank 1. This matters computationally because proposal development and terminal

recognition are separable: a system can construct a stronger candidate set without reliably placing its strongest measured member first.

RQ1 supports treating the sampled set as a first-class computational output rather than evaluating only its top-ranked member. The A5 advantage over the substantially matched iterative control is measured over the complete five-proposal pool, and the gain is not confined to one oracle-best candidate (Section 6.1.1). This result does not establish that artists benefit from receiving exactly five dossiers, that five is an optimal set size, or that every surrounding alternative deserves preservation. Those are artist-facing questions for subsequent study. The narrower implication is that the surrounding proposals contain experimentally relevant information and should not be treated as disposable scaffolding around one winner.

Across the evaluated briefs, the externally highest-utility candidate already present in each A5 pool has mean utility 0.8617, whereas the proposal surfaced first has mean Utility@1 0.6633 under Base and 0.6353 under Hybrid. In the secondary selected-output check, the A5–M5 comparison is near parity at 0.520 despite the strong complete-pool A5–M5 advantage. These results do not show that an artist would identify the externally highest-utility dossier more reliably than the model; they show that automatically collapsing the developed set to one system-selected output can discard measured value already present in the pool.

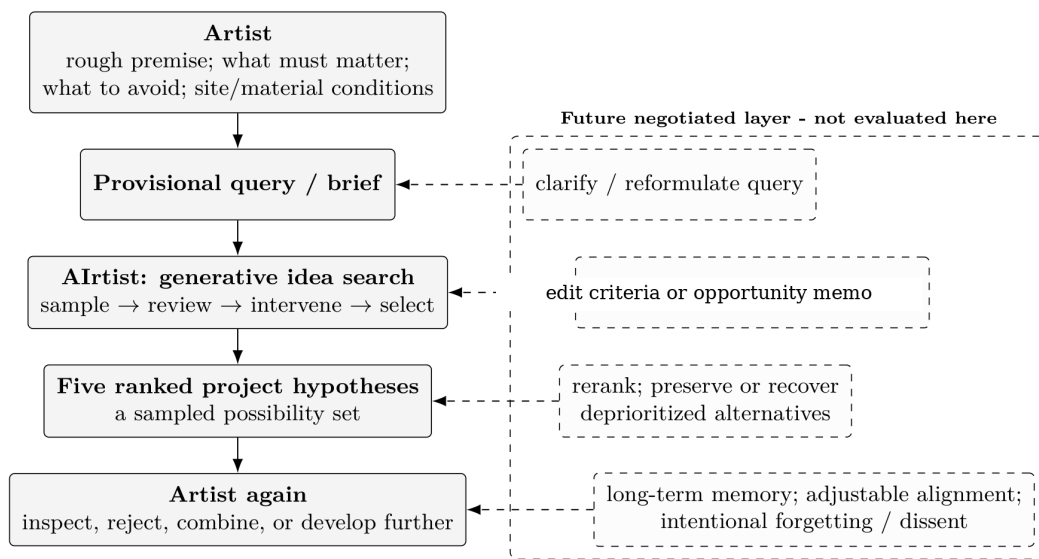


Figure 11: From autonomous idea search to a possible future negotiation layer. The evaluated system expands an artist-authored provisional brief into a ranked sampled possibility set of five complete project hypotheses without artist feedback inside the run. The dashed right-hand layer is a conceptual design sketch, not an implemented or evaluated interface: query reformulation, editable opportunity memos and criteria, branch recovery, reranking, and longitudinal personalization remain future interaction mechanisms.

For an artist-facing system, the resulting design hypothesis is deliberately modest: candidate quality and breadth may need to remain separately visible and separately controllable. A future interface could, for example, cluster closely related proposals, preserve one or more minority directions, or expose operation-, material-, or spectator-level differences rather than assuming that five branches are meaningfully distinct. The present experiment does not evaluate such policies or show that artists prefer them; it only motivates testing them instead of treating multiplicity itself as sufficient diversity.

9.3 Ranking as Local Visibility and Attention Allocation

Generating several alternatives creates a second problem familiar from information retrieval: not everything that exists in the candidate collection will receive equal attention. Within the sampled possibility set, ranking therefore determines local visibility: which generated direction is encountered first, which proposal anchors comparison with the others, and which candidates risk being overlooked. At larger cultural and institutional scales, ranking systems can govern visibility in much broader ways. The present experiment makes no such culture-wide claim; it studies the smallest instance of the same structural problem, namely visibility among five candidate project hypotheses produced for one artist-authored brief.

RQ3 supplies the empirical reason to keep this ordering provisional. Under the external

model-based measure, stronger candidates often already exist in the generated pool without being surfaced first. Neither terminal policy establishes superiority: Base has higher mean Utility@1 (0.6633 versus 0.6353) and lower regret, while Hybrid more often places an externally maximal candidate first (0.32 versus 0.26); the secondary ranking metrics are therefore mixed. Section 7.4 further shows that recognition error is not confined to one stage: a strong candidate can be excluded before the finalist tournament, can remain among the finalists without being recovered, or can be displaced when tournament aggregation overturns a better pointwise choice. These are different internal failure paths, but at the interface they all appear as the same event: a weaker proposal is shown first.

The artist-facing implication is correspondingly limited but important. Ranking provides one way to sequence attention across five developed dossiers; it should not be treated as an authoritative artistic verdict. A concise statement of the intended semantics is:

The ranking says which proposal AItist recommends reading first, not which proposal the artist should make.

This is also why the intended output remains the ranked set rather than rank 1 alone. Because recognition can fail at several stages, preserving access to lower-ranked proposals protects against more than an imperfect final sort: it keeps candidates recoverable when screening or comparison has allocated attention badly. The recognition gap makes this recoverability an empirical safeguard against automated attention-allocation error, not merely a diversity affordance. The artist can inspect, reject, revive, or further develop a deprioritized direction rather than inheriting the selector's first choice as a hidden decision about what deserves continuation.

This creates a direct connection between information retrieval and co-creative system design. Relevance is no longer only a property of documents retrieved for an information need; it can govern which generated possibility becomes creatively visible within an interaction. A future system could expose alternative orderings of the same sampled set according to different criteria—for example conceptual necessity, feasibility, public or ethical risk, or distance from a creator's habitual solutions—instead of collapsing all visibility into one fixed ordering. The present experiment intentionally does not evaluate such criteria-contestable ranking. It holds one bounded internal working model fixed so that ranking policies can be compared without simultaneously changing the criterion of preference.

9.4 From Contestable Search to Human-Negotiated Search

The evaluated system implements autonomous search, not human negotiation over that search. It provides no artist-side query clarification, opportunity-memo editing, criteria negotiation, branch recovery during a run, or personalization. Section 9.5 develops these as future interaction mechanisms rather than capabilities of the evaluated system. This boundary matters because empirical work with artists indicates both that domain expertise remains consequential when using generative AI and that artistic co-creative practice is shaped by ongoing control, intuition, and adaptation to system affordances [48, 49].

Competitive refinement nevertheless creates computational objects around which negotiation could later occur. Inside AItist, intervention is already contestable in a limited computational sense: the incumbent remains eligible and a proposed change must earn replacement. Human-negotiated search would expose a different level of contestability. The artist could contest not only a generated candidate but the interpretation of the problem and the search commitments used to generate alternatives.

The opportunity memo is a natural negotiation surface because it makes several steering commitments explicit: which incumbent property should be preserved, what is treated as the dominant liability, what transformation is proposed for testing, and which additive repairs are treated as insufficient. In the present experiment these commitments remain internal. In an artist-facing system they could become objects of disagreement. An artist could accept the diagnosis but reject the proposed transformation, protect a mechanism the model wants to remove, declare that an inefficient material process is essential, or deliberately pursue a direction the default quality model regards as too didactic, decorative, spectacular, or familiar.

This changes what AI initiative would mean. A proactive system could ask questions, point out contradictions, propose transformations, criticize a direction, or refuse an unsafe implementation. Each action can be useful, but each can also shift the formulation of the artistic problem. Human-negotiated search therefore requires more than adding conversational turns: the interface must make sufficiently visible what the system is proposing to preserve or change, where that steering comes from, and where the artist can accept, reject, or redirect it.

9.5 Negotiating the Search Process

The current architecture exposes several natural intervention points for future human negotiation without requiring every internal model call to become a conversational turn.

The first is query negotiation. If the initial brief is a provisional query rather than a complete specification, the system need not assume that the first formulation is sufficient. It could detect unresolved constraints, tensions between `must_address` conditions, or ambiguity about what is intended to remain open, then ask a small number of targeted questions before committing search resources. The important research question is not simply whether clarification improves proposal scores, but when clarification helps articulate an artist's intention and when it instead prematurely determines the problem for them.

The second is opportunity-memo negotiation. Before challenger generation, an artist could inspect, edit, or lock memo fields; request an alternative diagnosis; or ask different agents to formulate competing readings of the same incumbent. A disagreement over what deserves protection, which liability should organize the next move, or which transformation is worth testing could then produce genuinely different searches rather than another rewrite under one hidden interpretation.

The third is recoverable selection. Ranking need not permanently erase losing candidates. A future interface could preserve minority alternatives, permit criterion-specific reranking, display proposals whose pairwise ordering is unstable, and allow the artist to recover a branch that the model deprioritized. The current recognition gap makes this especially important: if selection is fallible, recoverability is not only an interface convenience but a safeguard against premature collapse of the sampled set.

These mechanisms support selective involvement rather than continuous micromanagement. An artist might delegate broad initial search, intervene only when the system surfaces a consequential conflict, and reclaim control when a diagnosis, transformation, or ranking decision would otherwise narrow the directions available for further development.

9.6 Longitudinal Collaboration, Memory, and Aesthetic Enclosure

The current Airtist has no persistent model of an individual artist. Every user receives the same fixed search machinery and bounded working criteria. Long-term creative collaboration, however, creates an obvious role for memory. Retrieval and personalization could make recurring concerns, materials, rejected clichés, philosophical commitments, past projects, and unresolved ideas available to later searches.

Such memory also creates a possible failure mode that we call *aesthetic enclosure*. A system that becomes increasingly good at predicting what an artist has previously accepted may progressively reduce the chance of encountering what the artist has not yet learned to want. Personalization can therefore become another route to fixation rather than simply better alignment. Adjacent studies already show that generative assistance can contribute to design fixation and homogenization of human ideation [10, 53], but they do not test longitudinal personalization or establish aesthetic enclosure as an observed effect.

The present experiment does not test longitudinal memory or personalization. A future personalized system should therefore test, rather than assume, that increasing fit to an artist's prior preferences preserves productive difference over time.

Possible countermeasures are design hypotheses rather than requirements established by the present study. They include different temporal layers of memory, intentional forgetting, adjustable degrees of alignment, or separate agents in which one remains closely aligned while others deliberately retrieve or generate dissenting and unfamiliar directions. Their purpose would be to let personalization accumulate useful continuity without silently turning an artist's past choices into a boundary on future search.

Longitudinal collaboration would also require visible creative-data governance. An artist should be able to inspect what personal material is being treated as precedent, decide what may be retained or forgotten, and distinguish persistent artistic commitments from temporary experiments that should not harden into a permanent profile. In this sense, memory policy becomes part of the creative interface rather than only a background data-management choice.

10 Limitations and Threats to Validity

10.1 Construct Validity: Model Preference Is Not Artistic or Curatorial Judgment

The dominant limitation is construct validity. The principal outcomes in this study are derived from blind holistic comparisons performed by GPT-5.6 Sol. Exact order reversal, consensus analysis, verbosity diagnostics, and leave-out tests probe the stability of this measurement procedure. None establishes that model preferences coincide with those of professional artists, curators, producers, institutions, or publics.

This boundary is particularly important because two model-mediated evaluative structures appear in the system and should not be conflated. AIArtist’s internal ten-dimensional working model shapes review and ranking through explicit concerns such as prompt fidelity, enacted tension, artistic-operation necessity, field distinctiveness, spatial and visual encounter, material necessity, spectator specificity, reception resilience, and production/institutional viability. The external evaluator, by contrast, performs a separate holistic A/B judgment of complete proposals. It is instructed to compare the artistic proposition as a whole rather than mechanically reproduce the internal scorecard. The two instruments overlap in the concerns they attend to, but they are not the same measurement.

All principal preference outcomes nevertheless come from one external evaluator model and configuration. The reversal and robustness analyses test stability within that evaluator under presentation and benchmark perturbations; they do not establish evaluator-model invariance. Replication with independently trained model judges and professional human panels is therefore needed to determine whether the measured ordering is specific to this judging regime.

Consequently, the main results establish comparative preference under a disclosed model-based evaluation instrument; they do not validate the internal quality model as a general theory of contemporary art. Separating the generator from the external evaluator is experimentally important because external utility is unavailable to the generator, lane selector, and terminal rankers, but it does not solve the larger construct-validity problem. Generator and evaluator can still share broader language-model priors about what a convincing contemporary-art proposal sounds like, including preferences for coherence, explanatory completeness, or familiar curatorial rhetoric.

The measurement is also text-bound. The judge compares written dossiers rather than installations, maquettes, spatial walkthroughs, sound fields, performances, or the post-hoc renderings. It may therefore privilege artistic relations that are unusually legible in language and under-represent qualities that depend on tacit material, visual, acoustic, performative, or spatial experience even before fabrication is considered.

The internal working model itself is intentionally normative. It favors operational necessity, material and spatial specificity, explicit spectator relations, field differentiation, resistance to generic additive spectacle, reception resilience, and plausible institutional realization. These are defensible pressures for the constrained installation-ideation problem studied here, but they are neither neutral nor exhaustive. Other artistic and curatorial positions may value, for example, decorative excess, explicit pedagogy, deliberate impracticality, or institutional antagonism differently. Making these commitments explicit improves inspectability; it does not turn them into universal criteria of artistic quality.

The same limitation applies to model-generated reasons. Judge rationales are useful in the Mechanism and Failure Analysis because they reveal what the evaluator says it attended to—for example, site integration, a feasibility problem, or excessive apparatus. They are not an independent source of expert interpretation and should not be treated as causal explanations of why one proposal is artistically superior. Verdict and explanation originate from the same model-mediated instrument. The post-hoc thematic coding is itself LLM-assisted and was not independently human-coded; no inter-rater reliability or human thematic consensus is claimed. The preserved coding artifact also does not retain the exact model/version or inference configuration used for that coding pass, so the analysis should be treated as an exploratory diagnostic rather than a fully specified independent replication protocol.

The strongest claim the study therefore supports is intentionally narrower than “AIArtist makes better art”: under the evaluated experimental setting and external holistic model evaluator, its candidate sets and selected lineage interventions exhibit the measured differences reported in Section 6. Establishing correspondence with professional artistic judgment requires a separate human validation study.

10.2 External and Ecological Validity: Synthetic Briefs and Unrealized Proposals

The study evaluates 50 held-out commission-style briefs, IDs 026–075, disjoint from the 25 briefs available during system development. The main matched-control direction survives removal of every individual brief and every complete coverage category. This supports the aggregate comparison on the held-out benchmark, but it should not be confused with evidence from a natural sample of contemporary-art commissions.

The benchmark briefs are LLM-generated commission-style tasks rather than briefs written by practicing artists during actual project development. They were constructed to cover five deliberately balanced forms of pressure: thematic/conceptual, site-specific, material-constrained, institutional/production-constrained, and anti-cliche/difficult. These categories are useful for coverage and sensitivity analysis, but they are not an empirical taxonomy of contemporary-art practice, and their equal frequency is artificial. The experiment therefore tests controlled generalization across unseen synthetic tasks, not ecological generalization to the idiosyncratic language, incompleteness, contradictions, private references, stylistic commitments, and evolving intentions that may appear in real artist-authored briefs.

The common commissioning context narrows the domain further. Proposals are interpreted within a temporary or medium-duration contemporary-art museum commission with a museum-controlled footprint and a shared production envelope. The results therefore do not establish performance under other commissioning regimes, such as public art, artist-run or commercial-gallery contexts, substantially lower-budget practice, or performance-centered production with different institutional constraints.

The benchmark size also places a specific boundary on subgroup interpretation. Each coverage category contains ten briefs. Leave-one-category-out analysis is useful as a robustness check of the overall matched-control direction; it is not evidence that AIArtist performs differently across artistic categories. Likewise, individual success and failure cases should not be interpreted as prevalence estimates for particular artistic mechanisms.

There are then two further representational gaps. First, the evaluated proposal is a structured textual dossier. Even before fabrication, a text can fail to carry qualities that would be evident through drawing, sound, scale model, embodied rehearsal, material sample, or spatial demonstration. The text-only external evaluation therefore measures the proposal as linguistically represented, not every artistic property the proposed work might possess.

Second, the evaluated objects are unrealized project hypotheses rather than fabricated installations. A dossier can make a coherent case for material behavior, scale, circulation, duration, maintenance, public encounter, staffing, and budget and still fail when produced. No engineering study, conservation review, accessibility review, fabrication process, installation period, maintenance cycle, commissioning decision, or audience observation validates the proposed works.

This text-to-work gap is unusually consequential in installation practice. Sound behaves differently in an actual room; materials wear unpredictably; visitors ignore intended routes; access and fire requirements alter spatial arrangements; natural light changes; repeated interaction modifies the work; and institutional labor that appears minor in prose may become constitutive in practice. Textual proposal evaluation can reason about such relations only counterfactually.

The post-hoc renderings do not close this gap. Their purpose is to make transformations in source dossiers visually legible—for example, a boundary moving from a frontal condition into a walkable field. Image generation can also supply convincing architecture, fabrication detail, lighting, scale, and finish that the text did not earn. Renderings are therefore explanatory visual hypotheses, were never shown to the evaluator, and provide no independent evidence that the represented work would function as depicted.

10.3 Internal Validity and Mechanism Attribution

M5 substantially matches the iterative scaffold but is not compute-identical, and several domain-structured components change together. The A5–M5 result is therefore a package-level comparison; component attribution requires targeted ablations. M5 is a deliberately strong matched control for the architecture studied here, not an exhaustive representative of generic multi-agent or iterative search. A different generic policy with different prompting, branching, memory, or selection could produce a different contrast.

RQ2 likewise supports only a conditional claim: adopted replacements are preferred on average to their exact parents, but no matched forced-replacement policy was generated. The experiment therefore cannot show that allowing retention itself causes higher quality or that com-

petitive refinement is superior to mandatory replacement. Role-specific adopted-replacement summaries are additionally conditioned on both generation and internal selection, so differences between adopted Focused and Structural replacements cannot identify an intrinsic causal advantage of one challenger role over the other.

RQ3 isolates terminal-ranking policy from candidate generation because Base and Hybrid receive identical candidate pools, but it does not isolate the pairwise tournament component alone: the two policies obtain separate stochastic pointwise assessments before producing their rankings. The Hybrid-minus-Base Utility@1 difference is not statistically resolved and its point estimate slightly favors Base (-0.0280 , 95% CI $[-0.0950, +0.0337]$, $p = .410$); secondary ranking metrics are mixed. The supported claim is therefore that selection is a separable recognition problem, not that either current ranking policy is reliably superior.

The three RQs are analytically distinct, but their external evidence is not wholly independent. RQ2 reuses exact-parent comparisons already contained in the exhaustive A5–B5-self matrices, and RQ3 derives candidate utilities from the same complete cross-pool judgments that underlie RQ1. Their convergence should therefore not be interpreted as three independent replications of one effect.

Finally, all core generation conditions use the same base generation model family. This is an advantage for the within-study comparison because variation in base-model capability does not confound the search-policy contrasts. It leaves model-generalization open. A stronger generator might reduce the marginal value of structured refinement; a weaker generator might benefit more from it; and a differently trained model might respond differently to opportunity-memo conditioning, preservation instructions, or Focused/Structural transformation permissions. Cross-model generation experiments are required before treating the architecture as model-independent.

10.4 Evaluation-Procedure Dependence and Exploratory Analyses

The primary external evaluation is exhaustive within each brief but not composed of independent judgments at the pair level. Each logical A5–control pair is presented in exact A/B-reversed orientations. Opposite orientations of the same logical pair are assigned to different fresh Temporary Chats, while each chat contains multiple comparisons. This makes order instability observable without eliminating the possibility of contextual carryover among comparisons processed in the same chat.

For this reason, the study treats the brief rather than the candidate pair as the inferential unit. The 7,500 physical judgments provide dense measurement of 50 held-out tasks; they do not constitute 7,500 independent benchmark observations. A stricter future design could place every logical comparison—or small randomized micro-batches—in an independent context.

Exact swaps expose rather than eliminate evaluator instability. Overall swap agreement is 78.37%, rising to 92.21% among 1,695 logical pairs where both orientations report confidence 4–5. Consensus-only analysis shows that the matched-control direction does not depend on the comparisons whose verdict visibly reverses under presentation order, but selecting only stable pairs changes the estimand and may preferentially retain easier distinctions. It should therefore remain a sensitivity analysis rather than be interpreted as a cleaner ground truth.

Relative-length analysis addresses another identifiable threat but should not be overread. A5 dossiers are somewhat longer than M5 dossiers, while the association between relative length and preference is weak and the A5 direction persists when A5 is not longer or is strictly shorter. This makes a simple monotonic verbosity explanation difficult to sustain. It does not establish that the holistic model judge is free of subtler stylistic biases involving fluency, curatorial vocabulary, structural completeness, tone, or apparent professional confidence.

Several analyses in the paper are explicitly post-hoc. The evaluator-rationale coding and the pointwise-to-tournament stage diagnostic were introduced to understand the evaluated experiment rather than as members of the planned primary endpoint family. Their role is mechanistic and hypothesis-generating. The separate neural-embedding semantic-dispersion diagnostic is retained only in Appendix A; it is not treated as evidence of artistic diversity and is not used to support the primary or mechanism claims.

The qualitative cases are selected for explanatory contrast rather than prevalence estimation. They illuminate mechanisms measured elsewhere and add no independent statistical evidence.

10.5 Field-Knowledge Validity: Model Recall Is Not Art-Historical Retrieval

AIrtist's field-context construction introduces a distinct information-quality boundary. The system uses model-mediated recall of artists, works, texts, and neighboring formats to construct a shared field map; it does not retrieve provenance-bearing evidence from a verified art-historical source. Such recall can be incomplete, inaccurate, culturally narrow, or anachronistic. The resulting field map should therefore be read as a working search prior that helps the system recognize nearby artistic attractors and formulate differentiation pressures, not as evidence that a recalled precedent is correctly described or that a proposal is objectively novel in contemporary art.

This matters because field awareness is one of AIrtist's distinctive search commitments. An erroneous or culturally narrow recalled neighborhood can steer review and opportunity synthesis toward the wrong comparison class even when later reasoning is internally coherent. Recalled precedents are accordingly used as internal prompts for search rather than as factual art-historical evidence in the external evaluation. A stronger future system should separate associative recall from verifiable retrieval, expose provenance when factual claims about precedent matter, and allow uncertainty or disagreement about the relevant artistic neighborhood to remain visible.

10.6 Human Co-Creative Validity Is Outside the Present Experiment

The intended artist-facing use model makes a final boundary especially important. The present study evaluates an autonomous concept-development backend, not negotiated human-AI co-creativity. It does not observe whether artists find the returned hypotheses generative, combine or reject them, experience ranking-induced fixation, benefit from seeing all five proposals, or find editable opportunity memos and criteria useful for articulating and redirecting their own intentions.

Improved sampled-set quality should therefore not be rewritten as improved human creativity, creative satisfaction, authorship, trust, agency, or co-creative experience. Nor does the present design establish that five is the right number of proposals, that ranking reduces cognitive load, or that one interface for exposing alternatives is preferable to another. The recognition results also do not establish that artists would identify the externally highest-utility candidate more reliably than the automatic ranker; retaining the full set is motivated by recoverability under ranking uncertainty, not by a demonstrated human-selection advantage.

Future validation should proceed at three levels. First, professional artists, curators, producers, and related practitioners should evaluate proposal sets and individual interventions, with analysis of disagreement among groups rather than assuming a single homogeneous human gold standard. Second, longitudinal interaction studies should test whether ranked alternatives, editable opportunity memos and criteria, and recoverable branches actually support exploration rather than fixation or cognitive overload. Third, selected proposals should be followed through technical review and, eventually, fabrication and situated encounter. These studies test different validity layers, and no larger model-judge benchmark can substitute for them.

11 Conclusion

A consequential part of contemporary-art practice occurs before an artwork has stabilized: an artist may know what problem, pressure, site, material condition, or institutional constraint matters without yet knowing what the work itself should become. This paper treats that interval as an explicit computational task. AIrtist develops an under-specified installation brief into several complete project hypotheses and models development as contestable transformation: proposed changes compete with the artistic state they are intended to replace rather than being accepted simply because additional inference has produced them.

AIrtist produces a stronger measured five-proposal set than all three planned controls. Against M5, the substantially matched generic-iteration condition that begins from the same five self-refined parents, the centered complete-pool effect is +0.2352 (95% CI [+0.1472, +0.3212], Holm $p < .0001$). Because A5 and M5 start from the same parents, the difference cannot be attributed to a luckier starting sample; it supports the full AIrtist development package without isolating which component or residual compute difference is responsible. Across 250 lineages, 22 final states remain incumbent-identical. Among 228 adopted replacements, challengers receive candidate-weighted exact-parent preference 0.6601 and a brief-balanced centered effect of +0.3347 (95% CI [+0.2143, +0.4520]). Thus structured development can improve the measured

candidate set while preserving non-intervention as an available realized outcome, without establishing the causal value of retention or any individual component.

Terminal selection remains less resolved. Base has higher mean Utility@1 than Hybrid (0.6633 versus 0.6353), while Hybrid more often places an externally highest-utility candidate first (0.32 versus 0.26); neither policy establishes superiority. Stronger measured candidates often already exist in the pool without being surfaced first. The experiment therefore establishes the recognition problem more clearly than the present solution. Candidate-set construction, adopted intervention quality, and attention allocation should be measured separately because failure at any one can produce a weak surfaced result. Post-hoc coding of stable evaluator rationales adds a mechanism-level qualification: A5 wins more often cite site or operational integration, whereas control wins more often cite feasibility or economy/restraint, suggesting both a strength and a recurrent over-development failure mode under the external evaluator.

Contemporary-art installation is central to this argument rather than merely a difficult benchmark. Changing material causality, spatial organization, spectator position, or institutional procedure can change a proposal's identity rather than simply improve its presentation. A diagnosis can therefore be correct while its repair is worse than the state it was meant to improve. Preservation, transformation, and selection should remain separately inspectable under such non-monotonic development.

The claims remain bounded by the evaluation setting. The benchmark briefs are synthetic commission-style tasks rather than project notes written by practicing artists; the evaluated outputs are textual proposals rather than realized works; and the principal quality signal comes from a single external model evaluator. No artist negotiates with the system inside the evaluated pipeline. The next empirical step is therefore twofold: test the measured preference signal against professional artistic and curatorial judgment, including patterns of disagreement, and study human-negotiated search around the same development backend so that artists can contest diagnoses, opportunity memos, candidate priorities, and recoverable branches.

The aim is not to automate the decision of what the artwork should be. It is to make the space in which that decision is formed richer, more inspectable, and less prematurely collapsed.

A Exploratory Neural-Embedding Diagnostic of Candidate-Pool Semantic Dispersion

This appendix records an exploratory representation-level diagnostic that is not used as evidence for RQ1–RQ3 or for the main-text mechanism claims. It measures semantic dispersion in textual proposal representations, not artistic, visual, art-historical, political, or realized-installation diversity.

The analysis uses the same 50 held-out briefs and the five A5 and five M5 proposals available for each brief (500 proposals total). From each dossier it uses only `artistic_operation`, `spatial_logic`, `material_logic`, and `spectator_encounter`. Texts are embedded with the frozen Qwen/Qwen3-Embedding-0.6B encoder without a task prompt, and embeddings are L2-normalized. For a pair of proposals, semantic distance is $1 - \cos(\mathbf{e}_i, \mathbf{e}_j)$; pool breadth is the mean distance across the ten unordered proposal pairs.

We compute two representations. In the concatenated-document representation, the four raw field values are joined in a fixed order with blank-line separators and embedded as one document. In the field-wise representation, each field is embedded separately; the four cosine distances for a proposal pair are averaged with equal weight before pool-level averaging. For each representation, A5 and M5 are compared within brief. Reported intervals are paired bootstrap 95% CIs over the 50 briefs (50,000 resamples), and p values are two-sided paired sign-flip Monte Carlo tests (200,000 random sign assignments).

Both aggregate representations point toward lower textual semantic dispersion for A5, but the effect is much smaller under equal-weight field-wise representation. The brief-level differences from the two representations are strongly associated (Pearson $r = .821$; Spearman $\rho = .806$), with the same sign on 80% of briefs. The field decomposition is heterogeneous: the negative difference is concentrated in `artistic_operation` and `material_logic`, while no directional difference is detected for `spatial_logic` or `spectator_encounter`.

A5 is also more verbose within this four-field representation: the mean concatenated length is 198.1 whitespace-delimited words per proposal for A5 and 146.1 for M5. However, the within-brief A5-minus-M5 length difference is not meaningfully associated with the corresponding breadth difference in this sample. For concatenated-document breadth, Pearson $r = .010$ and Spearman $\rho = -.045$; for field-wise breadth, Pearson $r = .144$ and Spearman $\rho = .083$.

Table 19: Exploratory neural-embedding diagnostic of within-pool semantic dispersion. Breadth is the mean of the ten pairwise cosine distances among the five proposals in each pool. Differences are paired by brief; intervals are paired bootstrap 95% CIs and p values are two-sided Monte Carlo sign-flip tests. Field-level tests are exploratory.

Representation	A5	M5	A5–M5	95% CI	p
Concatenated four-field document	0.3510	0.3964	−0.0454	[−0.0575, −0.0335]	< .00001
Equal-weight field-wise mean	0.3779	0.3889	−0.0109	[−0.0179, −0.0040]	.0033
<i>artistic_operation</i>	0.3988	0.4190	−0.0202	[−0.0321, −0.0083]	.0017
<i>spatial_logic</i>	0.3841	0.3791	+0.0050	[−0.0069, +0.0172]	.4237
<i>material_logic</i>	0.3912	0.4196	−0.0284	[−0.0415, −0.0154]	.00013
<i>spectator_encounter</i>	0.3375	0.3377	−0.0002	[−0.0107, +0.0102]	.9755

The result should therefore be read narrowly. It suggests modestly greater semantic concentration of A5 proposal texts under this frozen neural representation, especially in operation and material logic. It does not establish that A5 produces less diverse artworks, less visual or spatial variety, or less art-historical novelty. No independent encoder or human validation of this representation-level measure is included here, so the analysis remains hypothesis-generating and is intentionally separated from the primary results.

B Post-hoc Coding of External-Evaluator Rationales

This appendix documents the post-hoc analysis of pair-specific reasons produced by the same GPT-5.6 Sol evaluator that issued the pairwise preference judgments. The analysis therefore describes what the evaluator says it attended to; it is not an independent causal explanation of artistic superiority.

B.1 Sampling, blinding, and coding

The three primary RQ1 control campaigns contain 3,750 logical cross-pool comparisons, each presented in two exact A/B-reversed orientations. For compactness in this appendix, C1 denotes A5 versus B5-self, C2 denotes A5 versus M5, and C3 denotes A5 versus B5-raw. For rationale coding, 811 logical pairs whose mapped winner changed with orientation were excluded. Forty-seven further pairs were ties in both orientations and were excluded because the analysis asks about reasons for directional preference. This left 2,892 eligible swap-consistent directional logical pairs.

A stratified sample of 600 logical pairs was drawn with 100 pairs from each campaign $\times$ winner stratum (C1/C2/C3 $\times$ A5/control), spread across all 50 briefs. The first 25 pairs per stratum (150 total) formed a blinded codebook-development sample. A 12-category codebook was developed from those records and then frozen. The remaining 75 pairs per stratum (450 total) formed a separate frozen-codebook sample. During coding, the coder saw the two orientation-specific rationale texts for the logical pair but not system identity, canonical winner, candidate role, or experimental condition beyond the information already contained in the rationale text. The key was joined only after coding was complete.

Coding was LLM-assisted rather than independently human-coded. Each logical pair received one primary cited reason, up to two secondary reasons, a locus tag (winner strength, loser weakness, both, or unclear), and optional specific failure-mode tags. The preserved coding artifacts contain the complete blinded inputs, frozen codebook, output fields, and final labels, but they do not retain the exact coder model/version or inference configuration used for this post-hoc pass; we therefore do not attribute the coding to a specific model. The operational coding protocol is nevertheless recoverable from the frozen codebook: both orientation-specific rationales are read as one record; generic boilerplate is ignored; exactly one primary category (or UNCLEAR) and up to two secondary categories are assigned only from reasons explicitly stated in the rationale; the primary code reflects the most concrete substantive contrast supporting the verdict; and substantively irreconcilable rationales are coded UNCLEAR. Table 20 gives the frozen categories and the operational distinction each one was intended to capture.

Table 20: Frozen rationale-codebook categories used for the 450-pair blind coding. Categories describe the reason cited by the evaluator; coders were instructed not to infer properties that were not stated in the rationale.

Category	Operational meaning
Brief / constraint fit	Explicit compliance with or violation of a stated brief requirement or prohibition, such as required materials, wall-only installation, no surveillance, passive airflow, or a budget ceiling.
Feasibility / safety / maintenance	Physical or technical plausibility, safety, reliability, fabrication practicality, maintenance or reset burden, and operational viability when the issue is not primarily a formal brief violation.
Conceptual specificity / distinctiveness	Whether the artistic operation is specific, non-generic, and non-arbitrary rather than a familiar metaphor, decorative move, generic sensory effect, or interchangeable artistic format.
Conceptual legibility / directness	How clearly the proposed mechanism communicates the intended relation or proposition. This concerns intelligibility of the relation, not whether the idea is distinctive.
Material / causal consequence	Whether the cited strength or weakness concerns an actual material transformation or cause-and-consequence relation, rather than a merely illustrative, symbolic, atmospheric, or digital representation.
Spatial / bodily consequence	Room-scale or architectural consequence, bodily relation, circulation, passage, threshold, obstruction, choreography, or another spatial condition that is itself part of the artistic reason.
Temporal / process logic	Duration, accumulation, persistence, recurrence, sequence, reset, rhythm, irreversibility, or another process that unfolds over time.
Spectator encounter / access	The visitor's role, participation or non-participation, accessibility, forced disclosure, interaction burden, or another feature of how spectators encounter the work.
Site / operational integration	Integration of the artistic operation with real site or institutional processes such as staff workflow, service movement, irrigation, handling, maintenance, conservation, or other existing routines.
Economy / restraint / over-elaboration	Economy of means versus unnecessary apparatus, media, mechanisms, theatricality, gadget-like proliferation, polishing, or complexity that adds burden without comparable artistic gain.
Visual / perceptual force	Visual, sensory, atmospheric, or perceptual vividness when that quality itself is cited as the reason for preference, rather than merely described as part of the proposal.
Ethical / tonal framing	A cited problem of moralizing, didactic, therapeutic, confessional, spectacular, romanticized, or otherwise tonally inappropriate framing.

B.2 All-category results and stable contrasts in the frozen sample

The main-text Table 17 reports all twelve primary-reason categories, not only those with non-zero contrasts. Because campaign and winner strata were deliberately sampled in equal numbers, rates are first estimated within campaign $\times$ winner strata and then reweighted to the C1/C2/C3 mixture of the 2,892 eligible pairs separately within A5 wins and control wins. They therefore retain the intended “more often in A5 wins versus control wins” interpretation while avoiding an artificial equal-campaign mixture imposed by the qualitative sampling design. They remain sample-based estimates rather than direct coding of every eligible rationale.

Three primary-reason contrasts have 95% brief-cluster bootstrap intervals separated from zero. All three contrasts have the same direction separately in C1, C2, and C3 (Table 21). When a theme is counted if it appears as either the primary or a secondary cited reason, the corre-

Table 21: Campaign-specific A5-minus-control differences for the three stable primary-reason contrasts in the frozen 450-pair sample. The common sign across C1–C3 is a descriptive sensitivity check rather than a separately corrected confirmatory family.

Primary cited reason	C1	C2	C3
Site / operational integration	+6.7 pp	+9.3 pp	+14.7 pp
Feasibility / safety / maintenance	−13.3 pp	−5.3 pp	−5.3 pp
Economy / restraint / over-elaboration	−5.3 pp	−9.3 pp	−8.0 pp

sponding A5-minus-control differences become +14.1 percentage points for site/operational integration, −12.2 points for feasibility/safety/maintenance, and −18.0 points for economy/restraint/over-elaboration; all three intervals remain separated from zero. Restricting the sample to logical pairs for which both orientations report confidence 4–5 preserves the same three directions.

The other nine categories remain visible in Table 17. Their primary-reason intervals overlap zero, so they are not promoted as stable directional differences. This includes material/causal consequence, spatial/bodily consequence, and conceptual specificity/distinctiveness: all are substantively common in individual rationales, but the frozen sample does not support treating any of them as a distinctive explanation of the A5 advantage.

B.3 Illustrative rationale language

One A5-win rationale emphasizes operational integration: “The overhead cart tracking and wall ribbon avoid floor cuts and mist entirely while connecting service movement to real water flow.” By contrast, one control-win rationale emphasizes economy of means: “B is a clearer and simpler timed custody arrangement ... A adds sensor-triggered sound, tactile notes, and multiple lighting systems that dilute the core.” These examples illustrate the aggregate coding pattern; they are not additional observations beyond the coded sample.

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